



BPSC PRELIMS · BIOLOGY

Consolidated Biology Notes

Topic-first · duplicate-free · visual · PYQ-prioritised

33 source PDFs read	141 reviewed PYQs	20 topic-first units
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Every original figure used is captioned with its lecture PDF and page. External/editorial additions are blue; corrections are coral. Supplied-key conflicts remain visible rather than being silently rewritten.

Edition 1.0 · September 2026

How to use this book

This is a reconstruction—not a stack of lecture handouts. Repeated coverage was merged once, related material was brought together, and the notes were reordered around conceptual dependencies and BPSC recurrence.

PRIORITY A	Highest recurrence across the reviewed corpus; revise first.
EDITOR ADDITION	Assistant-supplied context/data beyond the source lectures.
EDITORIAL CORRECTION	A source typo, outdated statement or scientific ambiguity corrected explicitly.
PYQ LENS	A prior-question hook placed next to the concept it tests.

EXAM KEY · Coverage statement

All 33 supplied lesson PDFs were read end-to-end. The two L22 files are exact byte-identical duplicates, so they were counted once for content. Administrative slide furniture, repeated promotional text and exact duplicate facts were omitted; unique educational content was retained and re-organised.

Three-pass revision

Pass 1 · Priority A: plants, disease/immunity, nutrition, and ecology (when extended ecology is in scope).

Pass 2 · Systems: genetics, endocrine/reproduction, circulation, nerves, digestion and excretion.

Pass 3 · Retrieval: Rapid Revision Atlas → key-sensitive table → complete PYQ audit appendix.

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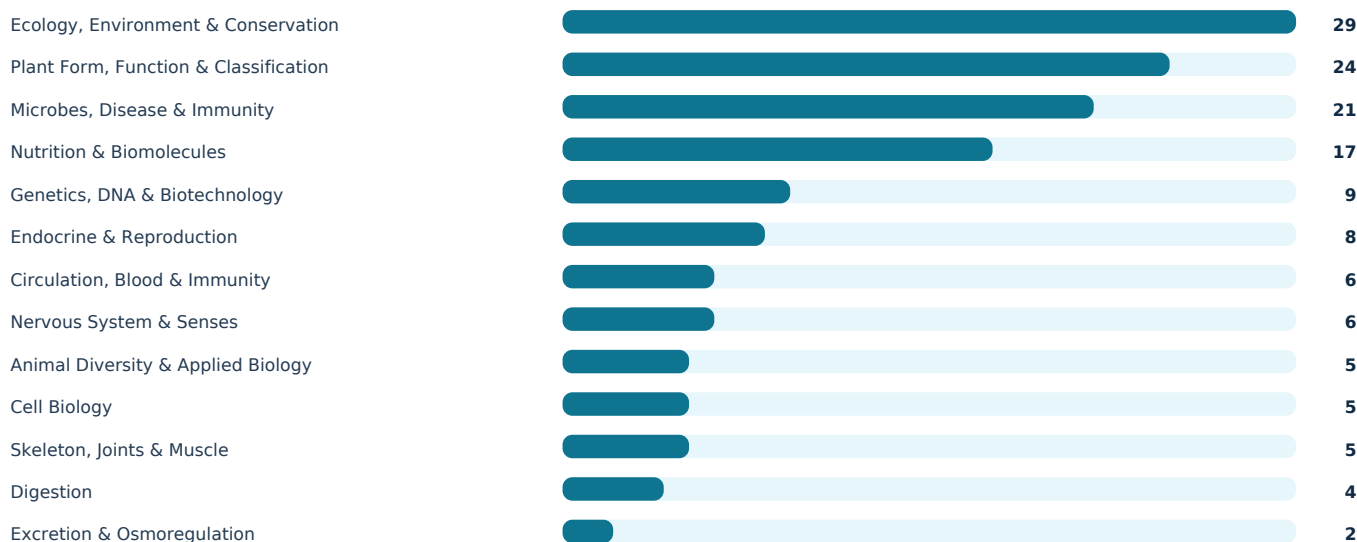
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PYQ-derived priority map

The two supplied workbooks yielded **141 manually reviewed biology-linked questions**: 111 core and 30 extended (ecology, conservation, agri-biology and public-health boundary). Keyword-only selection was not used.



Topic	All	Core	Latest 6	Papers	Tier
Ecology, Environment & Conservation	29	6	22	9	A
Plant Form, Function & Classification	24	22	20	8	A
Microbes, Disease & Immunity	21	19	11	9	A
Nutrition & Biomolecules	17	16	7	8	A
Genetics, DNA & Biotechnology	9	9	4	6	B
Endocrine & Reproduction	8	8	4	5	B
Circulation, Blood & Immunity	6	6	4	6	C
Nervous System & Senses	6	6	3	4	C
Animal Diversity & Applied Biology	5	3	4	4	C
Cell Biology	5	5	3	5	C
Skeleton, Joints & Muscle	5	5	4	4	C
Digestion	4	4	4	3	C
Excretion & Osmoregulation	2	2	0	2	C

KEY-SENSITIVE · How to read the ranking

Counting extended ecology makes Ecology/Environment rank first; in core biology alone, Plant Form/Function, Microbes/Disease and Nutrition lead. "Latest 6" means the 66th through 71st supplied examination groups. Frequency informs order but does not justify dropping foundational systems.

Latest-six signal

Ecology, Environment & Conservation 22 · Plant Form, Function & Classification 20 · Microbes, Disease & Immunity 11 · Nutrition & Biomolecules 7 · Animal Diversity & Applied Biology 4 · Circulation, Blood & Immunity 4 · Digestion 4 · Endocrine & Reproduction 4 · Genetics, DNA & Biotechnology 4 · Skeleton, Joints & Muscle 4 · Cell Biology 3 · Nervous System & Senses 3

UNIT 1

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

Biology Foundations, Classification & the Cell

Consolidated from L1, L32 · PYQ theme: Cell Biology

What biology studies

Biology is the science of life and living organisms—their structure, function, growth, origin, evolution and distribution. The name comes from Greek **bios** (life) + **logos** (study).

Branch	Scope	Founder / classic work
Botany	Plants	Theophrastus; Historia Plantarum
Zoology	Animals	Aristotle; Historia Animalium
Microbiology	Microscopic organisms: bacteria, viruses, fungi, protozoa	—
Genetics	Inheritance, genes and variation	Gregor Mendel is called the father of genetics
Ecology	Interactions of organisms with one another and the environment	Term associated with Ernst Haeckel
Anatomy	Physical structure of organisms and their parts	—
Physiology	How organisms, organs and systems function	—
Cell biology	Structure and function of cells	—
Molecular biology	DNA, RNA, proteins and information flow	—
Evolutionary biology	Change in species and populations over generations	—

Classification: the organising logic

- Aristotle grouped animals as **Enaima** (red blood present) and **Anaima** (red blood absent).
- Carl Linnaeus used the two-kingdom arrangement: **Plantae** and **Animalia**.
- R. H. Whittaker's five kingdoms are **Monera**, **Protista**, **Fungi/Mycota**, **Plantae** and **Animalia**. The stated bases are cell complexity, cellular organisation, cell wall and mode of nutrition.
- **Monera** includes prokaryotes such as bacteria and Mycoplasma. **Protista** includes unicellular eukaryotes such as protozoans, Chlamydomonas, Euglena, diatoms and yeast in the source scheme. Multicellular eukaryotes separate broadly into Plantae (cell wall; mainly autotrophic), Fungi/Mycota (cell wall; heterotrophic) and Animalia (no cell wall; heterotrophic).

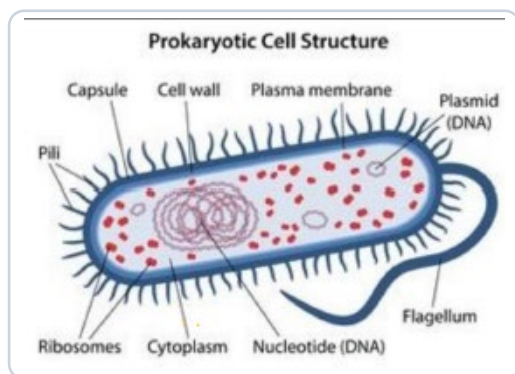
EDITOR ADDITION · Editorial update

Modern systems do not treat all “algae” or “protozoa” as single natural groups, and fungi are no longer classified as plants. For BPSC-style questions, first identify the classification system named in the stem.

Cell theory and discovery

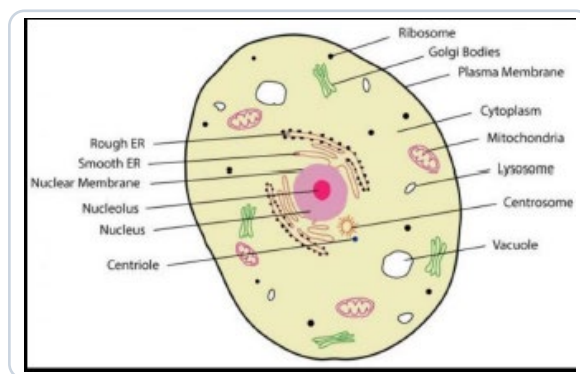
The **cell** is the structural and functional unit of life. All organisms are cellular, and new cells arise from pre-existing cells.

- Robert Hooke observed cork compartments and named “cells” in **1665**.
- Antonie van Leeuwenhoek first observed living cells/microorganisms.
- Cells are classified as **prokaryotic** or **eukaryotic** mainly by the presence of a membrane-bound nucleus and membrane-bound organelles.



Prokaryotic cell—nucleoid, ribosomes, wall/membrane, capsule and flagellum

Original source visual · L1. SYLLABUS ANALYSIS & CELL.pdf, p. 4



Eukaryotic animal cell—membrane-bound nucleus and organelles

Original source visual · L1. SYLLABUS ANALYSIS & CELL.pdf, p. 4

Prokaryote vs eukaryote — one-look table

Feature	Prokaryotic cell	Eukaryotic cell
Nucleus	No true nucleus; DNA lies in nucleoid	True membrane-bound nucleus
Typical size	About 1–10 μm	About 10–100 μm
Organisation	Usually unicellular	Unicellular or multicellular
Membrane-bound organelles	Absent	Present: mitochondria, ER, Golgi etc.
DNA	Usually circular; no histones in typical bacteria	Linear chromosomes associated with histones
Division	Binary fission	Mitosis; meiosis for germ-line formation
Ribosomes	70S	80S in cytosol; 70S in mitochondria/chloroplasts
Cell wall	Peptidoglycan in bacteria	Cellulose in plants; chitin in fungi; absent in animals
Examples	E. coli, Streptococcus, cyanobacteria	Amoeba, Paramecium, plants, animals, fungi
Recombination	Transformation, transduction or conjugation	Sexual reproduction / meiotic recombination

EDITORIAL CORRECTION · Source typo corrected

The source prints “70S” and “80S”; the intended ribosomal values are **70S** and **80S**. “S” is the Svedberg sedimentation coefficient and is not arithmetically additive.

Plant cell vs animal cell

Feature	Typical plant cell	Typical animal cell
Outer boundary	Cell wall outside plasma membrane	Plasma membrane only
Plastids	Present in photosynthetic/storage tissues	Absent
Centrosome/centrioles	Absent in most higher-plant somatic cells	Usually present
Vacuole	One large central vacuole may occupy most cell volume	Small, temporary vesicles/vacuoles
Nucleus position	Often peripheral after vacuole enlarges	Often more central
Nutrition	Commonly autotrophic in green tissues	Heterotrophic

PYQ focus

PYQ LENS · 64th: “Powerhouse of the cell”

Mitochondrion

PYQ LENS · 70th: Structural and functional unit of life

Cell

PYQ LENS · 56-59th: Smallest organism capable of autonomous growth and reproduction

Mycoplasma

EXAM KEY

Cell questions are often direct one-liners, but organelle composition and self-replication are increasingly used for statement-based questions.

UNIT 2

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

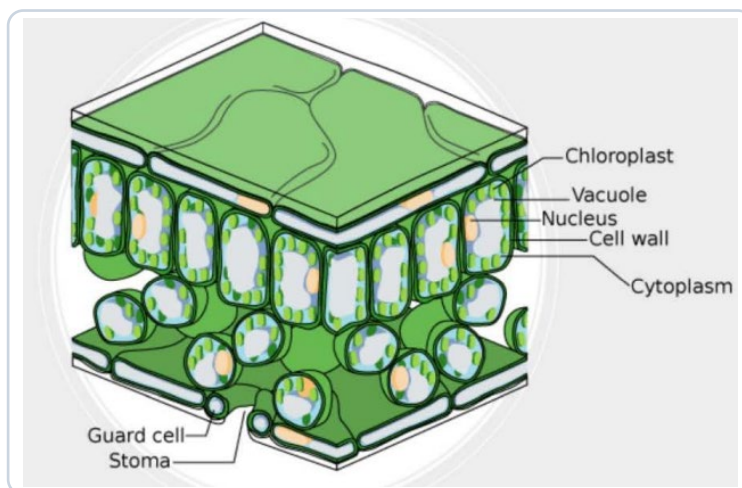
Cell Boundaries & Organelles

Consolidated from L2-L4 · PYQ theme: Cell Biology

Cell wall and plasma membrane

Cell wall

- A non-living, rigid layer outside the plasma membrane in plant cells; it provides shape, support and protection and resists osmotic bursting.
- **Plants:** cellulose, hemicellulose, pectins and proteins. **Algae:** composition varies and may include cellulose, galactans, mannans and calcium carbonate. **Fungi:** chitin and glucans.
- **Middle lamella:** mainly calcium pectate; cements adjacent cells. **Primary wall:** extensible wall formed during growth. **Secondary wall:** deposited inward after growth, often thick and strengthened.
- The wall protects against mechanical stress, physical shock and many pathogens while remaining porous to water and small solutes.

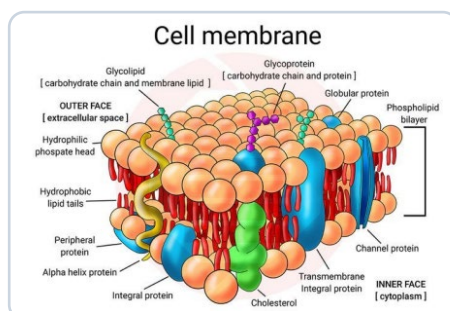


Leaf cross-section showing epidermis, mesophyll, vascular tissue and stomata

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 2

Plasma membrane — fluid mosaic model

- A thin, living, flexible and **selectively permeable** boundary found in every cell; it separates cytoplasm from the external environment and maintains homeostasis.
- Singer and Nicolson proposed the **fluid mosaic model (1972)**.
- Phospholipids form a bilayer: hydrophilic heads face aqueous surfaces; hydrophobic tails face inward. Cholesterol, proteins and carbohydrates occur within or on the bilayer.
- **Integral/intrinsic proteins** are partly or wholly embedded; **peripheral/extrinsic proteins** lie on the surface. Lateral mobility supports growth, junction formation, secretion and endocytosis.
- The membrane regulates movement of nutrients, oxygen, ions and wastes, receives signals and supports cell recognition.



Fluid-mosaic plasma membrane with phospholipids, cholesterol and proteins

Original source visual · L2. Cell Organelles.pdf, p. 2

PYQ LENS · 71st: Biological membranes are composed of—

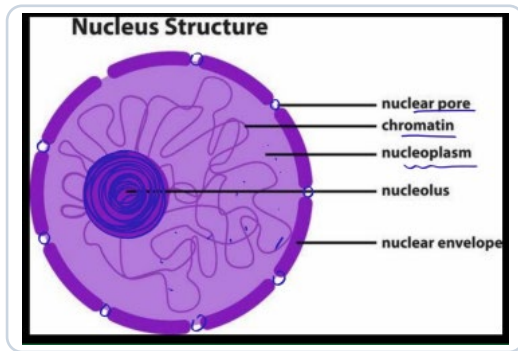
Lipids and proteins in proportions characteristic of cell type and species

The organelle dashboard

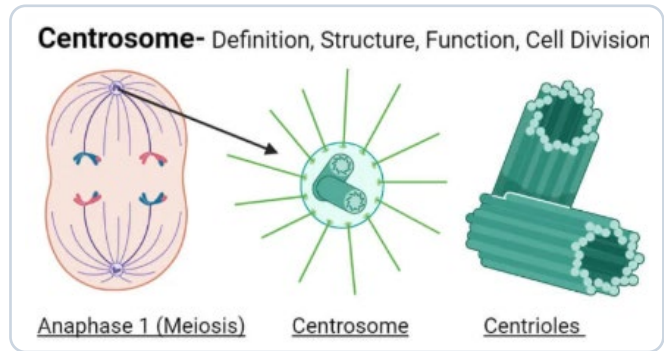
Organelle	Structure / location	Principal functions and exam hooks
Nucleus	Double membrane with pores; chromatin, nucleoplasm and nucleolus	Stores DNA; controls gene expression; nucleolus makes rRNA/ribosomal subunits
Mitochondrion	Double membrane; inner cristae and matrix; own circular DNA, RNA and 70S ribosomes	Aerobic respiration, ATP production, apoptosis, fatty-acid oxidation and heat production
Rough ER	Membranous cisternae continuous with outer nuclear membrane; ribosomes attached	Synthesis and folding of secreted/membrane proteins
Smooth ER	Tubules without ribosomes	Lipid and steroid synthesis; detoxification; selected metabolic roles
Golgi apparatus	Stacked flattened cisternae with cis and trans faces	Modifies, sorts and packages proteins/lipids; forms secretory vesicles and lysosomes
Lysosome	Single-membrane acidic vesicle with hydrolytic enzymes	Intracellular digestion and recycling; traditionally called “suicide bags”
Ribosome	Non-membranous rRNA-protein particle; 70S or 80S	Protein synthesis; assembled in nucleolus in eukaryotes
Plastid	Double membrane; plant/algal cells	Photosynthesis, pigment formation or storage
Vacuole	Tonoplast-bound compartment; large central form in mature plant cells	Storage, turgor, pH/ion balance and degradation
Centrosome	Two perpendicular centrioles in pericentriolar material; animal cells	Microtubule-organising centre; spindle and aster formation

Nucleus

- Robert Brown described the nucleus (1831/1833 in different source references). The outer nuclear membrane is continuous with ER.
- Nuclear pores control RNA and protein traffic. Nucleoplasm contains chromatin and one or more nucleoli.
- The **nucleolus** is non-membranous, rich in rRNA and protein, and is the site of rRNA synthesis and ribosomal-subunit assembly.
- Chromatin = DNA + histone and associated proteins**; it condenses into chromosomes during division.
- Ribosomes were characterised by George Palade and are non-membranous. Bacterial/organelar **70S = 50S + 30S**; eukaryotic cytosolic **80S = 60S + 40S**. The Svedberg values describe sedimentation and are not additive as ordinary numbers.
- A mature plant central vacuole may occupy around 90% of cell volume; small temporary vacuoles/vesicles occur in animal cells.
- Theodor Boveri is associated with discovery/description of the centrosome. It duplicates with the cell cycle and organises spindle microtubules and astral rays.



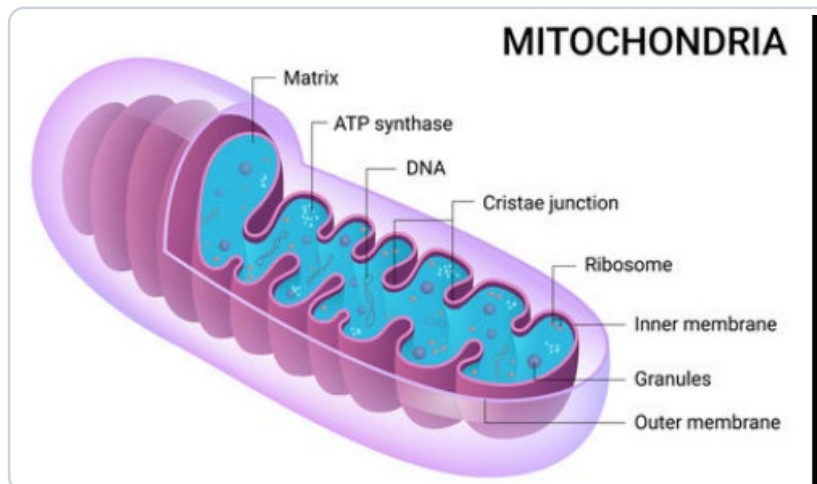
Nucleus—envelope, pores, chromatin, nucleolus and nucleoplasm
Original source visual · L4. Nucleus Centrosome, Pollination.pdf, p. 3



Centrosome and centrioles organising the spindle
Original source visual · L4. Nucleus Centrosome, Pollination.pdf, p. 2

Mitochondria

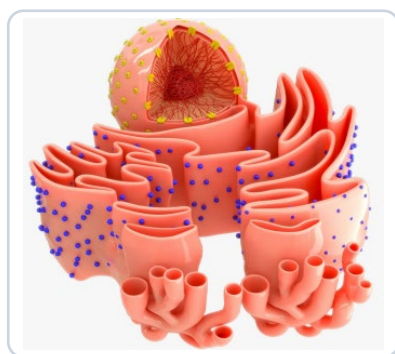
- Usually sausage-shaped or cylindrical; source dimensions are about 0.2–1.0 μm in diameter and 1.0–4.1 μm long.
- Smooth outer membrane surrounds an inner membrane folded into **cristae**, which increase area for electron transport and ATP synthesis. The **matrix** contains enzymes, mtDNA and ribosomes.
- Semi-autonomous: own circular DNA, RNA and 70S ribosomes, but depends heavily on nuclear genes.
- Discovery notes retained: Richard Altmann recognised “bioblasts” (1894); Carl Benda coined mitochondria (1898).



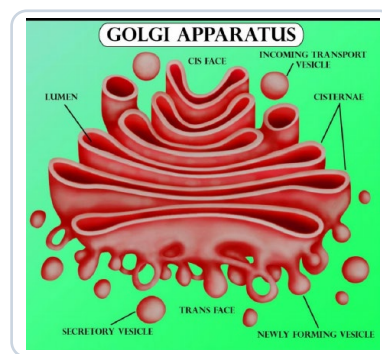
Mitochondrial membranes, cristae, matrix, mtDNA and ATP synthase
Original source visual · L2. Cell Organelles.pdf, p. 3

Endoplasmic reticulum, Golgi and lysosome

- ER partitions the cell into luminal and cytosolic compartments. It is absent in prokaryotes and mature mammalian RBCs.
- **RER**: abundant in protein-secreting cells. **SER**: abundant where lipids/steroids are made and drugs are detoxified.
- Golgi “post office”: the **cis face** receives ER vesicles; cisternae process cargo; the **trans face** dispatches it. Golgi also contributes to lysosomes, sperm acrosome and plant cell plate.
- Camillo Golgi observed the Golgi apparatus in 1897/1898. Lysosomes arise through the endomembrane system, contain many acid hydrolases and function near **pH 5**.



Rough and smooth endoplasmic reticulum
Original source visual · L2. Cell Organelles.pdf, p. 3

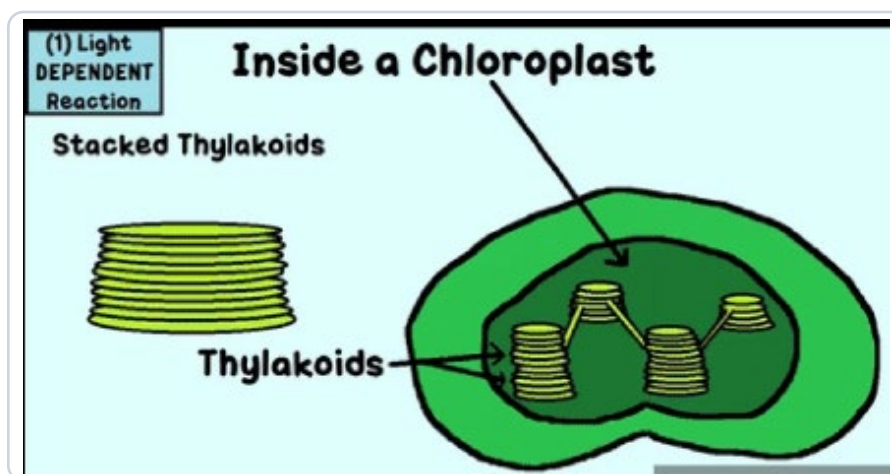


Golgi stack with cis and trans traffic
Original source visual · L3. Cell Organelles II.pdf, p. 2

Plastids

Type	Main role	Examples / details
Chloroplast	Photosynthesis	Green tissues; chlorophyll; thylakoids stacked as grana in stroma; semi-autonomous
Chromoplast	Coloured carotenoid pigment synthesis/storage	Flowers and fruits: tomato—lycopene; carrot—carotene; papaya—caricaxanthin
Leucoplast	Colourless storage plastid in non-green tissue	Amyloplast—starch; aleuroplast—protein; elaioplast—oil/lipid

- Plastids develop from **proplastids**. Ernest Haeckel is credited in the source with discovery and Schimper with observation/classification.
- Chlorophyll contains C, H, O, N and **Mg**; plants commonly contain chlorophyll **a** and **b**.
- Leucoplast examples retained: amyloplasts in potato/wheat/rice; aleuroplasts in legumes, pea, soybean and pulses; elaioplasts in castor, linseed and mustard seeds.



Chloroplast—double envelope, stroma, thylakoids and grana
Original source visual · L3. Cell Organelles II.pdf, p. 3

EDITORIAL CORRECTION · Pigment spellings corrected

The source prints “antholyanin” and “cathepsin” for colour. Standard terms are **anthocyanin** for many red/blue/purple colours and **capsanthin/capsorubin** for red chilli; turmeric is chiefly coloured by curcumin, not xanthophyll alone.

UNIT 3

PRIORITY B

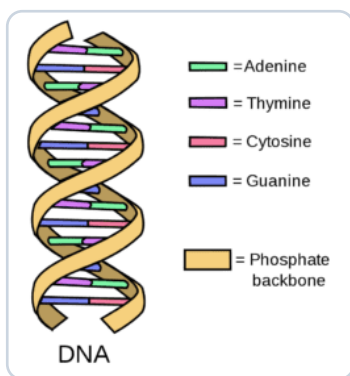
TOPIC-FIRST · DUPLICATES MERGED

Nucleic Acids, Genetic Code & Protein Synthesis

Consolidated from L6, L32 · PYQ theme: Genetics, DNA & Biotechnology

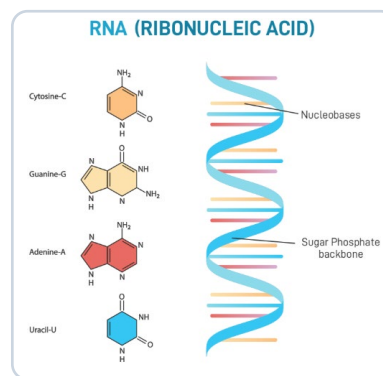
Building blocks

- Friedrich Miescher isolated an acidic nuclear substance from pus cells in 1869 and called it **nuclein**. Altmann popularised the term **nucleic acid**.
- Nucleic acids are nucleotide polymers. **Nucleoside = nitrogenous base + pentose sugar**; **nucleotide = nucleoside + phosphate**.
- Purines** have two rings: adenine (A), guanine (G). **Pyrimidines** have one ring: cytosine (C), thymine (T), uracil (U).
- A base joins carbon 1' of sugar: N1 of a pyrimidine or N9 of a purine. Phosphate is acidic and negatively charged.
- Pentoses are deoxyribose in DNA and ribose in RNA.



DNA double helix and base-pair legend

Original source visual · L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf, p. 2



RNA strand, sugar-phosphate backbone and bases

Original source visual · L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf, p. 3

DNA model and functions

- DNA is a long, negatively charged polymer of deoxyribonucleotides and is the hereditary material in almost all cellular organisms.
- Watson and Crick proposed the double-helix model in **1953**, drawing critically on X-ray evidence associated with Rosalind Franklin and Maurice Wilkins. Watson, Crick and Wilkins received the 1962 Nobel Prize.
- Two polynucleotide strands are **antiparallel** (5'→3' opposite 3'→5'), complementary and held by hydrogen bonds and base stacking.
- A pairs with T through **2 H-bonds**; G pairs with C through **3 H-bonds**. Pairing one purine with one pyrimidine keeps helix width nearly uniform.
- Functions: long-term information storage; self-replication before cell division; directing RNA and protein synthesis; mutation as a source of variation and evolution.

EDITOR ADDITION · Editorial context

"Who discovered DNA?" is key-sensitive. Miescher discovered the substance; Watson and Crick proposed the double-helical model; Franklin's diffraction data and Wilkins' work were crucial. Read the exact verb in the question.

PYQ LENS · 64th: "Who discovered DNA?" with Watson-Crick among options

Official-style answer may be "none" because Miescher discovered DNA; Watson-Crick described its structure

PYQ LENS · 69th: DNA double-helix structure

James Watson and Francis Crick

PYQ LENS · 56-59th: DNA synthesised in vitro

Arthur Kornberg

RNA and its forms

RNA	Approx. share	Key facts	Role
rRNA	About 80%	Made in nucleolus; most stable; structural/catalytic part of ribosome	Positions mRNA/tRNA and catalyses peptide-bond formation
tRNA	About 10-15%	Small “adapter” or soluble RNA; anticodon and amino-acid attachment site	Carries a specific amino acid to ribosome
mRNA	About 1-5%	Messenger name associated with Jacob and Monod; relatively short-lived	Carries codon sequence transcribed from DNA
Genomic RNA	—	Genetic material in many viruses; dsRNA in reovirus	Passes viral hereditary information

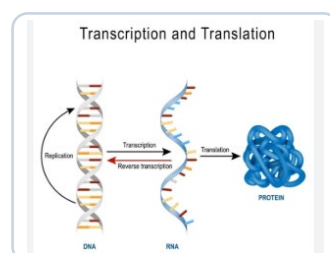
- RNA is usually single-stranded, contains ribose and uses **U** instead of T. Reovirus is a double-stranded RNA exception cited in the source.

Feature	DNA	RNA
Strands	Usually double helix	Usually single strand
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Stability	Generally more stable	Generally less stable
Main role	Long-term information storage	Expression of information and, in some viruses, heredity
Location	Nucleus; also mitochondria/chloroplasts	Nucleus and cytoplasm, including ribosomes

Central dogma and protein synthesis

DNA → RNA → protein summarises the usual flow of sequence information. Replication copies DNA; transcription makes RNA from DNA; translation reads mRNA to assemble a polypeptide.

- **Transcription:** RNA polymerase uses one DNA strand as template and makes complementary RNA 5'→3'.
- **Translation:** a ribosome reads mRNA codons; tRNAs match anticodons and deliver amino acids; peptide bonds form a polypeptide.
- Protein synthesis supplies enzymes, structural proteins and many peptide hormones; it underpins growth, repair and metabolic regulation.



Transcription of DNA to RNA followed by translation to protein

Original source visual · L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf, p. 4

Genetic code

- George Gamow advanced the triplet-code idea; Nirenberg, Matthaei and Khorana were central to deciphering the code.
- A **codon** is a three-base mRNA sequence specifying an amino acid or stop signal. Four bases in triplets yield $4^3 = 64$ **codons**, enough for 20 standard amino acids.
- **AUG** encodes methionine and commonly initiates translation. **UAA, UAG and UGA** are stop/nonsense codons.
- The code is degenerate (most amino acids have multiple codons), unambiguous for a given codon, nearly universal, non-overlapping and read in a fixed frame.

PYQ LENS · 71st: A nonsense codon among UAA/AUG/UUU/UGC

UAA

PYQ LENS · 69th: Expanded form of mRNA

Messenger ribonucleic acid

EDITOR ADDITION · Full codon quick map

Phe UUU/UUC; Leu UUA/UUG/CUN; Ile AUU/AUC/AUA; Met AUG; Val GUN; Ser UCN/AGU/AGC; Pro CCN; Thr ACN; Ala GCN; Tyr UAU/UAC; stop UAA/UAG/UGA; His CAU/CAC; Gln CAA/CAG; Asn AAU/AAC; Lys AAA/AAG; Asp GAU/GAC; Glu GAA/GAG; Cys UGU/UGC; Trp UGG; Arg CGN/AGA/AGG; Gly GGN. **N means any base.**

UNIT 4

PRIORITY B

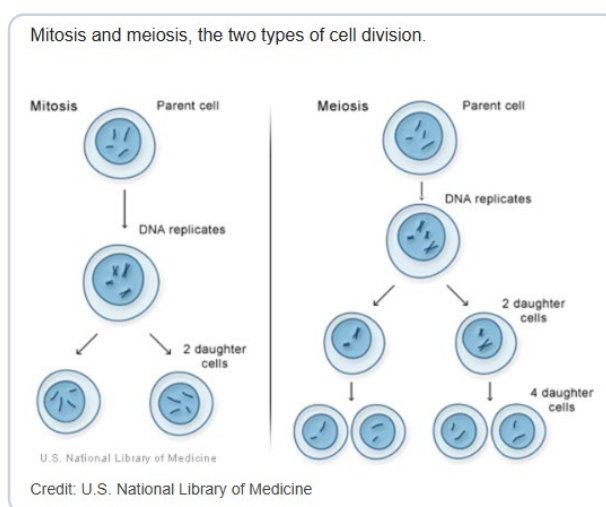
TOPIC-FIRST · DUPLICATES MERGED

Cell Division, Mendelian Genetics & Evolution

Consolidated from L7, L29, L32 · PYQ theme: Genetics, DNA & Biotechnology

Cell cycle overview

- Interphase is an active preparation period: G1 growth, S DNA replication, G2 preparation. Division phase includes nuclear division (**karyokinesis**) and cytoplasmic division (**cytokinesis**).
- **Mitosis** is equational division: one parent cell → two genetically similar cells with the same chromosome number.
- **Meiosis** includes one DNA replication followed by two divisions: one diploid germ cell → four genetically varied haploid cells.

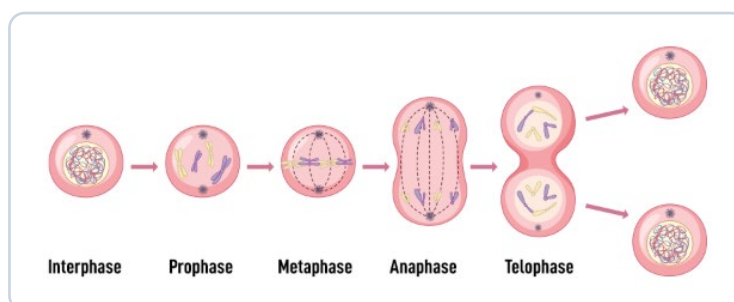


Mitosis produces two similar cells; meiosis produces four haploid cells
Original source visual · L7. Cell Division (Mitosis & meiosis).pdf, p. 2

Mitosis — PMAT

Stage	What happens
Prophase	Chromatin condenses; nucleolus and nuclear envelope disappear; centrosomes move poleward; spindle develops
Metaphase	Chromosomes align individually at the equator; spindle microtubules attach to kinetochores
Anaphase	Centromeres divide and sister chromatids move to opposite poles
Telophase	Chromosomes decondense and nuclear envelopes/nucleoli reform
Cytokinesis	Animal cell: inward cleavage furrow. Plant cell: central cell plate grows outward

- Significance: growth, replacement and repair, genetic stability and asexual reproduction in many organisms.



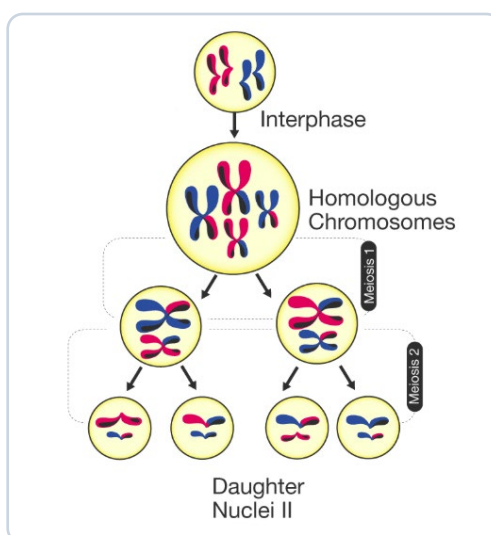
Mitosis from interphase through cytokinesis
Original source visual · L7. Cell Division (Mitosis & meiosis).pdf, p. 2

Meiosis

- **Meiosis I** is reductional: homologous chromosomes separate. **Meiosis II** is equational: sister chromatids separate.

Prophase-I stage	Memory hook	Defining event
Leptotene	Thin threads	Chromosomes become visible
Zygotene	Zippering	Homologues pair by synapsis; bivalents/tetrads form
Pachytene	Pair exchanges	Crossing over between non-sister chromatids; recombinase acts
Diplotene	Drift apart	Synaptonemal complex dissolves; chiasmata visible
Diakinesis	Done / terminalise	Chiasmata terminalise; chromosomes fully condense

- Metaphase I aligns homologous pairs; anaphase I separates homologues while centromeres remain intact; telophase I may form two haploid nuclei/cells.
- Meiosis II proceeds through prophase II, metaphase II, anaphase II and telophase II, yielding four haploid products.
- Importance: gamete formation, stable chromosome number across sexual generations, and variation through crossing over and independent assortment.



Homologue separation in meiosis I and chromatid separation in meiosis II
Original source visual · L7. Cell Division (Mitosis & meiosis).pdf, p. 4

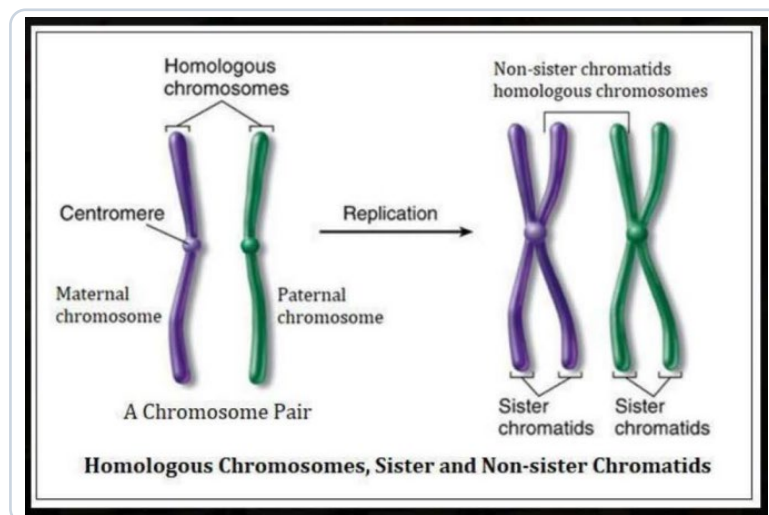
Mitosis vs meiosis

Feature	Mitosis	Meiosis
Cells	Somatic cells	Germ-line cells
Divisions	One	Two after one S phase
Products	2	Usually 4
Ploidy	Conserved	Halved
Genetic similarity	Largely identical	Different
Synapsis/crossing over	Absent	Present in prophase I
Main purpose	Growth, repair, asexual reproduction	Gametes, variation

Genetics vocabulary

Term	Meaning
Heredity	Transmission of biological traits from one generation to the next
Variation	Differences among individuals; genetic and/or environmental
Gene	DNA sequence that contributes to a functional product/trait; basic unit of heredity
Allele	Alternative form of a gene
Genotype	Allelic constitution
Phenotype	Observable outcome of genotype interacting with environment
Dominant allele	Expressed in a heterozygote in a simple Mendelian relation
Recessive allele	Masked in heterozygote; expressed when no dominant allele is present
Homozygous	Two identical alleles, e.g. TT or tt
Heterozygous	Two different alleles, e.g. Tt

- William Bateson coined **genetics** (1906); Gregor Mendel is the father of genetics. Human somatic cells have **46 chromosomes: 22 pairs of autosomes + one pair of sex chromosomes** (XX typically female, XY typically male).



Homologous chromosomes, replication and sister chromatids
Original source visual · L29. Genetics and Evolution.pdf, p. 3

Mendel's laws

- Mendel worked with garden pea (*Pisum sativum*) and seven pairs of contrasting traits. His work established particulate inheritance.
- **Dominance:** in a simple heterozygote, the dominant allele determines the F1 phenotype.
- **Segregation / purity of gametes:** two alleles of a gene separate during gamete formation; each gamete receives one.
- **Independent assortment:** alleles of different genes assort independently when genes are unlinked; a classic dihybrid F2 gives **9:3:3:1**.
- Pea contrasts retained: seed shape round/wrinkled; seed colour yellow/green; flower colour violet/white; pod shape inflated/constricted; pod colour green/yellow; flower position axial/terminal; stem tall/dwarf.

	Flower color	Seed shape	Seed color	Pod color	Pod shape	Plant height	Flower position
DOMINANT	 Purple	 Round	 Yellow	 Green	 Inflated	 Tall	 Axial
RECESSIVE	 White	 Wrinkled	 Green	 Yellow	 Constricted	 Short	 Terminal

Mendel's seven contrasting pea characters
Original source visual · L29. Genetics and Evolution.pdf, p. 4

Sex determination and modern applications

- In humans the ovum always carries X, whereas sperm carries X or Y. Fertilisation by X-bearing sperm usually produces XX and by Y-bearing sperm XY; therefore the sperm determines chromosomal sex in the textbook model.
- Mutation** is a change in DNA sequence and is a source of variation. **Biotechnology/genetic engineering** manipulates DNA for medicine, agriculture and research.
- DNA fingerprinting/profiling** compares highly variable DNA markers to support identity, forensic investigation and biological-parentage analysis; statistical interpretation is essential.

EDITOR ADDITION · Modern qualification

Chromosomal sex determination is more complex in development than “XX female/XY male” alone; the line above is the standard BPSC model and exceptions exist.

Sex-linked and chromosomal disorders

Condition	Genetic basis	High-yield description
Down syndrome	Trisomy 21	47 chromosomes; characteristic physical/developmental features
Edwards syndrome	Trisomy 18	47 chromosomes
Patau syndrome	Trisomy 13	47 chromosomes
Klinefelter syndrome	47, XXY	Phenotypic male; hypogonadism/infertility; some feminine secondary traits
Turner syndrome	45, X	Phenotypic female; gonadal dysgenesis and infertility
Haemophilia	Usually X-linked recessive	Deficiency of clotting factor; affected male X^hY ; carrier female $X^H X^h$
Red-green colour blindness	Usually X-linked recessive	Difficulty distinguishing red/green; affected male X^cY ; carrier female $X^N X^c$

- Autosomal-recessive examples retained: cystic fibrosis, sickle-cell disease, Tay-Sachs disease, phenylketonuria, thalassaemia and many forms of albinism.

EDITORIAL CORRECTION · Genetics corrections

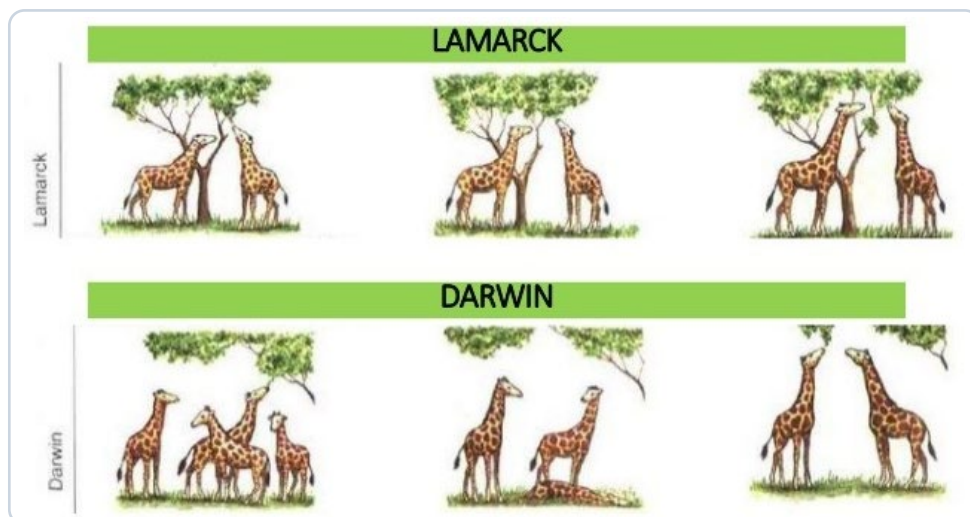
Affected homozygous females with haemophilia are rare but **not biologically impossible**; the source's claim that they always die as embryos is incorrect. Common inherited colour blindness is chiefly red-green, not inability to distinguish all primary colours.



Ishihara-style plates used to screen colour-vision deficiency
Original source visual · L29. Genetics and Evolution.pdf, p. 4

Evolution

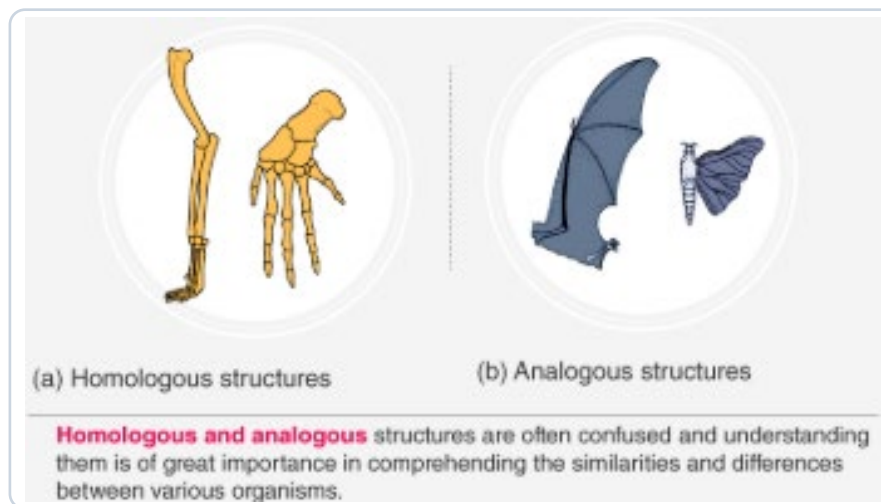
- Evolution is heritable change in populations over generations.
- **Lamarck:** use/disuse and inheritance of acquired characters; historically important but not the accepted general mechanism.
- **Darwin and Alfred Russel Wallace:** overproduction → variation → struggle for existence → differential survival/reproduction (natural selection) → adaptation.
- The modern synthesis combines natural selection with mutation, recombination, gene flow and genetic drift as causes of evolutionary change.



Lamarckian and Darwinian explanations illustrated with giraffes
Original source visual · L29. Genetics and Evolution.pdf, p. 5

Evidence

- **Fossils/palaeontology:** ordered remains in rock strata document succession and transitional forms.
- **Homologous structures:** common origin and basic plan, potentially different functions—forelimbs of human, whale, bat and cheetah; vertebrate hearts/brains; insect mouthparts; testes and ovaries; potato and ginger as shoots; radish and carrot as roots; molecular similarities.
- **Analogous structures:** different origin/plan but similar function—bird and insect wings; octopus and mammal eyes; penguin and dolphin flippers; sweet-potato root and potato stem as storage organs; bee and scorpion stings.
- **Vestigial structures:** reduced remnants—nictitating membrane, auricular muscles, vermiform appendix, coccyx, reduced canine/wisdom teeth, body hair, male nipples and segmented abdominal muscles.



Homologous forelimbs versus analogous wings
Original source visual · L29. Genetics and Evolution.pdf, p. 6

PYQ LENS · 65th: Author of On the Origin of Species

Charles Darwin

PYQ LENS · 56-59th: Reason for large-scale diversity among organisms (as keyed in supplied sheet)

Adaptation; remember mutation supplies heritable variation

UNIT 5

PRIORITY A

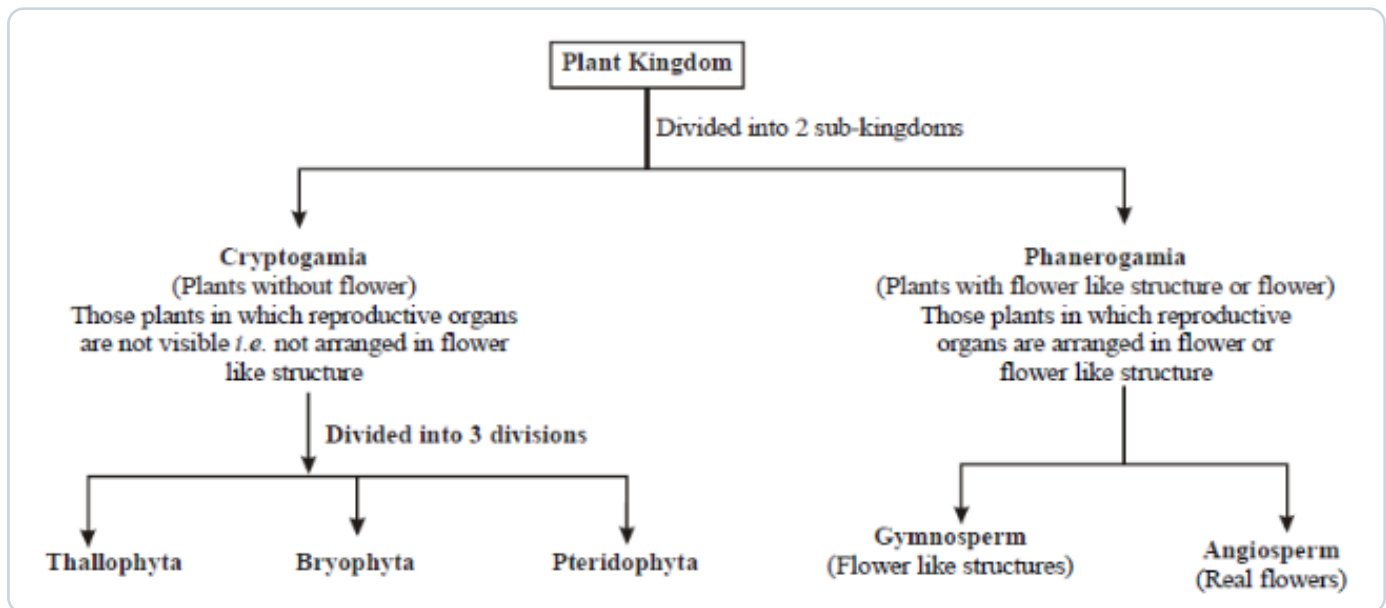
TOPIC-FIRST · DUPLICATES MERGED

Plant Kingdom & Major Groups

Consolidated from L24 · PYQ theme: Plant Form, Function & Classification

Classification frame

- Kingdom Plantae in the source comprises eukaryotic, multicellular, primarily autotrophic organisms. Botany studies plants; Theophrastus is its traditional father.
- A. W. Eichler divided plants into **Cryptogamae** (seedless/hidden reproductive structures) and **Phanerogamae** (seed plants) in an early phylogenetic arrangement.



Retained classification tree for plant groups
Original source visual · L24. Plant Kingdom.pdf, p. 2

Comparison of major groups

Group	Body / vascular tissue	Reproduction	Key hook	Examples
Algae	Thallus; no true root, stem, leaf; mostly aquatic	Spores, fragmentation and sexual modes	Phycology	Spirogyra, Chlamydomonas, Chlorella, Volvox
Bryophytes	Root absent; rhizoids and stem/leaf-like parts; no vascular tissue	Water needed for fertilisation	"Amphibians" of plant kingdom; moist/shady	Riccia, Marchantia, mosses
Pteridophytes	True roots, stem, leaves; xylem and phloem	Spores; water still needed for fertilisation	Vascular cryptogams	Psilotum, Lycopodium, Selaginella, Equisetum, ferns
Gymnosperms	Vascular, usually woody; exposed ovules/seeds	Seeds but no fruits	Naked-seeded plants	Cycas, Pinus, Cedrus, Gnetum, Ephedra
Angiosperms	Flowering vascular plants	Double fertilisation; seeds enclosed in fruits	Monocots + dicots	Wolffia to Eucalyptus

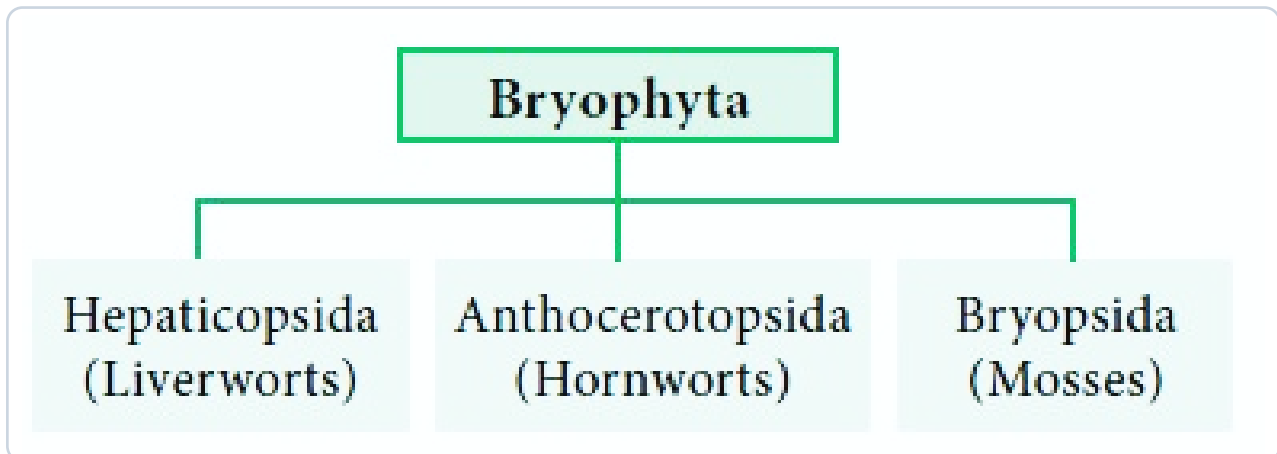
Algae

- Green algae: chlorophyll a and b; examples Chlorella, Chara, Cladophora, Volvox, Ulva and Acetabularia.
- Brown algae: fucoxanthin dominant; Fucus, Dictyota, Laminaria and kelps (commercial iodine source).

- Red algae: phycoerythrin prominent; Porphyra, Gracilaria and Gelidium.

Bryophytes

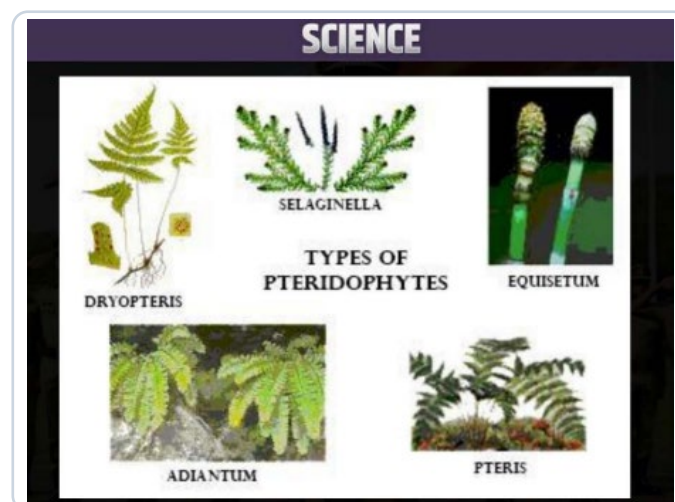
- Earliest land-plant lineages derived from green-algal ancestors; dependent on water for fertilisation because male gametes are motile.
- Lack true roots and vascular tissue, so remain small; water conduction is largely cell-to-cell/parenchymatous. They favour moist, shady places (**sciophytes**).
- Bryophyta term credited in the source to Robert Braun; study is **bryology**. Hedwig is widely called father of bryology; the source also mentions Cavers, and Prof. Shiv Ram Kashyap as father of Indian bryology.
- Main classes shown: liverworts (Hepaticopsida), hornworts (Anthocerotopsida), mosses (Bryopsida).



Bryophyte classes—liverworts, hornworts and mosses
Original source visual · L24. Plant Kingdom.pdf, p. 3

Pteridophytes

- Called vascular cryptogams and sometimes “reptiles of the plant kingdom.” They have roots and vascular tissue but still need water for fertilisation.
- Typical xylem lacks vessels and phloem lacks companion cells. Primary root is often short-lived and replaced by adventitious roots; underground stem may be a rhizome.
- **Psilopsida**: primitive vascular plants; Psilotum. **Lycopsida**: club mosses with microphylls and strobili; Lycopodium, Selaginella. **Sphenopsida**: horsetails with jointed stems and strobili; Equisetum. **Pteropsida**: ferns with large fronds, young leaves circinnately coiled.
- Fern examples: Azolla (small aquatic fern, biofertiliser); Adiantum (maidenhair/walking fern; leaf-tip propagation); Alsophila (tree fern); Pteridium, Dryopteris, Marsilea, Pteris, Salvinia. Marsilea, Azolla and Salvinia form sporocarps.
- Giant ancient pteridophyte vegetation contributed substantially to coal-forming deposits.



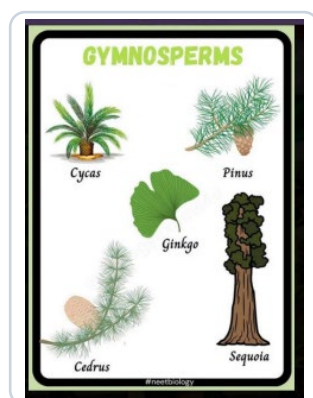
Representative pteridophytes and forms
Original source visual · L24. Plant Kingdom.pdf, p. 3

EDITORIAL CORRECTION · Leaf terminology

Selaginella has **microphylls**; ferns have **megaphylls/macrophylls**. The source reverses one of these labels in a sentence.

Gymnosperms

- Ovules are not enclosed by an ovary, so seeds remain “naked” and fruits do not form. Roots are usually tap roots; stem may be unbranched (Cycas) or branched (Pinus, Cedrus).
- Pinus has mycorrhizal roots; Cycas has coralloid roots with nitrogen-fixing cyanobacteria.
- Conifers reduce water loss with needle leaves, thick cuticle and sunken stomata. Many are xerophytic and occur on Himalayan slopes.
- Typical gymnosperm xylem lacks vessels and phloem lacks sieve tubes/companion cells; **Gnetum, Ephedra and Welwitschia** are vessel-bearing exceptions cited.
- Secondary growth makes most woody/arborescent; Ephedra may be shrubby and Gnetum ula a woody climber.



Common gymnosperm forms—Cycas, Ginkgo, conifers and Sequoia
Original source visual · L24. Plant Kingdom.pdf, p. 5

Angiosperms

- Flowers bear pollen and ovules; ovules become seeds enclosed by ovary-derived fruits. Angiosperms range from tiny Wolffia to Eucalyptus over 100 m and supply food, fodder, fuel, medicines and commercial products.
- Two classic classes: monocotyledons and dicotyledons.

PYQ LENS · 67th: Pine, fir, spruce, cedar, larch and cypress belong to

Gymnosperms

PYQ LENS · 71st: First to classify plants using vascular-tissue characters

A. P. de Candolle

PYQ LENS · 70th: Spore formation occurs in which listed organisms?

Question is multi-correct in biological reality; fungi, algae, yeast and ferns can form spores

UNIT 6

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

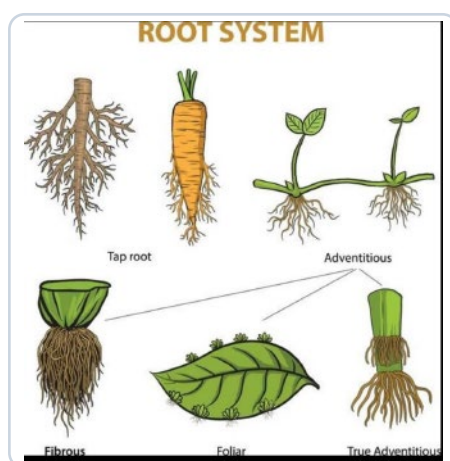
Plant Morphology, Pollination, Fertilisation & Fruits

Consolidated from L4, L5, L25 · PYQ theme: Plant Form, Function & Classification

Root

- Develops from the radicle. Tap root retains the primary root; fibrous systems replace it with many similarly sized roots; adventitious roots arise from stem/leaves.
- Tropisms: roots are generally **positively geotropic**, **positively hydrotropic** and **negatively phototropic**.

Modification	Function	Examples
Storage roots	Food reserve	Carrot, cassava, sweet potato
Prop roots	Mechanical support from branches/stem	Banyan
Stilt/adventitious support roots	Brace lower stem	Maize
Pneumatophores	Upward respiratory roots in waterlogged soil	Rhizophora / mangroves



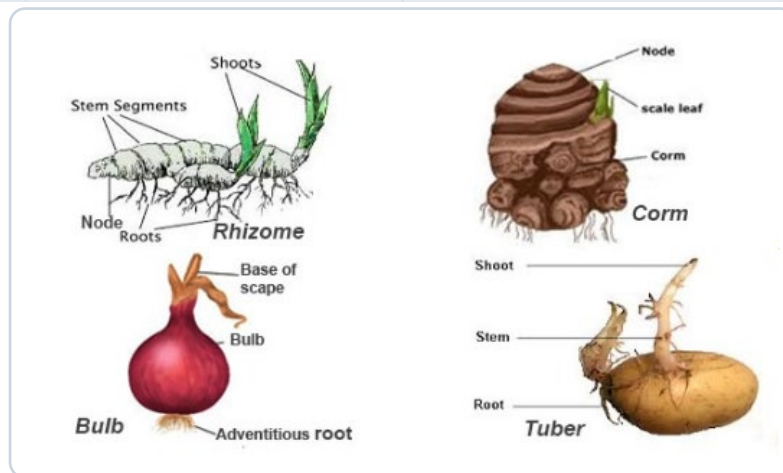
Tap, fibrous and adventitious root systems
Original source visual · L25. Plant Morphology.pdf, p. 2

Stem and modifications

- Stem develops from plumule, bears nodes and internodes, leaves, buds and flowers. The presence of nodes/internodes distinguishes ginger as a stem.

Category	Type	Examples / function
Underground	Rhizome	Ginger, turmeric; source also lists banana
Underground	Tuber	Potato
Underground	Corm	Colocasia, Amorphophallus (zamikand), saffron, Alocasia
Underground	Bulb	Onion, garlic
Sub-aerial	Runner	Doob grass, Oxalis, strawberry; roots below and shoots/leaves above nodes
Sub-aerial	Stolon	Wild strawberry, jasmine, peppermint; arching branch roots on contact
Sub-aerial	Sucker	Mint, pineapple, chrysanthemum, banana; underground basal branch emerges

Category	Type	Examples / function
Sub-aerial	Offset	Pistia, Eichhornia; short aquatic runner with rosette and roots
Aerial	Stem tendril	Grapevine, Passiflora and gourds; climbing
Aerial	Stem thorn	Carissa, Bougainvillea, pomegranate, Citrus; defence
Aerial	Phylloclade/cladode	Opuntia/Euphorbia; green photosynthetic stem



Rhizome, corm, bulb and tuber compared
Original source visual · L25. Plant Morphology.pdf, p. 3

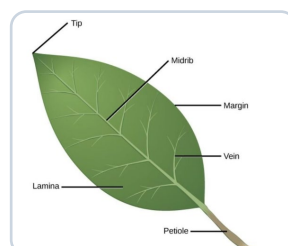
EDITORIAL CORRECTION · Example correction

Garlic is a **bulb**, not a rhizome. Bermuda/doob grass is classically a runner; strawberry can form runners/stolons depending on terminology.

Leaf

- Leaf is a flattened lateral appendage borne at a node. Parts: **leaf base**, optional **stipules**, **petiole**, and photosynthetic **lamina** with midrib and veins. An axillary bud lies in its axil.

Modification	Role	Examples
Leaf tendril	Climbing	Lathyrus aphaca; leaflet tendril in garden/sweet pea
Leaf spine	Defence and reduced water loss	Opuntia, cacti, Argemone
Insectivorous pitcher	Trapping insects	Nepenthes; lamina forms pitcher
Water-storage pitcher	Storage, not insectivory	Dischidia; whole leaf modified
Leaf bladder	Aquatic prey capture	Utricularia
Phyllode	Flattened photosynthetic petiole	Australian Acacia, Parkinsonia
Leaflet hook	Climbing	Cat's nail, Bignonia unguis-cati
Snap trap	Insect capture	Dionaea / Venus flytrap



Leaf parts—base, petiole, lamina, midrib, margin and veins
Original source visual · L25. Plant Morphology.pdf, p. 4

PYQ LENS · 66th: Ginger is a stem rather than a root because

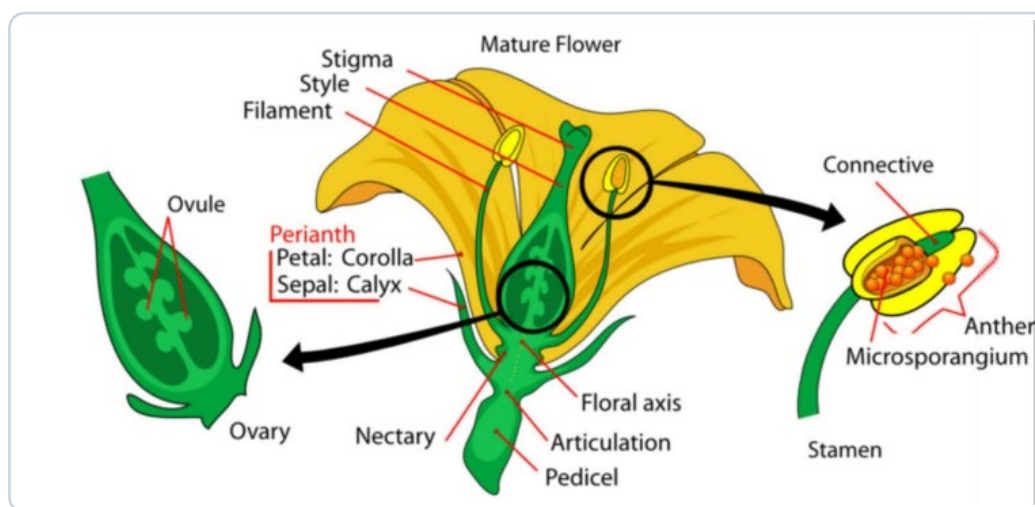
It has nodes and internodes

PYQ LENS · 68th: Petiole

Attaches leaf/lamina to stem

Flower architecture

- A flower is a modified reproductive shoot with four whorls: **calyx (sepals)**, **corolla (petals)**, **androecium (stamens)** and **gynoecium (carpels/pistil)**.
- Sepals protect the bud; petals attract pollinators; a stamen consists of filament + pollen-producing anther; a carpel consists of receptive stigma, style and ovary containing ovules.



Stamen, carpel and accessory floral parts
Original source visual · L28. Plant Physiology.pdf, p. 3

Pollination — de-duplicated

Pollination is transfer of pollen from **anther to stigma**. Compatibility in the pollen-pistil interaction determines whether pollen germinates and its tube grows.

Mode	Definition	Genetic/agent note
Autogamy	Same flower	Self; requires synchrony or mechanisms ensuring contact
Geitonogamy	Different flowers on the same plant	Needs a pollinating agent but genetically self
Xenogamy	Flower on a different plant of the same species	True cross-pollination; introduces different parental genotype

- Cleistogamous** flowers remain closed and ensure selfing: *Viola*, *Oxalis* and *Commelina* (source spelling “Camelina”). **Chasmogamous** flowers open and expose reproductive parts.
- Anemophily (wind)**: abundant light, non-sticky pollen; exposed stamens; large feathery stigma.
- Hydrophily (water)**: rare, especially in some aquatic monocots—*Vallisneria*, *Hydrilla* and *Zostera*/seagrasses.
- Zoophily**: bees, butterflies, beetles, birds and bats; flowers often conspicuous, scented and/or nectar-bearing.

PYQ LENS · 67th: Best definition of pollination

Transfer of pollen grain from anther to stigma

Double fertilisation and post-fertilisation

- Pollen tube reaches the ovule and releases two male gametes.

- **Syngamy:** male gamete (n) + egg (n) → zygote (2n).
- **Triple fusion:** second male gamete (n) + two polar nuclei (n+n) → primary endosperm nucleus (3n).
- Together they constitute **double fertilisation**, characteristic of angiosperms.
- Post-fertilisation: zygote → embryo; ovule → seed; ovary → fruit; ovary wall → pericarp.

Fruit layer	Position	Example / exam point
Epicarp	Outer pericarp	Fruit skin
Mesocarp	Middle pericarp	Edible fleshy part of mango
Endocarp	Inner pericarp	Hard “stone” around mango seed
Seed	Mature ovule	Embryo plus stored food/covering

PYQ LENS · 71st: Edible part of mango

Mesocarp

PYQ LENS · 66th: Which listed fruit is not a true fruit?

Apple—an accessory/false fruit because thalamus contributes greatly

PYQ LENS · 56-59th and 66th: Clove is obtained from

Dried flower bud

UNIT 7

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

Plant Tissues, Transport & Annual Rings

Consolidated from L8 · PYQ theme: Plant Form, Function & Classification

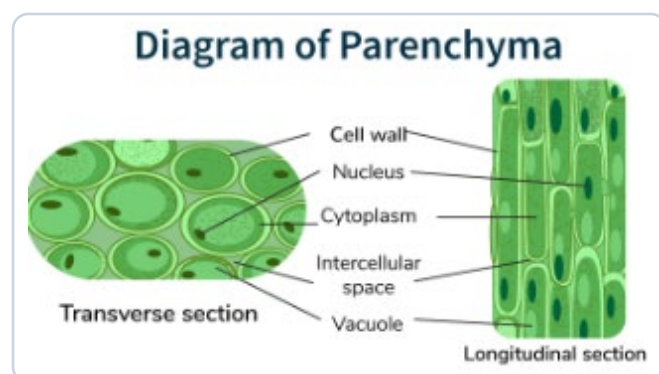
Meristematic tissue

- Meristems contain actively dividing, small, thin-walled cells with dense cytoplasm, prominent nuclei and little/no intercellular space. Nägeli is associated with the meristem concept.

Meristem	Location	Growth effect
Apical	Root and shoot tips	Primary growth / length
Intercalary	Bases of leaves or internodes, especially grasses	Elongation and regrowth
Lateral	Sides of stems/roots: vascular and cork cambia	Secondary growth / girth

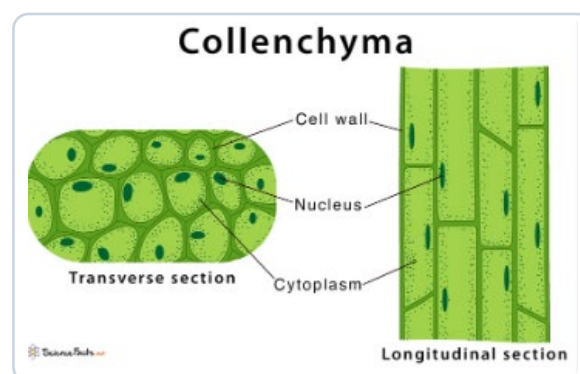
Permanent tissues

Tissue	Cells / wall	Functions and location
Parenchyma	Living, thin-walled, large vacuole, intercellular spaces	Storage; photosynthesis as chlorenchyma; air spaces as aerenchyma in aquatic plants
Collenchyma	Living, unevenly thickened at corners	Flexible support in young stems, petioles and below epidermis
Sclerenchyma	Dead at maturity; thick lignified wall; fibres or sclereids	Strength, hardness and rigidity



Parenchyma in transverse and longitudinal view

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 2

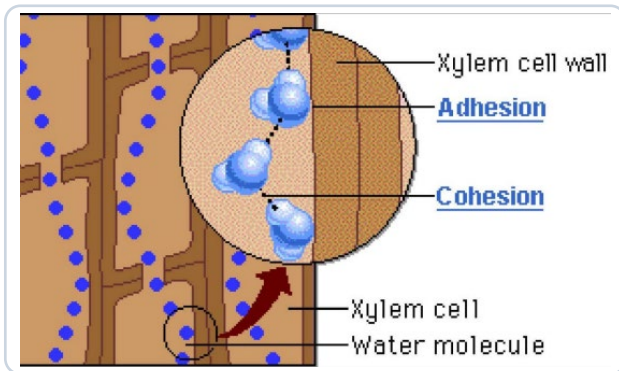


Uneven corner thickening in collenchyma

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 2

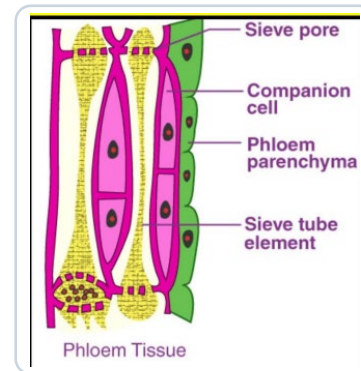
Complex permanent tissue

System	Elements	Transport
Xylem	Tracheids, vessels, xylem parenchyma, xylem fibres	Water/minerals, mainly root to shoot; transpiration pull is major long-distance driver
Phloem	Sieve-tube elements, companion cells, phloem parenchyma, phloem fibres	Organic food from source to sink; direction can change with source-sink relation



Cohesion and adhesion in a xylem conduit

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 3



Phloem sieve tube, companion cell and sieve plate

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 3

- Xylem flow is usually described as unidirectional upward; phloem translocation is physiologically bidirectional across the plant, though a given sieve tube carries bulk flow in one direction at a time.

PYQ LENS · 71st: Xylem function

Transport of water and dissolved minerals

PYQ LENS · 68th: Water reaches great height in trees because of suction pull caused by

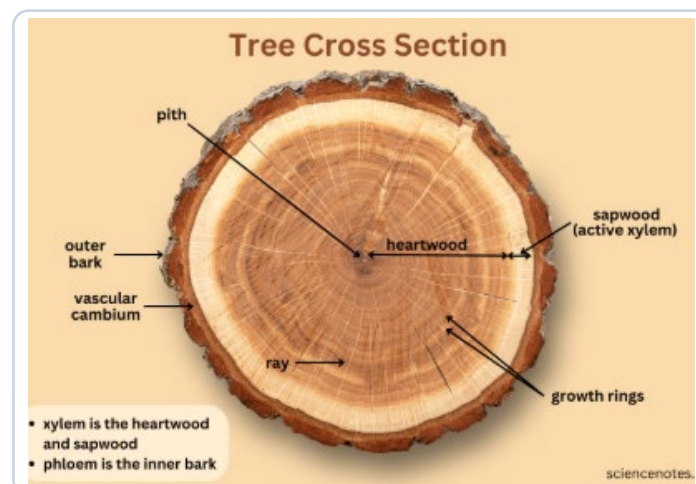
Transpiration

PYQ LENS · 68th: Stomata open/close mainly due to change in

Guard-cell water content/turgor

Secondary growth and annual rings

- Vascular cambium adds secondary xylem inward and secondary phloem outward. Seasonal activity creates visible **annual rings** in many woody stems.
- One year's early wood + late wood generally forms one ring. Ring counts estimate tree age; ring width/density can record climate. Study of dated rings is **dendrochronology**.



Tree cross-section showing growth rings, sapwood, heartwood and cambium

Original source visual · L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf, p. 4

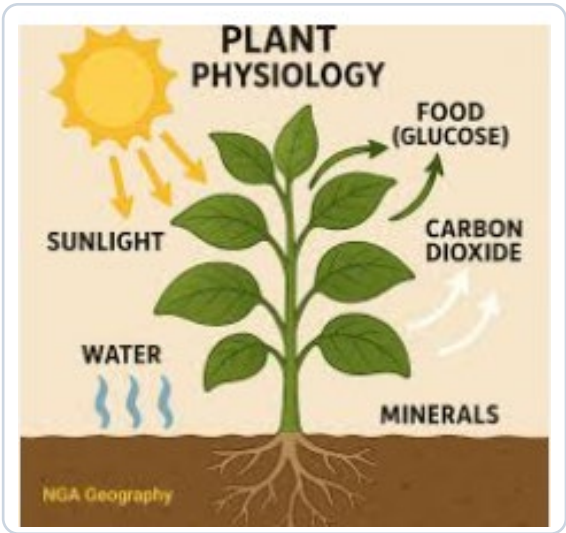
Plant Physiology, Mineral Nutrition & Diseases

Consolidated from L8, L26, L27 · PYQ theme: Plant Form, Function & Classification

Photosynthesis

Green plants convert light energy to chemical energy, reducing CO₂ to carbohydrate and releasing O₂ from water. It occurs principally in chloroplasts.

- Overall classroom equation: $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$ in the presence of chlorophyll.
- Joseph Priestley showed that plants restore air; Jan Ingenhousz established the role of light and green parts. Melvin Calvin mapped the carbon-reduction cycle; Robert Hill demonstrated light-driven oxygen release in isolated chloroplasts.
- **Light reactions** on thylakoid membranes use photosystems, electron carriers and ATP synthase; water is split, O₂ is released, and ATP/NADPH form.
- **Calvin cycle** in stroma fixes CO₂ using Rubisco, reduces intermediates and regenerates RuBP; ATP and NADPH are consumed.



Original plant-physiology visual retained from the photosynthesis lecture
Original source visual · L28. Plant Physiology.pdf, p. 2

C₃, C₄ and CAM

EDITOR ADDITION · Added-scope marker

This comparison is an assistant-added completion prompted by recent plant-physiology PYQs; the source lecture states basic photosynthesis but does not develop all three carbon-fixation pathways.

Feature	C ₃	C ₄	CAM
First stable product	3-PGA (3C)	Oxaloacetate/malate (4C)	Organic acid at night
Initial fixation	Rubisco in mesophyll	PEP carboxylase in mesophyll	PEP carboxylase at night
Separation	None	Spatial: mesophyll → bundle sheath	Temporal: night → day
Photorespiration	Higher, especially hot/dry	Low	Low
Examples	Wheat, rice, many temperate plants	Maize, sugarcane, sorghum	Cacti, pineapple, many succulents

- C₄ plants display **Kranz anatomy**. They spend additional ATP but concentrate CO₂ around Rubisco and are efficient in high light, high temperature and water stress.
- CAM plants open stomata chiefly at night, storing CO₂ as organic acids and using it by day while stomata remain mostly closed.

PYQ LENS · 71st: A C₄ plant among pineapple/pea/sugarcane/barley

Sugarcane

Respiration in plants

Plants respire continuously. Gas exchange is by diffusion through stomata, lenticels and root surfaces; no specialised respiratory organ or long-distance gas-transport system is required.

- **Glycolysis:** cytosol; glucose → 2 pyruvate + net 2 ATP + 2 NADH.
- **Aerobic respiration:** pyruvate oxidation and Krebs/TCA cycle in mitochondrial matrix; electron transport and oxidative phosphorylation on inner membrane. O₂ is final electron acceptor.
- Approximate educational yield is commonly stated as 36–38 ATP per glucose; actual cellular yield varies and modern accounting is often ~30–32 ATP in eukaryotes.
- **Fermentation:** anaerobic regeneration of NAD⁺. Yeast produces ethanol + CO₂; muscles under oxygen limitation produce lactate.

EDITORIAL CORRECTION · Source equation clarified

The lecture equation reverses products in places. Aerobic respiration **consumes O₂ and releases CO₂ + H₂O while conserving energy in ATP**.

Water relations and movement

EDITOR ADDITION · Added-scope marker

The cohesion-tension, water-potential and stomatal-control synthesis below is assistant-added to connect the source xylem material with supplied PYQs.

- Water potential is greatest in pure water; solutes lower it. Water enters root hairs osmotically, crosses cortex via apoplast and symplast, is forced through endodermal membranes by the Casparian strip, and enters xylem.
- **Transpiration** is water-vapour loss, mainly through stomata. It cools leaves, helps mineral ascent and drives cohesion-tension flow, but excessive loss causes wilting.
- **Cohesion** holds water molecules together; **adhesion** attracts them to xylem walls. Together with negative pressure from evaporation, they maintain the water column.
- **Root pressure** can contribute, especially at night/small plants, and may cause guttation; it cannot by itself explain ascent in the tallest trees.
- Stomata are epidermal pores controlled by paired guard cells. High turgor generally opens the pore; turgor loss closes it.

Mineral nutrition

- **Ash analysis** identifies mineral elements remaining after plant material is burned. More than 60 elements can be detected across plants, but 17 are generally considered essential; C, H and O are obtained chiefly from CO₂/water and are termed non-mineral nutrients, while most others enter as mineral ions. Beneficial elements cited in the source include Na, Si, Se and Co.
- Plants require elements for structure, enzymes, osmotic/charge balance and metabolism. **Macronutrients** are needed in relatively large amounts; **micronutrients** in trace amounts.

Element	Class	Major role	Indicative deficiency
N	Macro	Amino acids, nucleic acids, chlorophyll; vegetative growth	General chlorosis, older leaves first
P	Macro	ATP, nucleic acids, phospholipids; roots, flowering and seed	Stunting, dark/purplish foliage

Element	Class	Major role	Indicative deficiency
K	Macro	Enzyme activation, stomata, osmotic balance	Marginal chlorosis/necrosis, weak stems
Ca	Macro	Cell wall/membrane and signalling	Death of shoot/root tips
Mg	Macro	Central atom in chlorophyll; enzymes	Intervinal chlorosis of older leaves
S	Macro	Cysteine/methionine and coenzymes	Chlorosis, often younger leaves
Fe	Micro	Electron carriers and chlorophyll formation	Intervinal chlorosis, young leaves
Mn	Micro	Enzymes and photosystem-II water splitting	Chlorosis/necrotic spotting
Zn	Micro	Enzymes and auxin-related metabolism	Short internodes, small leaves
B	Micro	Cell wall, sugar transport, meristem/reproduction	Death of tips, poor fruit/seed set
Mo	Micro	Nitrate reduction/nitrogen fixation	Nitrogen-deficiency-like symptoms
Cl	Micro	Osmotic/ionic roles and photosynthesis	Wilting/chlorosis (rare)
Ni	Micro	Urease cofactor	Urea accumulation (rare)

PYQ LENS · 70th: Nutrient commonly keyed for rapid early vegetative development

Nitrogen; distinguish from phosphorus and root establishment

Nitrogen fixation and biofertilisers

- Atmospheric N_2 is unavailable to most plants until converted to ammonia/nitrate. Biological nitrogen fixation uses the oxygen-sensitive enzyme **nitrogenase**.
- Rhizobium** forms nodules on legumes such as pea, gram and groundnut; **Frankia** associates with selected non-legumes.
- Free-living/asymbiotic fixers include *Azotobacter* (aerobic), *Clostridium* (anaerobic) and *Azospirillum* (associative, especially grasses/cereals).
- Cyanobacteria *Anabaena* and *Nostoc* fix nitrogen. The **Azolla-Anabaena** association is used in paddy fields.
- Mycorrhiza is a fungus-root mutualism: fungi improve mineral/water absorption, notably phosphorus, while plants supply carbohydrate; it can protect against root-borne pathogens and stress.

PYQ LENS · 71st: Mycorrhiza helps plants through

Increased mineral absorption and protection from disease

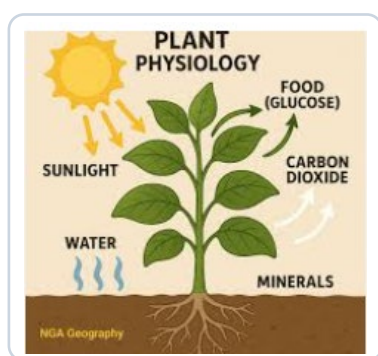
EDITOR ADDITION · Crop-specific correction

For groundnut, the crop-specific inoculant is **Rhizobium/Bradyrhizobium**, not *Azospirillum*. *Azospirillum* is mainly associated with grasses and cereals.

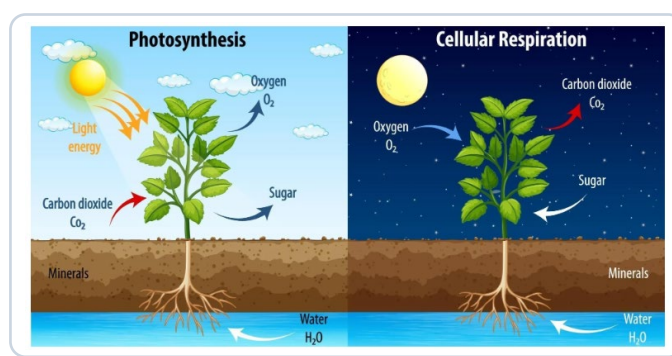
Plant hormones and movements

Hormone	Origin / classic clue	Actions
Auxin (IAA)	Shoot tip, young leaves and seeds	Cell elongation, apical dominance, rooting, tropisms, parthenocarp; synthetic 2,4-D is a weedicide
Gibberellin	Discovered from “bakanae” disease of rice caused by Gibberella	Stem/internode elongation, bolting, seed germination, fruit growth
Cytokinin	Dividing tissues, roots, fruits/seeds	Cell division, lateral-bud growth, nutrient mobilisation and delayed senescence
Abscissic acid (ABA)	Mature leaves, roots under stress	Dormancy, stomatal closure, growth inhibition and stress response
Ethylene	Gaseous hormone in ripening/senescent tissues	Fruit ripening, abscission/senescence; epinasty and triple response

- **Tropism** is directional growth according to stimulus: phototropism, geotropism/gravitropism, hydrotropism, thigmotropism and chemotropism.
- Shoots generally show positive phototropism and negative gravitropism; roots generally reverse these. Pollen-tube growth toward the ovule is chemotropic; tendril coiling is thigmotropic.
- **Nastic movement** depends on stimulus intensity, not direction: touch closure of Mimosa and sleep movements are examples.



Original overview of major plant-physiology processes
Original source visual · L28. Plant Physiology.pdf, p. 2



Photoperiodism and the effect of day length on flowering
Original source visual · L28. Plant Physiology.pdf, p. 5

Asexual and sexual plant reproduction

Mode	Description	Examples
Vegetative propagation	New plant from root, stem or leaf; clone of parent	Potato tuber, sweet-potato root, Bryophyllum leaf, rose cutting, onion bulb, strawberry runner
Budding	Outgrowth enlarges and separates	Yeast (a fungus used in the source table)
Fragmentation	Each fragment grows independently	Spirogyra
Spore formation	Resistant dispersal units germinate in favourable conditions	Ferns, mosses and fungi
Sexual reproduction	Gamete fusion creates recombined offspring	Flowering plants through pollination and fertilisation

- **Photoperiodism** is a plant's developmental response to relative day/night length, especially flowering. Short-day, long-day and day-neutral plants respond differently; phytochrome participates in light sensing.

EDITOR ADDITION · Added detail

The phytochrome mechanism and explicit short-/long-/day-neutral grouping are assistant-added context to explain the retained photoperiodism visual.

PYQ LENS · 56-59th: "Sleep movement" of plants

Nastic movement

PYQ LENS · 67th: Hormone associated with ripening

Ethylene

Plant diseases — agent-first revision

Disease / crop	Causal agent	Exam-safe classification
Citrus canker	Xanthomonas citri	Bacterial
Black rot of cabbage/crucifers	Xanthomonas campestris pathovar campestris	Bacterial
Crown gall	Agrobacterium tumefaciens	Bacterial
Red stripe of sugarcane	Acidovorax avenae subsp. avenae	Bacterial
Little-leaf of brinjal	Phytoplasma	Mollicute/phytoplasmal
Tobacco mosaic	Tobacco mosaic virus	Viral
Leaf curl	Several plant viruses, often whitefly transmitted	Viral
Late blight of potato	Phytophthora infestans	Oomycete; fungus-like
Early blight of potato/tomato	Alternaria solani	Fungal
Rusts	Puccinia spp.	Fungal
Smut	Ustilago spp.	Fungal
Powdery mildew	Erysiphales	Fungal
Red rot of sugarcane	Colletotrichum falcatum	Fungal
Ergot of rye	Claviceps purpurea	Fungal
Wilt diseases	May be Fusarium/Verticillium, bacteria or other agents	Read crop + agent in stem
Root-knot	Meloidogyne nematodes	Nematode

EDITORIAL CORRECTION · Disease-table corrections

Citrus canker is bacterial, not fungal. Little-leaf of brinjal is phytoplasmal, not a conventional virus. Late blight is caused by an oomycete; exam books often group it loosely with fungal diseases.

UNIT 9

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

Biomolecules, Nutrition, Vitamins & Minerals

Consolidated from L14, L28 · PYQ theme: Nutrition & Biomolecules

Modes of nutrition

Mode	How food is obtained	Examples
Autotrophic	Makes organic food from CO ₂ and water using light or chemical energy	Green plants, cyanobacteria and chemosynthetic bacteria
Holozoic heterotrophic	Ingestion → digestion → absorption → assimilation → egestion	Humans, dogs, Amoeba
Saprotrophic	Extracellular digestion of dead/decaying matter followed by absorption	Mushrooms, many fungi and bacteria
Parasitic	Obtains nutrients from a living host and harms it	Cuscuta, tapeworm, leech

Biomolecules at a glance

Class	Building block	Main roles	Energy value
Carbohydrate	Monosaccharide	Immediate energy, storage, structural cellulose	4 kcal/g
Protein	Amino acid	Enzymes, structure, transport, defence, many hormones	4 kcal/g
Lipid	Glycerol + fatty acids (typical fat)	Dense energy, membranes, insulation, steroid precursors	9 kcal/g
Nucleic acid	Nucleotide	Heredity and information transfer	Not a routine energy source

- Carbohydrates contain C, H and O, often roughly (CH₂O)_n. Monosaccharides include glucose, fructose and galactose; disaccharides include sucrose, lactose and maltose; polysaccharides include starch, glycogen and cellulose.
- Starch** stores carbohydrate in plants; **glycogen** stores it chiefly in animal liver and muscle; **cellulose** forms plant cell walls and is dietary fibre for humans.
- Proteins are polypeptides built from amino acids; shape determines function. Essential amino acids must be supplied by diet; the body can synthesise non-essential amino acids.
- Lipids include triglycerides, phospholipids and steroids. Unsaturated fatty acids contain double bonds; essential fatty acids include linoleic and alpha-linolenic acids. Stored adipose fat is an energy reserve, insulates and cushions organs, and aids absorption/transport of fat-soluble vitamins.

Fat type	Structure / health clue	Common sources
Saturated	No C=C bond; solid fats common; excess can raise LDL	Butter, ghee, meat fat, coconut/palm oils
Trans	Unsaturated configuration formed especially in partial hydrogenation; raises cardiovascular risk	Some fried/packaged foods; restrict intake
Monounsaturated (MUFA)	One C=C bond	Olive/groundnut oils, avocado, almonds and other nuts
Polyunsaturated (PUFA)	Two or more C=C bonds; omega-3/omega-6 families	Vegetable oils, walnuts, flaxseed and oily fish

EDITORIAL CORRECTION · Fat oversimplification

The source treats all saturated fatty acids as non-essential and all unsaturated fatty acids as essential, and implies unsaturated fat cannot contribute to obesity. Correctly, only particular polyunsaturated fatty acids are essential; **all fats supply ~9 kcal/g and excess energy can be stored.**

PYQ LENS · 56-59th: Energy reserve in liver and muscle

Glycogen

PYQ LENS · 64th: Storage carbohydrate in fungi

Glycogen

PYQ LENS · 66th: Major food-energy source

Carbohydrates

Vitamins

Vitamins are organic micronutrients needed in small amounts. **Fat-soluble: A, D, E, K. Water-soluble: B-complex and C.** Fat-soluble vitamins can accumulate; many water-soluble vitamins are not extensively stored, with B₁₂ a major exception.

Vitamin	Chemical name	Core function	Deficiency / sources
A	Retinol; beta-carotene precursor	Visual pigments, epithelium, growth and immunity	Night blindness/xerophthalmia; liver, fish-liver oil, egg, milk; carrot/leafy vegetables as carotene
B ₁	Thiamine	TPP coenzyme in carbohydrate metabolism and nerves	Beriberi; whole grains, pulses, nuts, yeast, pork
B ₂	Riboflavin	FAD/FMN redox coenzymes	Cheilosis, glossitis, dermatitis; milk, eggs, green vegetables
B ₃	Niacin	NAD/NADP	Pellagra: dermatitis, diarrhoea, dementia; meat, fish, groundnut, whole grains
B ₅	Pantothenic acid	Coenzyme A	Rare: fatigue/neurological symptoms; widely distributed
B ₆	Pyridoxine	Amino-acid metabolism, neurotransmitters, haem synthesis	Microcytic anaemia/dermatitis/neuropathy; meat, grains, banana, legumes
B ₇	Biotin	Carboxylase coenzyme	Rare dermatitis/alopecia; egg yolk, nuts, gut microbes
B ₉	Folate	One-carbon transfer, DNA and RBC formation	Megaloblastic anaemia; neural-tube risk; leafy greens, pulses
B ₁₂	Cobalamin	DNA/RBC formation and myelin	Megaloblastic/pernicious anaemia, neuropathy; animal foods/fortified products
C	Ascorbic acid	Collagen hydroxylation, antioxidant, improves iron absorption	Scurvy; citrus, amla, guava, tomato
D	Calciferol	Ca/P absorption and bone mineralisation	Rickets in children, osteomalacia in adults; sunlight, fish-liver oil, egg/fortified foods
E	Tocopherol	Lipid antioxidant; membrane protection	Rare neuromuscular/haemolytic effects; plant oils, nuts, seeds
K	Phylloquinone / menaquinone	Activation of clotting proteins and bone proteins	Bleeding tendency; green leafy vegetables, gut microbes

PYQ LENS · 60–62nd: Vitamin A deficiency

Night blindness

PYQ LENS · 60–62nd: Vitamin D function

Calcium and phosphorus absorption / bone health

PYQ LENS · 64th: Vitamin B₁₂ contains

Cobalt

PYQ LENS · 65th: Ascorbic acid

Vitamin C

PYQ LENS · 67th: Rickets

Vitamin D deficiency

EDITORIAL CORRECTION · Vitamin nomenclature

Vitamin B₁₂ is **cobalamin**, not “cyanide”; cyanocobalamin is one stable medicinal form. Vitamin C deficiency causes scurvy, whereas folate/B₁₂ deficiency causes megaloblastic anaemia.

Minerals

Mineral	Principal role	Deficiency / food clues
Calcium	Bone/teeth, contraction, signalling and clotting	Weak bones/tetany; milk, ragi, sesame, greens
Phosphorus	Bone, ATP, nucleic acids, phospholipids	Rare alone; dairy, pulses, meat
Iron	Haemoglobin, myoglobin and electron carriers	Microcytic anaemia; liver/meat, legumes, greens; vitamin C aids absorption
Iodine	Thyroid hormones	Goitre and hypothyroidism; iodised salt, seafood
Sodium	Extracellular fluid, nerve and muscle	Hyponatraemia; common salt
Potassium	Intracellular fluid, nerve/muscle	Weakness/arrhythmia; fruits, vegetables, pulses
Magnesium	Enzyme cofactor, bone, neuromuscular function	Cramps/weakness; nuts, grains, greens
Zinc	Enzymes, growth, immunity, wound healing	Poor growth/healing; meat, seeds, pulses
Fluoride	Strengthens tooth enamel at appropriate dose	Low: caries; excess: fluorosis

PYQ LENS · 66th: Haemoglobin metal

Iron

PYQ LENS · 60–62nd: Iodine deficiency

Goitre

Balanced diet and deficiency disorders

EDITOR ADDITION · Added-scope marker

Kwashiorkor, marasmus and BMI interpretation below are assistant-added exam context; the source chapters focus chiefly on macro-/micronutrient chemistry and deficiency tables.

- A balanced diet supplies adequate carbohydrate, protein, fat, vitamins, minerals, fibre and water according to age, sex, work, health and physiological state.
- **Kwashiorkor:** severe protein deficiency, often with oedema, fatty liver, hair/skin changes and growth failure.
- **Marasmus:** severe energy deficiency with marked wasting and loss of subcutaneous fat, usually without oedema.
- Roughage/fibre improves stool bulk and gut health; water supports reactions, transport, temperature regulation and waste removal.
- BMI = mass in kilograms / height² in metres; it is a population screening tool, not a direct measure of body fat or health in every person.

PYQ LENS · 66th: Development of sex organs/secondary characters depends especially on

Proteins (also hormones and adequate total nutrition)

PYQ LENS · 60–62nd: Insulin is chemically

A peptide/protein hormone

UNIT 10

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

Animal Tissues, Skeleton, Joints & Muscle

Consolidated from L3, L12, L13 · PYQ theme: Skeleton, Joints & Muscle

Animal tissues

Tissue	Types	Functions / locations
Epithelial	Simple squamous/cuboidal/columnar /ciliated; stratified; glandular	Covering, lining, absorption, secretion and protection
Connective	Areolar, adipose, cartilage, bone, blood, tendon, ligament	Support, binding, storage, transport and defence
Muscular	Skeletal, smooth and cardiac	Movement, posture, pumping and visceral motility
Nervous	Neurons + neuroglia	Rapid communication, integration and support

Epithelia

- Epithelial cells are tightly packed on a basement membrane and are usually avascular. Diffusion from underlying tissue supplies nutrients.
- Squamous epithelium allows exchange (alveoli, capillaries); cuboidal secretion/absorption (kidney tubules, ducts); columnar absorption/secretion (gut); ciliated columnar transport (respiratory tract, oviduct).

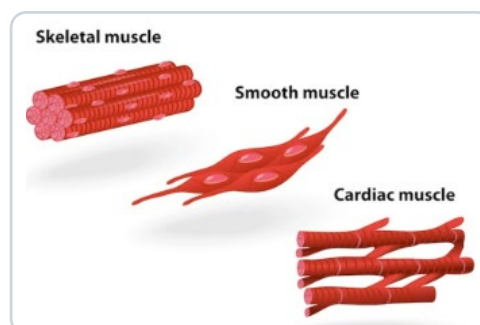
Connective tissue

- Areolar tissue packs organs; adipose stores fat and insulates; cartilage provides flexible support; bone supplies rigid mineralised support.
- Tendon** connects muscle to bone and is relatively inelastic; **ligament** connects bone to bone and is more elastic. Blood is a fluid connective tissue.

Muscle tissue

Muscle	Appearance/location	Control	Special points
Skeletal/striated	Long cylindrical multinucleate fibres; attached to bones	Voluntary	Fast contraction; fatigues
Smooth/unstriated	Spindle-shaped single-nucleus cells; viscera/vessels	Involuntary	Slow sustained contraction; no visible striations
Cardiac	Branched striated cells, mostly one nucleus; heart	Involuntary	Intercalated discs; rhythmic; fatigue-resistant

- Skeletal muscle contracts by sliding of actin over myosin within sarcomeres; Ca^{2+} exposes binding sites and ATP powers cross-bridge cycling.

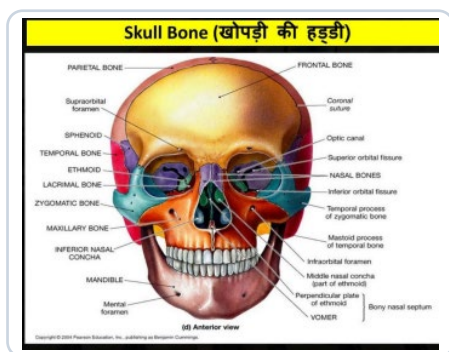


Smooth, skeletal and cardiac muscle compared

Original source visual · L9. Animal tissues and Skeleton system.pdf, p. 3

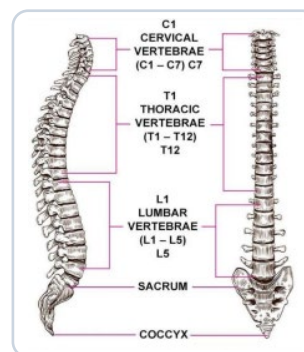
Human skeleton

- Adult human skeleton usually contains **206 bones**. Axial skeleton (80) includes skull, vertebral column, ribs and sternum; appendicular skeleton (126) includes girdles and limbs.
- Skull is commonly described as 22 bones: 8 cranial + 14 facial. Auditory ossicles contribute 6 and hyoid 1 to a broader 29-bone head count.
- Vertebral formula: 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral and usually 4 fused coccygeal vertebrae.
- Twelve pairs of ribs: first 7 true, 8-10 false (vertebrochondral), 11-12 floating. Pectoral girdle = scapula + clavicle each side; pelvic girdle = two hip bones.



Major bones and regions of the human skull

Original source visual · L9. Animal tissues and Skeleton system.pdf, p. 5



Regions and curvatures of the vertebral column

Original source visual · L9. Animal tissues and Skeleton system.pdf, p. 5

Major limb bones

- Upper limb: humerus; radius and ulna; 8 carpals; 5 metacarpals; 14 phalanges. Lower limb: femur; patella; tibia and fibula; 7 tarsals; 5 metatarsals; 14 phalanges.
- **Femur** is longest and strongest; **stapes** is smallest. Mandible is the largest/strongest movable facial bone.

PYQ LENS · 68th: Number of bones in a human foot

26 (7 tarsals + 5 metatarsals + 14 phalanges)

PYQ LENS · 66th: Bones of a new-born child versus adult

About 300 at birth, many later fuse; adult usually 206

PYQ LENS · 70th: Stapes is found in

Middle ear

Joints

EDITOR ADDITION · Added-scope marker

The joint classification is assistant-added to integrate the supplied shoulder-joint PYQ with the source skeleton chapter.

Class	Mobility	Examples
Fibrous	Immovable or nearly so	Skull sutures
Cartilaginous	Slight movement	Intervertebral discs, pubic symphysis
Synovial	Freely movable, joint cavity and synovial fluid	Ball-and-socket shoulder/hip; hinge elbow/knee; pivot atlas-axis; gliding wrist/ankle; saddle thumb; condyloid wrist

PYQ LENS · 65th: Shoulder joint

Ball-and-socket joint

UNIT 11

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

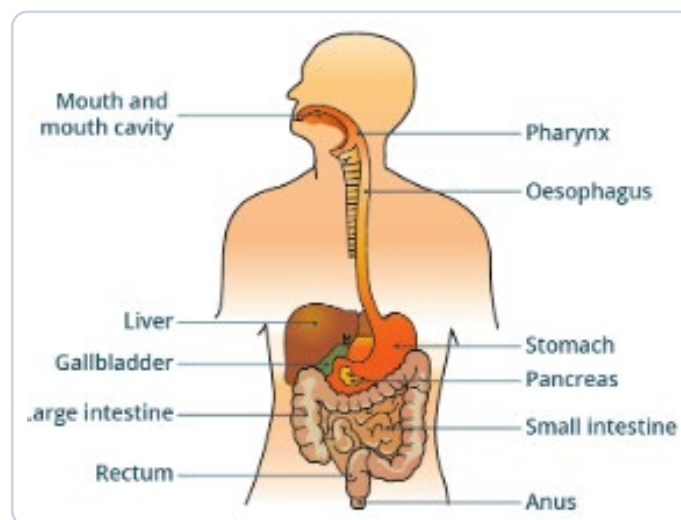
Digestion, Absorption & Energy Release

Consolidated from L14, L32 · PYQ theme: Digestion

Digestive tract map

Digestion converts complex, insoluble food into small absorbable molecules by mechanical and chemical processes.

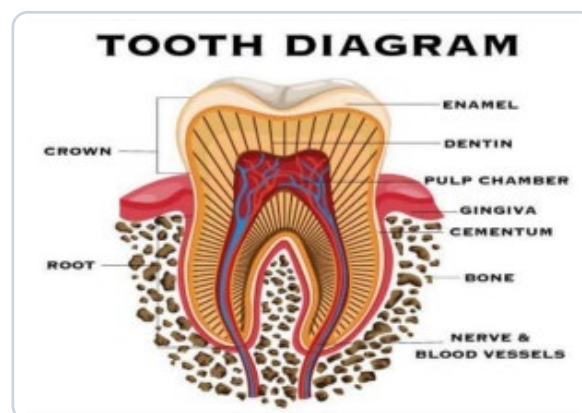
- Route: **mouth** → **pharynx** → **oesophagus** → **stomach** → **small intestine (duodenum, jejunum, ileum)** → **large intestine (caecum, colon, rectum)** → **anus**.
- Accessory glands/organs: salivary glands, liver with gall bladder, and pancreas.



Human digestive tract and accessory organs
Original source visual · L13. Digestive System, Part - 01.pdf, p. 3

Teeth and the oral cavity

- Humans are **thecodont** (teeth in sockets), **heterodont** (different types) and **diphyodont** (deciduous and permanent sets). An adult usually has 32 permanent teeth.
- Dental formula per half-jaw is **2 incisors : 1 canine : 2 premolars : 3 molars**, written 2123/2123. Incisors cut, canines tear, and premolars/molars crush and grind. Elephant tusks are modified incisors.
- Enamel is the hardest substance in the human body. Saliva from paired parotid, submandibular and sublingual glands lubricates food; amylase acts best near pH 6.8 and lysozyme contributes antimicrobial defence.



Longitudinal structure of a human tooth
Original source visual · L13. Digestive System, Part - 01.pdf, p. 4

Organ-by-organ

Site	Secretion / event	What happens
Mouth	Saliva; salivary amylase/ptyalin	Chewing; starch begins to maltose/dextrins; mucus lubricates
Oesophagus	Mucus; peristalsis	Bolus transport; essentially no enzyme digestion
Stomach	HCl, pepsinogen, mucus; gastric lipase minor	Acid kills many microbes/activates pepsin; proteins begin digestion; chyme forms
Liver/gall bladder	Bile salts, no digestive enzyme	Emulsifies fats and aids absorption of lipids/fat-soluble vitamins
Pancreas	Bicarbonate; trypsin/chymotrypsin precursors, amylase, lipase, nucleases	Neutralises acid and digests protein, carbohydrate, fat and nucleic acid
Small intestine	Brush-border enzymes; segmentation/peristalsis	Completes digestion; villi/microvilli absorb most nutrients
Large intestine	Mucus; bacterial activity	Absorbs water/electrolytes; forms faeces; microbes make some vitamins

Enzyme quick sheet

Enzyme	Substrate	Product	Main site
Salivary amylase	Starch	Maltose/dextrins	Mouth
Pepsin	Protein	Peptides	Stomach
Trypsin/chymotrypsin	Proteins/polypeptides	Smaller peptides	Small intestine
Peptidases	Peptides	Amino acids	Small-intestinal surface
Pancreatic amylase	Starch	Maltose/oligosaccharides	Small intestine
Maltase/sucrase/lactase	Disaccharides	Monosaccharides	Small-intestinal surface
Pancreatic lipase	Triglycerides	Fatty acids + monoglycerides/glycerol	Small intestine
Nucleases	DNA/RNA	Nucleotides	Small intestine

- **HCl is not an enzyme.** Bile has no digestive enzyme; bile salts emulsify lipids and form micelles.
- Villi contain blood capillaries that receive sugars/amino acids and a **lacteal** that receives chylomicron-rich lipid products. Absorbed nutrients first reach the liver via hepatic portal circulation, except much dietary fat enters lymph first.

The Summary of Absorption in Different Parts of Digestive System

Mouth	Stomach	Small Intestine	Large Intestine
Certain drugs coming in contact with the mucosa of mouth and lower side of the tongue are absorbed into the blood capillaries lining them.	Absorption of water, simple sugars, and alcohol etc. takes place.	Principal organ for absorption of nutrients. The digestion is completed here and the final products of digestion such as glucose, fructose, fatty acids, glycerol and amino acids are absorbed through the mucosa into the blood stream and lymph.	Absorption of water, some minerals and drugs takes place.

Original visual summary of intestinal digestion and absorption
Original source visual · L13. Digestive System, Part - 01.pdf, p. 5

PYQ LENS · 67th: Site of complete digestion and greatest absorption

Small intestine

PYQ LENS · 70th: Mumps primarily affects

Salivary glands

PYQ LENS · 70th: Bile is secreted by

Liver (stored/concentrated in gall bladder)

Liver, pancreas and control

- Liver is the largest gland (~1.5 kg in an adult). Hepatic lobules are its structural units. Bile contains bile salts and pigments including bilirubin/biliverdin; hepatic ducts and the cystic duct form the common bile pathway.
- Pancreas is a mixed/heterocrine gland: exocrine acini provide alkaline digestive juice, while endocrine islets make hormones.
- Gastric secretion and motility are coordinated neurally and hormonally. **Secretin**, released from duodenal mucosa in response to acid, stimulates bicarbonate-rich pancreatic/biliary secretion; sight, smell and oral food can stimulate salivation.

Common digestive disorders

EDITOR ADDITION · Added-scope marker

This concise disorder table is assistant-added context; the source digestive lectures focus on normal anatomy and digestion.

Condition	Mechanism	Key management idea
Diarrhoea	Frequent watery stool; infection or other causes	Oral rehydration replaces water and electrolytes; seek care for danger signs
Constipation	Difficult/infrequent stool	Fluid, fibre, activity; investigate persistent cases
Peptic ulcer	Often H. pylori or NSAID-related injury	Acid suppression and cause-specific treatment
Jaundice	Bilirubin accumulation; sign, not one disease	Evaluate liver, bile obstruction or haemolysis
Gallstones	Crystallised bile components	May obstruct ducts; management depends on symptoms
Lactose intolerance	Low lactase	Limit lactose/use alternatives according to tolerance

UNIT 12

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

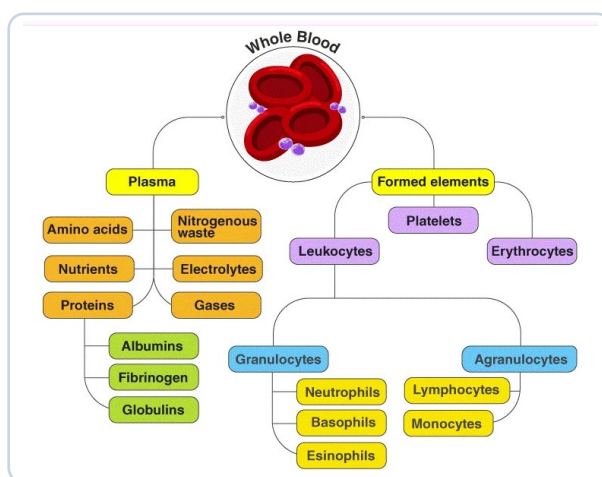
Blood, Circulation, Lymph & Immunity Basics

Consolidated from L9, L10 · PYQ theme: Circulation, Blood & Immunity

Blood components

Blood is a fluid connective tissue composed of plasma and formed elements. An adult has roughly 5 L, varying with body size and sex.

Component	Form	Function / high-yield fact
Plasma	Pale-yellow fluid; ≈55%, mostly water with proteins, salts and solutes	Transport; albumin maintains osmotic pressure; globulins include antibodies; fibrinogen aids clotting
RBC / erythrocyte	Biconcave; mammalian mature cell lacks nucleus; haemoglobin	O ₂ transport and part of CO ₂ transport; life about 120 days; formed in red marrow
WBC / leukocyte	Nucleated; granular or agranular	Defence and immunity; can leave capillaries by diapedesis
Platelet / thrombocyte	Megakaryocyte fragment	Primary haemostatic plug and clotting support; life about 7–10 days



Formed elements—RBCs, WBCs and platelets

Original source visual · L16. Circulatory System, Part - 02.pdf, p. 3

WBC differential

- **Granulocytes:** neutrophils—phagocytose bacteria, most numerous; eosinophils—parasites and allergy; basophils—histamine/heparin, least numerous.
- **Agranulocytes:** monocytes—largest, become macrophages; lymphocytes—B cells, T cells and natural killer cells.
- Source numerical ranges vary by laboratory; typical exam order is neutrophils > lymphocytes > monocytes > eosinophils > basophils.

Blood groups

ABO type	RBC antigen	Plasma antibody	Can receive RBCs from	Can donate RBCs to
A	A	Anti-B	A, O	A, AB
B	B	Anti-A	B, O	B, AB
AB	A and B	Neither	A, B, AB, O	AB
O	Neither	Anti-A and Anti-B	O	A, B, AB, O

- Karl Landsteiner discovered the ABO system. The Rh system is especially important in transfusion and pregnancy.
- The phrases “universal donor” (O negative) and “universal recipient” (AB positive) are simplified for **red-cell transfusion**. Safe transfusion always requires compatibility testing; plasma compatibility reverses important ABO considerations.
- Rh-negative mother carrying an Rh-positive fetus can become sensitised; anti-D immunoglobulin prevents haemolytic disease of the fetus/newborn in appropriate pregnancies.

PYQ LENS · 65th: Universal red-cell donor among common textbook choices

O negative

PYQ LENS · 67th: Major blood-group systems used in routine matching

ABO and Rh

Clotting and haemostasis

- Vessel injury causes constriction, platelet adhesion/aggregation and activation of clotting factors.
- In the simplified cascade, prothrombin becomes thrombin; thrombin converts soluble **fibrinogen to insoluble fibrin**; the fibrin mesh traps cells and stabilises the clot.
- Vitamin K supports synthesis/activation of several clotting proteins; Ca^{2+} is needed in multiple steps.

PYQ LENS · 68th: Fibrinogen is present in

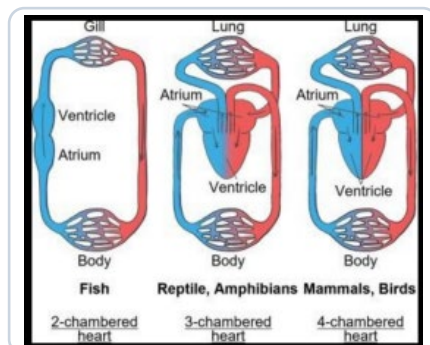
Blood plasma

PYQ LENS · 56-59th: Substance essential in coagulation but not ordinarily present as an active plasma component

Thrombin; it is generated from prothrombin

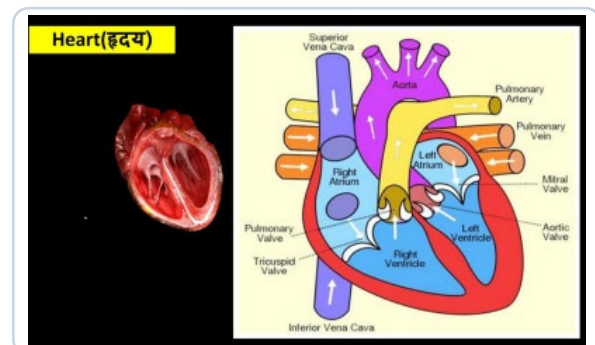
Circulatory designs and heart wall

- William Harvey established the circulation of blood and is associated in the source with angiology. **Open circulation** releases haemolymph into sinuses, typical of arthropods and most molluscs; **closed circulation** keeps blood in vessels, typical of annelids and chordates.
- Fish show single circulation with a two-chambered heart; amphibians and most reptiles have incomplete double circulation with three chambers; crocodilians, birds and mammals have four chambers and double circulation.
- Heart wall: outer **epicardium**, thick contractile **myocardium**, inner **endocardium**. A double-layered pericardium and fluid reduce friction. The heart is myogenic and mesodermal in origin.



Heart chambering across vertebrate classes

Original source visual · L15. Circulatory System, Part - 01.pdf, p. 3



Human heart chambers, valves and great vessels

Original source visual · L15. Circulatory System, Part - 01.pdf, p. 3

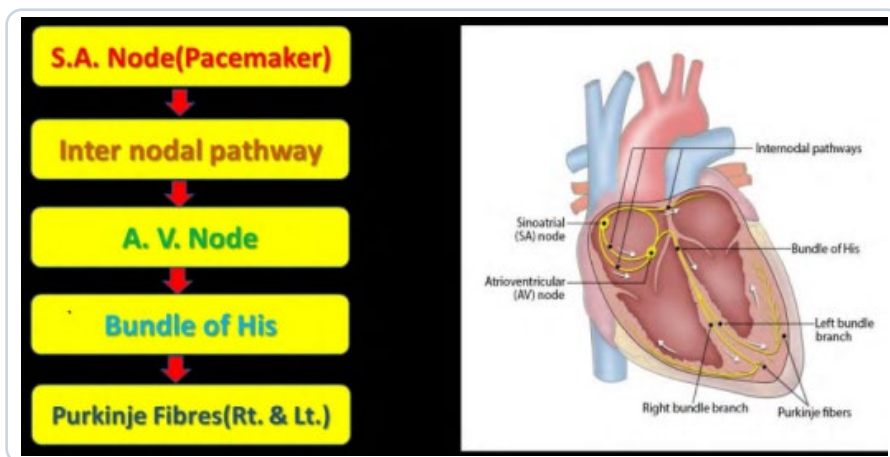
Heart and double circulation

- Four chambers: right atrium and ventricle carry deoxygenated blood; left atrium and ventricle carry oxygenated blood. Septa limit mixing.

- Valves ensure one-way flow: tricuspid (right AV), mitral/bicuspid (left AV), pulmonary semilunar and aortic semilunar.
- Flow: body → venae cavae → right atrium → tricuspid → right ventricle → pulmonary valve/artery → lungs → pulmonary veins → left atrium → mitral → left ventricle → aortic valve/aorta → body.
- Pulmonary artery** carries deoxygenated blood and **pulmonary veins** oxygenated blood—the key vessel-name exceptions. Vessel names are based on direction relative to heart, not oxygen content.

Conducting system and cycle

- Sinoatrial (SA) node initiates the heartbeat and is the normal pacemaker; atrioventricular (AV) node delays conduction; bundle of His and Purkinje fibres distribute it through ventricles.
- One cardiac cycle comprises atrial systole, ventricular systole and joint diastole. Normal resting adult heart rate averages ~72/min but varies widely.
- “Lub” is mainly closure of AV valves; “dub” closure of semilunar valves.
- Blood pressure is written systolic/diastolic; a conventional textbook reference is ~120/80 mm Hg, not a universal personal target. A **sphygmomanometer** measures it. Persistent elevation is hypertension; tachycardia is increased heart rate and bradycardia decreased rate.
- Cardiac output** = **heart rate** × **stroke volume**; textbook resting values are about 72/min × 70 mL ≈ 5 L/min. A cardiac cycle at 72/min lasts about 0.8 s.
- ECG: P wave—atrial depolarisation; QRS—ventricular depolarisation; T—ventricular repolarisation.



Cardiac conduction pathway from pacemaker to ventricular fibres
Original source visual · L17. Circulatory System, Part - 03.pdf, p. 2

Vessels and lymph

Vessel	Wall / pressure	Main role
Artery	Thick, elastic, muscular; high pressure; usually no valves	Away from heart
Vein	Thinner, wider lumen; lower pressure; valves common	Toward heart
Capillary	Single-cell endothelial wall	Exchange of gases, nutrients and wastes
Lymph vessel	Blind-ended capillaries → larger valved channels	Returns tissue fluid/proteins; transports absorbed intestinal fat; immune surveillance

- Lymph resembles interstitial fluid with fewer proteins and no normal RBC population. It passes through lymph nodes rich in immune cells.

PYQ LENS · 56-59th: Pulse can be felt in

Arteries

PYQ LENS · 67th: Lymph chiefly transports

Absorbed fats, tissue fluid and immune cells (not oxygen as its main role)

UNIT 13

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

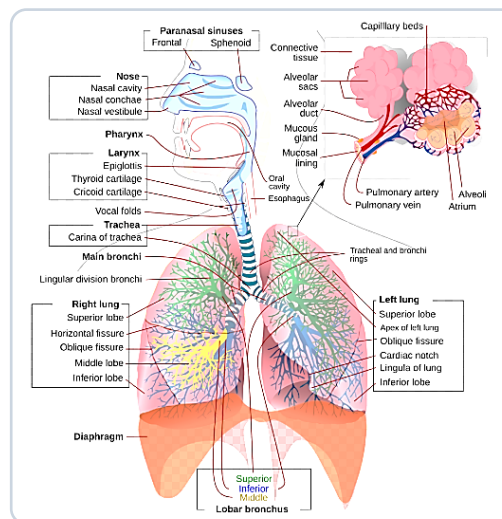
Respiration & Gas Exchange

Consolidated from L15 · PYQ theme: Circulation, Blood & Immunity

Respiratory pathway

Air moves through **nostrils** → **nasal cavity** → **pharynx** → **larynx** → **trachea** → **bronchi** → **bronchioles** → **alveoli**.

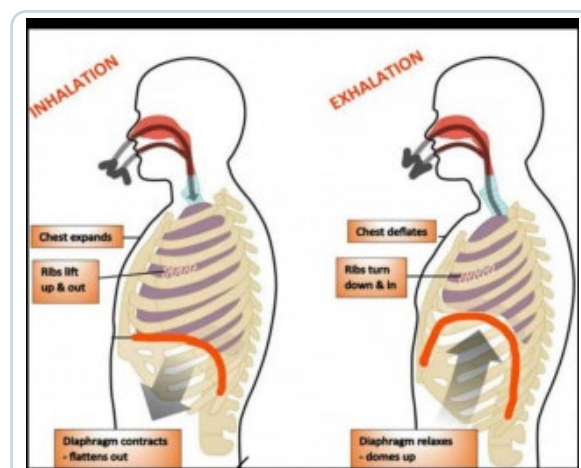
- Nose filters with hair/mucus and warms/humidifies air. Epiglottis protects the airway during swallowing. Cartilage supports trachea/bronchi.
- Lungs are covered by double pleura with lubricating pleural fluid. The right lung has three lobes; left has two and a cardiac notch.



Respiratory passages, lungs, diaphragm and alveoli
Original source visual · L18. Respiratory System- 01.pdf, p. 2

Mechanics of breathing

- Inspiration at rest: diaphragm contracts/flattens and external intercostals lift ribs → thoracic volume rises → intrapulmonary pressure falls → air enters.
- Expiration at rest is mostly passive: muscles relax and elastic recoil reduces thoracic volume, raising pressure. Forced expiration uses internal intercostals and abdominal muscles.
- **Surfactant** from type-II alveolar cells lowers surface tension and prevents alveolar collapse.
- Intrapleural pressure remains negative relative to atmospheric pressure in normal breathing, helping keep lungs expanded.



Pressure and volume changes during inhalation and exhalation
Original source visual · L18. Respiratory System- 01.pdf, p. 3

Gas exchange and transport

- Alveoli provide enormous area, thin diffusion barrier, moist surface and rich capillary network. O₂ diffuses down partial-pressure gradient into blood; CO₂ diffuses oppositely.
- Most O₂ travels reversibly bound to haemoglobin as oxyhaemoglobin; a small fraction dissolves in plasma.
- CO₂ travels chiefly as bicarbonate, with smaller fractions as carbaminohaemoglobin and dissolved gas. Carbonic anhydrase in RBCs accelerates $\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \text{H}^+ + \text{HCO}_3^-$.
- Tissue conditions with higher CO₂/H⁺/temperature promote oxygen unloading (Bohr effect). Oxygenation of haemoglobin in lungs facilitates CO₂ release (Haldane effect).

Volumes and control

Term	Meaning	Typical educational value
Tidal volume	Air in quiet breath	≈500 mL
Inspiratory reserve	Additional inspired after normal inspiration	≈2500–3000 mL
Expiratory reserve	Additional expired after normal expiration	≈1000–1100 mL
Residual volume	Air left after forceful expiration	≈1100–1200 mL
Vital capacity	IRV + TV + ERV	≈4–5 L, varies by person

- Medullary centres generate rhythm; pons modifies it. Chemoreceptors respond strongly to CO₂/H⁺; rising CO₂ is the main routine stimulus for ventilation.
- Smoking damages cilia/alveoli and raises risks of COPD, cancer and cardiovascular disease. Pneumonia inflames/fluid-fills alveoli; asthma produces reversible airway narrowing and inflammation; emphysema destroys alveolar walls.

EDITORIAL CORRECTION · Respiration versus breathing

Breathing is ventilation; **cellular respiration** is biochemical ATP release. Plants and animals both respire day and night.

UNIT 14

PRIORITY B

TOPIC-FIRST · DUPLICATES MERGED

Excretion, Osmoregulation & Human Reproduction

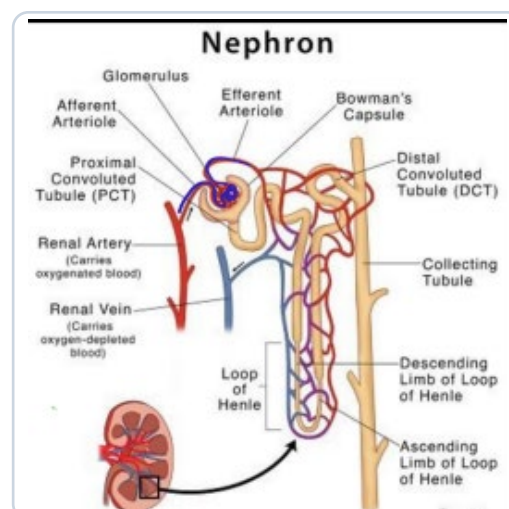
Consolidated from L16, L17 · PYQ theme: Excretion & Osmoregulation; Endocrine & Reproduction

Nitrogenous wastes

Type	Toxicity / water need	Examples
Ammonotelic	Ammonia; very toxic; much water needed	Many aquatic invertebrates, bony fishes, aquatic amphibian larvae
Ureotelic	Urea; moderately toxic	Mammals, adult amphibians, cartilaginous fishes
Uricotelic	Uric acid; least water loss, paste/crystals	Birds, reptiles, insects, land snails

Urinary system and nephron

- Human urinary system: paired kidneys, ureters, urinary bladder and urethra. Kidneys are retroperitoneal; each has outer cortex, inner medulla, pyramids and a renal pelvis.
- The **nephron** is the structural and functional unit: glomerulus within Bowman's capsule → proximal convoluted tubule (PCT) → loop of Henle → distal convoluted tubule (DCT) → collecting duct.



Nephron segments, vessels and filtrate path

Original source visual · L20. Excretory System & Reproductive System.pdf, p. 4

Urine formation

EDITOR ADDITION · Added-detail marker

The source describes filtration, reabsorption and secretion; counter-current concentration and hormonal fine control below are assistant-added physiology.

- Glomerular filtration:** blood pressure drives water and small solutes into Bowman's capsule; cells and most proteins remain in blood.
- Selective reabsorption:** PCT reclaims most filtered water/NaCl, all normal glucose and amino acids, and much bicarbonate. Loop, DCT and collecting duct further adjust fluid.
- Tubular secretion:** H^+ , K^+ , ammonia, drugs and other substances may be secreted into tubule.
- Counter-current arrangement of Henle's loop and vasa recta helps create medullary gradient and concentrate urine.

- **ADH/vasopressin** increases water permeability of late DCT/collecting duct; **aldosterone** increases Na⁺ reabsorption and K⁺ secretion; atrial natriuretic peptide favours Na⁺ loss.
- Typical urine is ~95% water plus urea, ions, uric acid, creatinine and other wastes; volume/composition vary with intake, climate, hormones and health.

PYQ LENS · 56-59th: Main nitrogenous waste excreted by humans

Urea

PYQ LENS · 63rd: Nephron is the functional unit of

Kidney

EDITORIAL CORRECTION · Renal correction

Urea is made mainly in the **liver** by the urea cycle and excreted by kidneys. The source line implying urine is formed in the liver is incorrect.

Dialysis

EDITOR ADDITION · Added-scope marker

Dialysis is assistant-added renal-failure context and is not developed in the source lecture.

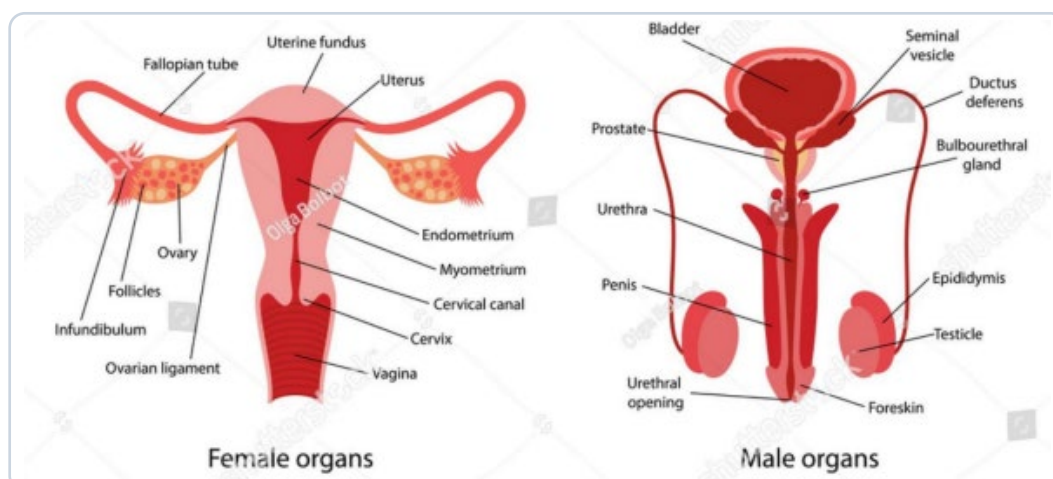
- Haemodialysis passes blood beside dialysate across a semipermeable membrane; small wastes/excess ions diffuse out and cleaned blood returns. Anticoagulant prevents circuit clotting.
- Dialysis replaces selected excretory/homeostatic functions but not all endocrine/metabolic roles of healthy kidneys. Kidney transplantation may be definitive treatment in selected end-stage disease.

Male reproductive system

EDITOR ADDITION · Added-scope marker

The source introduces reproductive organs and gametes; the detailed pathways, menstrual physiology and contraception from this point through the end of the unit are assistant-added completion for PYQ readiness.

- Testes lie in scrotum, which keeps them cooler than core body temperature. Seminiferous tubules produce sperm; Sertoli cells support developing germ cells; Leydig cells make testosterone.
- Sperm path: seminiferous tubules → rete testis → vasa efferentia → epididymis (maturation/storage) → vas deferens → ejaculatory duct → urethra.
- Seminal vesicles contribute fructose-rich alkaline fluid; prostate adds enzymes/secretions; bulbourethral glands provide lubricating mucus. Sperm head contains nucleus and acrosome; middle piece contains mitochondria; tail propels.



Original human reproductive-system visual retained from the source lecture
Original source visual · L20. Excretory System & Reproductive System.pdf, p. 6

Female reproductive system

- Ovaries produce oocytes, oestrogens and progesterone. Oviduct regions: fimbriae/infundibulum, ampulla, isthmus. Fertilisation normally occurs near the **ampullary-isthmic junction**.
- Uterus has perimetrium, muscular myometrium and cyclic endometrium; cervix opens to vagina. Mammary glands support lactation.

Gametogenesis and menstrual cycle

- Spermatogenesis begins at puberty: spermatogonia → primary spermatocytes → secondary spermatocytes → spermatids → spermatozoa. One primary spermatocyte yields four sperm.
- Oogenesis begins before birth; primary oocytes arrest in prophase I, and typically one resumes per cycle. One primary oocyte ultimately yields one ovum plus polar bodies.

Phase	Approx. days in 28-day model	Hormonal/ovarian event
Menstrual	1–5	Fall in progesterone/oestrogen sheds endometrium
Follicular/proliferative	1–13	FSH supports follicles; oestrogen rebuilds endometrium
Ovulation	≈14	LH surge ruptures mature follicle
Luteal/secretory	15–28	Corpus luteum secretes progesterone; without pregnancy it regresses

- Cycle length varies; “day 14” applies only to an idealised 28-day cycle. Implantation begins roughly a week after fertilisation; placenta exchanges gases/nutrients/wastes and acts as endocrine tissue.
- Parturition involves fetal/placental signals, oxytocin and positive feedback. Prolactin supports milk production; oxytocin causes milk ejection.

Contraception and reproductive health

Method	Examples	Main action
Natural / behavioural	Fertility-awareness, withdrawal, lactational amenorrhoea under strict criteria	Avoids intercourse/fecund window; effectiveness varies
Barrier	Condom, diaphragm/cervical cap	Prevents sperm entry; condoms also reduce STI transmission
Intrauterine device	Copper IUD; hormone-releasing IUS	Impairs fertilisation; hormonal forms thicken mucus/endometrium
Hormonal	Combined/progestin-only pills, injection, implant	Suppresses ovulation and/or alters mucus/endometrium
Emergency contraception	Levonorgestrel; ulipristal; copper IUD	Delays/inhibits ovulation or prevents fertilisation; not an abortifacient
Sterilisation	Vasectomy, tubectomy	Blocks gamete transport; intended permanent

PYQ LENS · 71st: Common active ingredient of emergency contraceptive pills

Levonorgestrel

EDITOR ADDITION · STI safeguard

Only barrier methods such as condoms reduce transmission risk for many sexually transmitted infections; hormonal pills and IUDs do not.

UNIT 15

PRIORITY B

TOPIC-FIRST · DUPLICATES MERGED

Endocrine Coordination

Consolidated from L18 · PYQ theme: Endocrine & Reproduction

Hormone principles

- **Endocrinology** studies endocrine glands; the source names Thomas Addison as its father. Ernest Starling coined the term **hormone**, and secretin is commonly credited as the first hormone discovered.
- Endocrine glands are ductless; hormones enter blood and act on receptor-bearing target cells. Exocrine glands deliver secretions through ducts.
- Peptide/protein hormones generally bind cell-surface receptors; steroid and thyroid hormones generally bind intracellular receptors and regulate transcription.
- Negative feedback stabilises hormone levels. The hypothalamus integrates neural/endocrine control and regulates pituitary secretion.

PYQ LENS · 71st: Hormones are secreted by

Endocrine glands

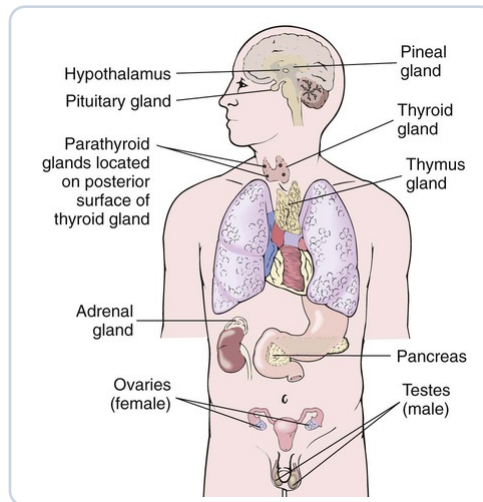
Gland-hormone map

EDITOR ADDITION · Added-scope marker

This complete gland-hormone map supplements the shorter source list; entries not explicitly developed in the lecture are assistant-added standard physiology.

Gland / tissue	Hormone	Major action	Disorder hook
Hypothalamus	Releasing/inhibiting hormones; makes ADH and oxytocin	Controls anterior pituitary; ADH/oxytocin stored in posterior pituitary	—
Anterior pituitary	GH	Growth and metabolism	Low child: pituitary dwarfism; high child: gigantism; high adult: acromegaly
Anterior pituitary	TSH, ACTH, FSH, LH	Controls thyroid, adrenal cortex and gonads	Tropic hormones
Anterior pituitary	Prolactin	Milk production	—
Posterior pituitary	ADH, oxytocin	Water retention; uterine contraction/milk ejection	Low ADH: diabetes insipidus
Thyroid	T ₄ thyroxine, T ₃ ; calcitonin	Metabolic rate/growth; calcitonin lowers blood Ca ²⁺	Iodine deficiency goitre; hypo/hyperthyroidism
Parathyroids	PTH	Raises blood Ca ²⁺	Tetany with severe hypoparathyroidism
Adrenal cortex	Cortisol, aldosterone, androgens	Stress metabolism; Na ⁺ /K ⁺ balance	Cushing/Addison patterns
Adrenal medulla	Adrenaline/noradrenaline	Fight-or-flight	Raises heart rate and glucose availability
Pancreatic islets	Insulin (β cells), glucagon (α cells)	Lower vs raise blood glucose	Diabetes mellitus
Pineal	Melatonin	Circadian rhythm	—
Thymus	Thymic hormones	T-cell maturation, prominent in childhood	—
Testes	Testosterone	Male reproductive function and secondary traits	—

Gland / tissue	Hormone	Major action	Disorder hook
Ovaries	Oestrogen, progesterone	Cycle, pregnancy support and female secondary traits	—



Location of major human endocrine glands

Original source visual · L21. Human Endocrine System.pdf, p. 2

Pancreas and diabetes

- Pancreas is heterocrine: exocrine acini release digestive enzymes; endocrine islets release hormones.
- Insulin promotes glucose uptake/storage and generally lowers blood glucose; glucagon promotes glycogen breakdown/glucose release and raises it.
- Type 1 diabetes: autoimmune β -cell destruction and absolute insulin deficiency. Type 2: insulin resistance with progressive relative deficiency; lifestyle and genetics both contribute.
- Diabetes mellitus may cause hyperglycaemia, excessive urination/thirst and long-term vascular, renal, neural and retinal complications. It differs from **diabetes insipidus**, an ADH-related water-balance disorder.

PYQ LENS · 60–62nd: Insulin

Protein/peptide hormone

PYQ LENS · 66th: Gland with both endocrine and exocrine function

Pancreas

PYQ LENS · 65th: Insulin is produced by

Beta cells of islets of Langerhans

Thyroid and iodine

- Thyroid follicles store thyroglobulin and use iodine to make T_3/T_4 , which support basal metabolism, growth and nervous-system development.
- Childhood severe thyroid-hormone deficiency impairs growth and neurodevelopment; adult hypothyroidism may cause fatigue, weight gain and cold intolerance. Excess causes weight loss, heat intolerance, tremor and rapid heartbeat.
- Iodised salt prevents iodine-deficiency disorders. Calcitonin has a relatively minor Ca^{2+} -lowering role in humans compared with PTH/vitamin D control.

Stress and adrenal glands

- Adrenal medulla gives rapid sympathetic fight-or-flight output. Cortex provides slower steroid responses: cortisol for metabolism/stress and aldosterone for salt/water balance.
- Addison disease = adrenal-cortex insufficiency; Cushing syndrome = prolonged glucocorticoid excess. Pheochromocytoma is a catecholamine-secreting tumour of adrenal medulla.

EDITORIAL CORRECTION · Pituitary label

The “master gland” phrase is exam-useful, but hypothalamus regulates much pituitary activity and endocrine control is distributed through feedback loops.

UNIT 16

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

Nervous System, Brain & Sense Organs

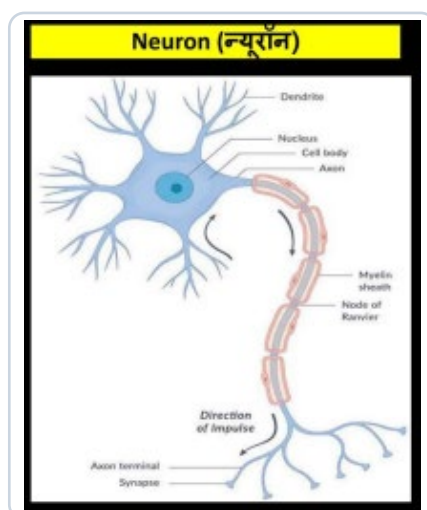
Consolidated from L19, L20 · PYQ theme: Nervous System & Senses

Neuron and impulse

EDITOR ADDITION · Added-scope marker

The source gives neuron parts, synapse and myelin basics; membrane-potential, channel and glial-cell detail in this subsection is assistant-added explanation.

- Neuron is the structural/functional unit: dendrites receive, cell body integrates, axon conducts away, terminals communicate at synapses.
- Resting membrane potential reflects ion gradients and selective permeability. An action potential is all-or-none depolarisation/repolarisation propagated along axon.
- Myelin insulates axons; nodes of Ranvier allow **saltatory conduction**, speeding impulses. Schwann cells myelinate peripheral axons; oligodendrocytes myelinate central axons.
- At a chemical synapse, arriving action potential opens Ca^{2+} channels; vesicles release neurotransmitter; receptors alter postsynaptic activity; transmitter is removed by breakdown, diffusion or reuptake.



Neuron—dendrites, soma, myelin, nodes and axon terminals
Original source visual · L23. Nervous System - 02.pdf, p. 2

Divisions

- Humans have **12 pairs of cranial nerves** and **31 pairs of spinal nerves**. A spinal nerve is mixed after sensory dorsal and motor ventral roots join; dorsal-root ganglia contain sensory neuron cell bodies.

Division	Parts	Role
Central nervous system	Brain and spinal cord	Integration, processing and command
Peripheral nervous system	Cranial and spinal nerves	Links CNS and body
Somatic	Sensory pathways and motor supply to skeletal muscle	Conscious sensation and voluntary action; reflexes
Autonomic	Sympathetic, parasympathetic, enteric	Involuntary control of smooth/cardiac muscle and glands

Sympathetic vs parasympathetic

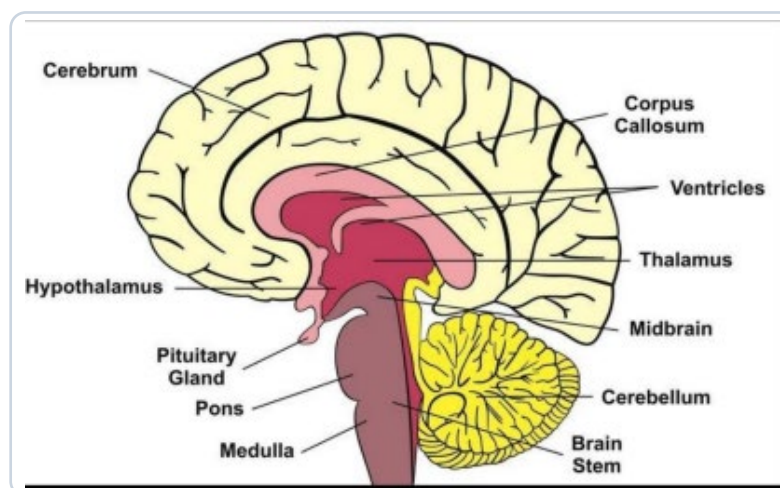
Effect	Sympathetic	Parasympathetic
Overall	Fight/flight	Rest/digest
Pupil	Dilates	Constricts
Heart	Rate/force increase	Rate decreases
Bronchi	Dilate	Constrict toward baseline
Gut	Motility/secretion decrease	Motility/secretion increase
Bladder	Urine storage favoured	Voiding favoured

Brain

- Brain and spinal cord are protected by three meninges: tough outer **dura mater**, web-like **arachnoid mater**, and delicate vascular **pia mater** attached to nervous tissue. Cerebrospinal fluid fills ventricles and subarachnoid space; arachnoid villi/granulations return it to venous blood.

Part	Major structures	Functions
Forebrain	Cerebrum, thalamus, hypothalamus	Perception, voluntary action, memory/language; relay; homeostasis/endocrine control
Midbrain	Tectum and tracts	Visual/auditory reflexes, movement pathways
Hindbrain	Pons, medulla, cerebellum	Relay and breathing modulation; vital autonomic centres; balance/coordination

- Cerebrum has two hemispheres, folded cortex with frontal, parietal, temporal and occipital lobes. Corpus callosum connects hemispheres.
- Frontal:** planning, voluntary motor control, speech production. **Parietal:** somatic sensation and spatial processing. **Temporal:** hearing, memory and language comprehension. **Occipital:** vision.
- Thalamus is major sensory relay except olfaction. Hypothalamus controls temperature, hunger, thirst, autonomic activity and pituitary.
- Cerebellum coordinates precision, posture and balance; medulla controls breathing, cardiovascular activity and reflexes such as swallowing/coughing.



Major regions and lobes of the human brain
Original source visual · L22. Human Endocrine System.pdf, p. 4

PYQ LENS · 64th: Human-brain part with two hemispheres

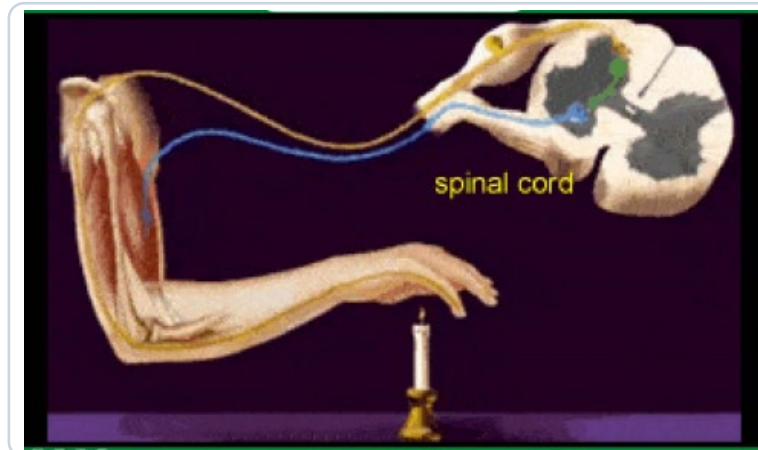
Cerebrum

PYQ LENS · 69th: Speech production centre

Broca's area, usually left frontal lobe

Spinal cord and reflex

- Spinal cord continues from medulla through vertebral canal. It carries ascending sensory and descending motor tracts and integrates many reflexes.
- Reflex arc: receptor → sensory neuron → integration centre/interneuron → motor neuron → effector. It is rapid and automatic; brain can receive/modify information without initiating the immediate response.



Reflex arc from receptor to spinal cord and effector
Original source visual · L23. Nervous System - 02.pdf, p. 3

Eye

EDITOR ADDITION · Added-scope marker

The eye and ear subsections are assistant-added system completion driven by supplied sensory PYQs; the nervous-system lectures do not contain a dedicated sense-organ chapter.

- Outer fibrous coat: sclera and transparent cornea. Middle vascular coat: choroid, ciliary body and iris. Inner sensory coat: retina with rods and cones.
- Cornea provides most refraction; lens fine-focuses by accommodation. Iris controls pupil. Ciliary muscle changes lens curvature.
- **Rods:** dim light, no colour, peripheral retina. **Cones:** bright light, colour/high acuity, concentrated in fovea. Optic disc has no photoreceptors—blind spot.
- Photoreceptor signal passes through retinal neurons to optic nerve. Image on retina is real, inverted and diminished; brain interprets perception.

Refractive defects

Defect	Problem	Correction
Myopia	Distant image focuses in front of retina; eye too long/power too high	Concave/diverging lens
Hypermetropia	Near image would focus behind retina; eye too short/power too low	Convex/converging lens
Presbyopia	Age-related accommodation loss	Convex reading/bifocal/progressive lens as needed
Astigmatism	Unequal curvature causes directional blur	Cylindrical/toric lens

PYQ LENS · 69th: Myopia is corrected by

Concave lens

PYQ LENS · 64th: Eye colour depends mainly on

Pigment in iris

Ear

- Outer ear: pinna + auditory canal directs sound to tympanic membrane. Middle ear ossicles **malleus, incus, stapes** transmit/amplify vibration; Eustachian tube equalises pressure.
- Inner ear cochlea contains organ of Corti; hair cells transduce sound. Vestibule and semicircular canals provide linear and rotational balance.

PYQ LENS · 65th: Organ of Corti is in

Cochlea

PYQ LENS · 70th: Stapes belongs to

Middle ear

Selected nervous-system disorders

Disorder	Core description	Exam-safe clue
Stroke	Sudden brain injury from blocked vessel or bleeding	FAST symptoms require emergency care
Epilepsy	Recurrent unprovoked seizures from abnormal neural activity	Not every single seizure is epilepsy
Parkinson disease	Degeneration of dopamine-producing circuits	Resting tremor, rigidity and slowed movement are classic
Alzheimer disease	Progressive neurodegenerative dementia	Memory and cognition decline beyond normal ageing

EDITORIAL CORRECTION · Sensory correction

Taste receptors are in taste buds, not every papilla; olfactory receptors are specialised neurons in olfactory epithelium. Balance depends chiefly on vestibular apparatus, vision and proprioception together.

UNIT 17

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

Microbes, Disease, Immunity, Vaccines & Cancer

Consolidated from **L21, L22, L23, L30** · PYQ theme: Microbes, Disease & Immunity

Bacteria, viruses, viroids and prions

- Antonie van Leeuwenhoek observed bacteria/“animalcules” in the 17th century. **Bergey's Manual of Determinative Bacteriology** is a classic identification/classification reference.
- Bacterial nutrition is metabolically diverse: photoautotrophs use light (e.g., purple/green sulphur bacteria), chemoautotrophs oxidise inorganic compounds, saprotrophs decompose dead matter, and parasites/pathogens feed in living hosts.
- Useful activities retained from the source: ammonification; nitrification by *Nitrosomonas*/*Nitrobacter*; nitrogen fixation; curd/fermentation and vinegar by *Acetobacter*; fibre retting; vitamin/antibiotic production; *E. coli* as an indicator of faecal water contamination; *E. coli* and *Agrobacterium* as genetic-engineering workhorses.
- The term **antibiotic** is associated with S. A. Waksman; streptomycin from *Streptomyces griseus* was the first antibiotic discovered from a bacterium/actinomycete in the source framing.
- A virus is acellular genetic material (DNA **or** RNA) inside a protein capsid, sometimes with an envelope. It is inert/crystallisable outside cells and an obligate intracellular parasite inside. Plant viruses are often RNA viruses; bacteriophages infect bacteria.
- Discovery milestones: Ivanowsky linked a filterable agent to tobacco mosaic (1892); Beijerinck called it *contagium vivum fluidum* (1898); W. M. Stanley crystallised TMV (1935).
- **Viroid**: small naked infectious RNA, discovered by T. O. Diener. **Prion**: infectious misfolded protein causing transmissible neurodegeneration.

EDITORIAL CORRECTION · Source terminology

The source prints “prisons”; the intended word is **prions**. A virus is not a nucleoprotein in every simplistic structural sense—enveloped viruses also contain lipid membrane and accessory proteins.

Pathogen groups

EDITOR ADDITION · Added-detail marker

This unified agent table merges the source disease lists and adds prion/treatment-logic context; those additions are assistant-supplied.

Agent	Cellular status	Examples	Treatment logic
Bacteria	Prokaryotic cells	TB, typhoid, cholera, diphtheria, tetanus	Antibiotics can work; resistance/diagnosis matter
Virus	Acellular; nucleic acid + capsid, sometimes envelope	Influenza, measles, polio, rabies, HIV/AIDS, COVID-19	Antibiotics do not work; antivirals exist for selected viruses
Protozoan	Unicellular eukaryote	Malaria, amoebiasis, kala-azar, sleeping sickness	Specific antiprotozoal therapy
Fungus	Eukaryotic; chitinous wall	Ringworm, candidiasis, athlete's foot	Antifungals
Helminth	Multicellular parasitic worm	Ascariasis, filariasis, taeniasis	Anthelmintics
Prion	Misfolded infectious protein	CJD and related encephalopathies	Not a cellular pathogen; exceptionally resistant

Bacterial disease map

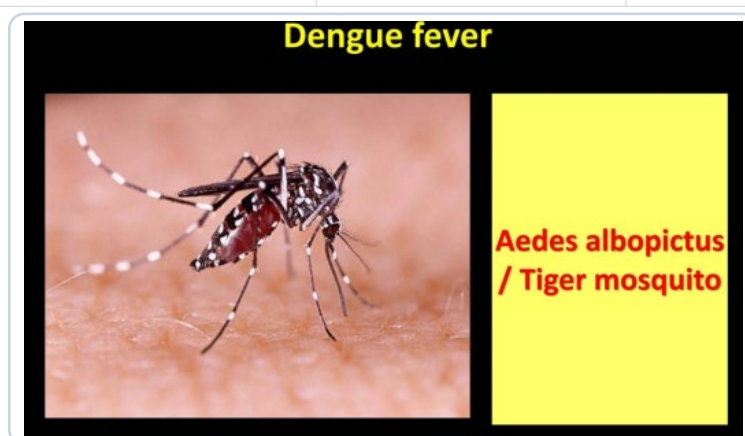
Disease	Agent	Spread	Core clue
Tuberculosis	Mycobacterium tuberculosis	Airborne droplets/aerosols	Lungs common; BCG helps protect children from severe TB
Typhoid	Salmonella enterica serovar Typhi	Faeco-oral	Prolonged fever; Widal is imperfect, culture preferred
Cholera	Vibrio cholerae	Contaminated water/food	Profuse watery diarrhoea; ORS is lifesaving
Diphtheria	Corynebacterium diphtheriae	Respiratory	Pseudomembrane; toxin-mediated; DPT-containing vaccine
Whooping cough	Bordetella pertussis	Respiratory	Paroxysmal cough; pertussis vaccine
Tetanus	Clostridium tetani spores through wounds	Not person-to-person	Spastic paralysis; toxoid vaccine
Plague	Yersinia pestis	Flea/rodent or respiratory depending form	Buboes in bubonic plague
Leprosy	Mycobacterium leprae	Prolonged close contact	Skin/peripheral nerves; multidrug treatment
Gonorrhoea	Neisseria gonorrhoeae	Sexual/perinatal	STI; antibiotic resistance important
Syphilis	Treponema pallidum	Sexual/transplacental	Stages; penicillin treatment
Pneumonia	Often Streptococcus pneumoniae; also viruses/others	Respiratory	Alveoli inflamed and fluid-filled

Viral disease map

Disease	Virus	Transmission/vector	High-yield point
AIDS	HIV, a retrovirus	Sexual, blood, shared needles, mother-to-child	Targets CD4 T cells; not spread by casual contact
Dengue	Dengue virus	Aedes aegypti/albopictus, mainly daytime biting	Severe disease may cause bleeding/shock
Chikungunya	Chikungunya virus	Aedes	Severe joint pain common
Yellow fever	Yellow fever virus	Aedes/other mosquitoes	Jaundice; vaccine available
Japanese encephalitis	JE virus	Culex mosquitoes, pig/bird cycle	Vaccine available in risk settings
Polio	Poliovirus	Faeco-oral	Can damage anterior-horn motor neurons; OPV/IPV
Rabies	Rabies lyssavirus	Bite/saliva of infected mammal	Nearly invariably fatal after symptoms; prompt wound care + PEP
Measles	Measles virus	Airborne	Highly contagious; MMR-containing vaccine
Mumps	Mumps virus	Respiratory	Parotid salivary-gland swelling
Rubella	Rubella virus	Respiratory/transplacental	Congenital rubella risk; MMR
Smallpox	Variola virus	Respiratory/contact	Eradicated globally; Jenner pioneered vaccination using cowpox
COVID-19	SARS-CoV-2	Respiratory particles	Vaccines reduce severe disease; variants emerge by mutation/evolution

Protozoan, fungal and helminth diseases

Disease	Agent	Vector / entry	Clue
Malaria	Plasmodium spp.	Female Anopheles mosquito	Parasite infects liver then RBCs; intermittent fever patterns
Kala-azar	Leishmania donovani	Female sandfly	Visceral leishmaniasis; enlarged spleen/liver
Sleeping sickness	Trypanosoma brucei	Tsetse fly	African trypanosomiasis, neurological sleep disturbance
Amoebiasis	Entamoeba histolytica	Faeco-oral cysts	Dysentery; liver abscess possible
Giardiasis	Giardia duodenalis	Contaminated water/food	Greasy diarrhoea/malabsorption
Ringworm	Dermatophyte fungi	Contact/fomites	Itchy annular skin lesion; not a worm
Ascariasis	Ascaris lumbricoides	Ingested eggs	Intestinal roundworm
Filariasis	Wuchereria/Brugia	Mosquito	Lymphatic obstruction/elephantiasis
Taeniasis	Taenia spp.	Undercooked meat / eggs depending species	Intestinal tapeworm; pork-tapeworm eggs can cause cysticercosis



Aedes mosquito—the vector highlighted in the source lecture
Original source visual · L27. Infectious and Non-Infectious Diseases, Part - 02.pdf, p. 4

PYQ LENS · 65th: Malaria vector

Female Anopheles mosquito

PYQ LENS · 60-62nd: Sleeping sickness vector

Tsetse fly

PYQ LENS · 67th: Kala-azar vector

Sandfly

PYQ LENS · 69th: AIDS is caused by

Human immunodeficiency virus

Antibiotics and antimicrobial resistance

- Antibiotics target bacterial structures/processes such as cell-wall synthesis, ribosomes, DNA metabolism or folate pathways. They do not treat ordinary viral infections.
- Penicillin was observed by Alexander Fleming (1928); Florey and Chain helped develop it therapeutically. Penicillins inhibit bacterial cell-wall synthesis.

- Chloramphenicol inhibits the bacterial 50S ribosomal subunit. Quinine is an antimalarial drug, not an antibiotic. Sulfonamides are synthetic antibacterial agents and are often distinguished from classic microbe-derived antibiotics.
- Resistance arises through mutation or acquisition of resistance genes and is selected by misuse/overuse. Complete prescribed courses as clinically directed; avoid self-medication and use diagnostics/stewardship.

PYQ LENS · 60–62nd: Antibiotic among chloramphenicol/quinine/etc.

Chloramphenicol; supplied workbook key is corrected

PYQ LENS · 63rd: Antibiotic single-choice item containing penicillin and sulfadiazine

More than one is defensible under broad wording; use the official key but remember both are antibacterial drugs

Immune system

EDITOR ADDITION · Added-scope marker

The innate/adaptive immune map and immunoglobulin comparison are assistant-added completion prompted by the supplied PYQs; the source lectures mention antibodies but not this full framework.

Arm	Components	Character
Innate immunity	Skin/mucosa, secretions, phagocytes, inflammation, complement, NK cells	Immediate, broad, no classic antigen-specific memory
Adaptive humoral	B lymphocytes → plasma cells → antibodies	Antigen-specific; memory; extracellular pathogens/toxins
Adaptive cell-mediated	T helper, cytotoxic and regulatory cells	Antigen-specific; infected cells, intracellular pathogens and coordination

- Primary lymphoid organs: bone marrow and thymus. Secondary organs: lymph nodes, spleen, tonsils and mucosa-associated lymphoid tissue.
- Antibody/immunoglobulin classes: **IgM** first prominent in primary response and strong agglutinator; **IgG** most abundant in serum and crosses placenta; **IgA** mucosal secretions/breast milk; **IgE** allergy and parasites; **IgD** B-cell receptor role.

PYQ LENS · 71st: Antibody first produced in a primary immune response

IgM

PYQ LENS · 65th: B lymphocytes mature in

Bone marrow in humans

Vaccines and public health programmes

EDITOR ADDITION · Added-scope marker

The vaccine-platform comparison and U-WIN synthesis are assistant-added from standard immunology and the recent PYQ/programme context.

- Vaccination exposes the immune system to a safe antigen or genetic instruction, producing antibodies, T-cell responses and memory without the risk of full natural disease.

Platform	What is delivered	Examples / caution
Live attenuated	Weakened replicating organism	MMR, BCG, OPV; strong response but unsuitable in some immunocompromised states
Inactivated	Killed whole organism	IPV, inactivated influenza, some rabies vaccines
Subunit/recombinant/conjugate	Selected antigen, often with carrier/adjuvant	Hepatitis B, HPV, Hib and pneumococcal conjugates
Toxoid	Inactivated bacterial toxin	Tetanus, diphtheria
Viral vector	Modified harmless carrier with antigen gene	Some COVID-19 and Ebola vaccines
mRNA	mRNA encoding antigen in lipid particles	Some COVID-19 vaccines

- **Active immunity:** infection or vaccination stimulates person's own response; slower but memory. **Passive immunity:** ready-made antibodies, immediate but temporary—maternal antibodies, antiserum or immunoglobulin.
- Herd/community protection occurs when enough people are immune to reduce transmission; threshold depends on pathogen and vaccine effectiveness.
- **U-WIN** is India's digital vaccination platform. Features in the 71st item include QR-based e-vaccination certificates, ABHA/Child ABHA creation and guidance in 11 Indian languages; therefore none of the three statements is incorrect.

PYQ LENS · 69th: Viral-vector vaccine mechanism

Modified harmless carrier virus delivers genetic instructions for an antigen

PYQ LENS · 71st: U-WIN “not correct” statements

None; all three are correct

EDITOR ADDITION · Public-health note

A vaccine's purpose is primarily to prevent infection and/or severe disease; it is not an antibiotic and does not directly kill bacteria. Follow current national schedules because recommendations change.

Cancer

EDITOR ADDITION · Added-scope marker

The cancer mechanism and treatment overview is assistant-added to support the isotope and platinum-compound PYQs; it is not a full source-lecture chapter.

- Cancer is uncontrolled clonal cell proliferation caused by accumulated genetic/epigenetic changes. **Benign** tumours remain local; **malignant** tumours invade and can metastasise through blood/lymph.
- Carcinogens include tobacco smoke, certain chemicals, ionising/UV radiation and selected infections (e.g., HPV, HBV/HCV). Inherited susceptibility and random replication errors also contribute.
- Warning signs are not diagnostic; screening and biopsy establish early disease. Treatment may include surgery, radiotherapy, chemotherapy, targeted therapy, hormone therapy and immunotherapy.
- Platinum compounds such as **cisplatin** form DNA cross-links and are classic anticancer drugs. Radioisotopes are chosen by tumour and radiation properties.

PYQ LENS · 71st: Widely used anticancer compound among common metals

Platinum compound, especially cisplatin

PYQ LENS · 56-59th: Radioisotope keyed for blood cancer

Cobalt-60 in the supplied/BPSC-oriented key; P-32 is a scientifically important haematological nuance

EDITORIAL CORRECTION · Radioisotope nuance

Cobalt-60 is a major external radiotherapy source and is the keyed answer in the supplied context. Phosphorus-32 has been used for selected myeloproliferative/haematological conditions. Do not turn this item into a universal clinical rule.

UNIT 18

FOUNDATION

TOPIC-FIRST · DUPLICATES MERGED

Animal Diversity & Applied Zoology

Consolidated from L24, L32 · PYQ theme: Animal Diversity & Applied Biology

Basis of animal classification

Criterion	States	Representative progression
Organisation	Protoplasmic → cellular → tissue → organ → organ-system	Protozoan model → Porifera → Cnidaria/Ctenophora → Platyhelminthes → roundworms onward
Symmetry	Asymmetrical, radial, bilateral	Most sponges; cnidarians/adult echinoderms; annelids/arthropods/chordates
Germ layers	Diploblastic vs triploblastic	Cnidaria/Ctenophora vs Platyhelminthes onward
Body cavity	Acoelomate, pseudocoelomate, true coelomate	Flatworms; roundworms; annelids onward
Segmentation	Unsegmented vs metameric	Annelids show clear metamerism
Notochord	Absent vs present at some stage	Non-chordates vs chordates

Non-chordate overview

Phylum	Body-plan clue	Examples	Exam hook
Porifera	Pores, canal system, choanocytes; no true tissues	Sycon, Spongilla, Euspongia	Sponges; sessile aquatic
Cnidaria/Coelenterata	Diploblastic, radial, cnidocytes; polyp/medusa	Hydra, Obelia, jellyfish, corals	Stinging cells
Ctenophora	Marine, diploblastic/radial; eight comb plates	Pleurobrachia, Ctenoplana	Comb jellies; bioluminescence common
Platyhelminthes	Triploblastic, bilateral, acoelomate, flat	Planaria, Taenia, Fasciola	Flame cells; many parasites
Nematoda/Aschelminthes	Pseudocoelomate, unsegmented round body	Ascaris, Wuchereria, hookworm	Complete gut; sexes often separate
Annelida	True coelom, metameric segmentation	Earthworm, leech, Nereis	Closed circulation in many
Arthropoda	Jointed appendages, chitin exoskeleton	Insects, crustaceans, arachnids	Largest animal phylum; open circulation
Mollusca	Soft unsegmented body, mantle and usually shell	Pila, Unio, Octopus	Muscular foot; radula often present
Echinodermata	Spiny skin, water-vascular system; marine	Starfish, sea urchin	Adults radial; larvae bilateral
Hemichordata	Proboscis, collar, trunk	Balanoglossus	Stomochord, not true notochord

Chordates and vertebrates

- Chordate diagnostic features at some life stage: **notochord, dorsal hollow nerve cord, pharyngeal slits and post-anal tail.**

Class	Covering / breathing	Heart / temperature	Examples
Cyclostomata	Jawless; gill pouches	2 chambers; ectothermic	Petromyzon, Myxine
Chondrichthyes	Placoid scales; gills; cartilage skeleton	2 chambers; ectothermic	Shark, ray
Osteichthyes	Bony skeleton; gills; operculum common	2 chambers; ectothermic	Rohu, catla, seahorse
Amphibia	Moist skin; gills in larva, lungs/skin adult	3 chambers; ectothermic	Frog, toad, salamander
Reptilia	Dry keratinised scales; lungs	Usually 3 chambers; crocodile 4; ectothermic	Lizard, snake, turtle, crocodile
Aves	Feathers; lungs + air sacs	4 chambers; endothermic	Pigeon, ostrich, penguin
Mammalia	Hair, mammary glands; lungs	4 chambers; endothermic	Human, bat, whale, platypus

- Fish gills use counter-current exchange; amphibians link aquatic and terrestrial life; reptiles have shelled amniotic eggs (with some live-bearing exceptions); birds have lightweight skeleton/air sacs; mammals typically have a diaphragm and differentiated teeth.

EDITORIAL CORRECTION · Vertebrate exceptions

Crocodylians have a four-chambered heart. Platypus and echidna are egg-laying mammals. Whale and bat are mammals, not fish/birds. Snakes are not literally deaf: they detect substrate vibration and airborne sound through inner-ear pathways despite lacking an external ear.

Applied zoology

EDITOR ADDITION · Added-scope marker

This applied-zoology table is assistant-added from recurring BPSC applications; the source animal-diversity lecture ends with classification.

Practice	Organism / product	Key point
Apiculture	Honey bees; honey, wax, pollination	Queen reproductive female, drones males, workers sterile females in common honey-bee colonies
Sericulture	Silkworm Bombyx mori; silk	Larva feeds on mulberry and spins cocoon
Lac culture	Kerria lacca	Resinous lac secretion on host plants
Pisciculture	Fish farming	Composite culture combines compatible feeding niches
Poultry	Chicken/duck production	Broilers for meat; layers for eggs
Vermiculture	Earthworms	Composting and soil improvement
Pearl culture	Pearl oysters	Irritant/nucleus induces nacre deposition

PYQ LENS · 56-59th: Lac is produced by

An insect

PYQ LENS · 68th: Apiculture

Bee keeping

PYQ LENS · 69th: Milk is a poor source of

Iron

Selected useful/harmful animal links

- Earthworms aerate/mix soil and recycle organic matter. Bees pollinate crops. Silkworm and lac insect provide economic products.
- Disease vectors are not pathogens themselves: female Anopheles transmits malaria; Aedes transmits dengue/chikungunya/yellow fever; Culex transmits filariasis/JE in relevant settings; sandfly transmits kala-azar; tsetse fly transmits sleeping sickness; rat flea transmits plague.
- Pests include locusts, termites and crop borers; biological control uses predators, parasitoids or pathogens while integrated pest management minimises ecological/health harm.

UNIT 19

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

Ecology, Biodiversity & Conservation

Consolidated from L31, L32 · PYQ theme: Ecology, Environment & Conservation

Levels and terms

EDITOR ADDITION · Added-scope marker

The population-level terms and growth equations below extend beyond the source ecology lecture and are assistant-added for exam completeness.

Organism → population → community → ecosystem → biome → biosphere. Ecology studies interactions at these levels.

- **Habitat:** where an organism lives. **Niche:** its functional role and resource/environmental relationships. Two species cannot indefinitely occupy precisely the same niche under stable limiting conditions (competitive-exclusion principle).
- **Population attributes:** density, natality, mortality, age structure, sex ratio, immigration and emigration.
- Exponential growth forms J-curve under unlimited resources; logistic growth forms S-curve and approaches carrying capacity **K**.

Species interactions

Interaction	Species A	Species B	Example
Mutualism	+	+	Lichen; mycorrhiza; pollinator-flower
Commensalism	+	0	Cattle egret with grazing cattle; epiphyte on tree
Parasitism	+	–	Tapeworm in human; Cuscuta on host
Predation	+	–	Tiger-deer; insectivorous plant-insect
Competition	–	–	Species sharing a limiting resource
Amensalism	–	0	Antibiosis where mould inhibits bacteria
Neutralism	0	0	No measurable direct effect (idealised)

- Obligate intracellular pathogens such as viruses are **endoparasites**; if an MCQ offers both parasitism and endoparasitism, the more specific label may be intended, making wording key-sensitive.

PYQ LENS · 71st: Interaction of virus with host cell

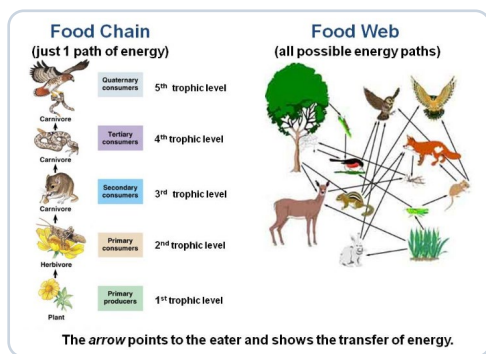
Parasitism, specifically intracellular endoparasitism; options overlap

PYQ LENS · 65th: Lichen relation

Mutualism/symbiosis of fungus and photosynthetic partner

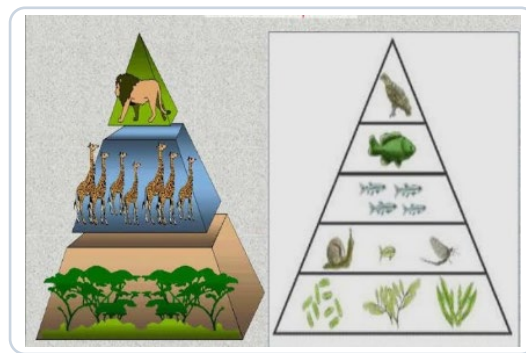
Ecosystem components and energy

- Abiotic: light, temperature, water, air, soil and mineral nutrients. Biotic: **producers, consumers and decomposers**.
- Producers fix energy; primary consumers eat producers; secondary/tertiary consumers eat other consumers. Detritivores fragment dead matter; decomposers mineralise organic matter.
- Energy flow is unidirectional: sun → producers → consumers → decomposers/heat. Nutrients cycle.
- Lindeman's **10% law** is an approximate transfer rule, not an exact universal value. Ecological pyramids show number, biomass or energy; the energy pyramid is always upright.



Food chain and food-web connections

Original source visual · L31. Ecology Microbes(Bacteria & Virus).pdf, p. 2



Pyramids of number, biomass and energy

Original source visual · L31. Ecology Microbes(Bacteria & Virus).pdf, p. 3

Productivity and decomposition

EDITOR ADDITION · Added-scope marker

Productivity and decomposition are assistant-added ecosystem context; the source focuses on food chains, pyramids and species interactions.

- Gross primary productivity (GPP) is total photosynthetic fixation; net primary productivity **NPP = GPP – plant respiration** and is available for growth/consumers.
- Decomposition involves fragmentation, leaching, catabolism, humification and mineralisation. Warm, moist, oxygenated conditions generally accelerate it; lignin/chitin resist it.

Biogeochemical cycles

EDITOR ADDITION · Added-scope marker

The cycle comparison is assistant-added to fill a source gap and support environment-linked BPSC questions.

Cycle	Main reservoir / flow	Human perturbation
Water	Ocean, atmosphere, soil and life via evaporation, transpiration, condensation and precipitation	Groundwater depletion, dams, land-use/climate change
Carbon	Atmospheric/oceanic CO ₂ , biomass, soils, rocks/fossil fuels	Fossil burning and deforestation raise greenhouse CO ₂
Nitrogen	Atmospheric N ₂ → fixation → nitrification → assimilation → ammonification → denitrification	Fertiliser runoff, eutrophication, N ₂ O
Phosphorus	Rocks → soil/water → organisms → sediment; no major gaseous phase	Mining/fertiliser runoff and eutrophication

- Nitrogen-cycle bacteria: Rhizobium/Azotobacter fix; Nitrosomonas oxidises ammonia to nitrite; Nitrobacter oxidises nitrite to nitrate; Pseudomonas and related bacteria denitrify.

Biodiversity

EDITOR ADDITION · Added-scope marker

Biodiversity levels, hotspots, threats and conservation through the Bihar map are assistant-added PYQ-oriented material beyond the source ecology lecture.

- Biodiversity has genetic, species and ecosystem levels. Species richness and evenness both influence diversity.
- India is megadiverse and overlaps four biodiversity hotspots: Himalaya, Indo-Burma, Western Ghats–Sri Lanka and Sundaland (Nicobar component).
- Hotspot criteria emphasise exceptional endemic vascular plants plus massive habitat loss. Endemic means geographically restricted; endangered means high extinction risk—these are not synonyms.

- Ecosystem services include provisioning (food/water), regulating (climate/flood/pollination), supporting (soil/nutrient cycling) and cultural values.

Threats — “evil quartet”

- Habitat loss/fragmentation; overexploitation; alien-species invasion; co-extinctions. Pollution and climate change intensify all four.
- Invasive examples in the source: Parthenium, Lantana and Eichhornia. Nile perch caused severe losses of native cichlids in Lake Victoria.
- Passenger pigeon and Steller's sea cow illustrate overexploitation; pollinator/host losses can trigger co-extinction.

Conservation

Approach	Meaning	Examples
In situ	Conserve species in natural ecosystem	National parks, wildlife sanctuaries, biosphere reserves, sacred groves
Ex situ	Conserve outside natural habitat	Zoos, botanical gardens, seed/gene banks, cryopreservation and captive breeding
National park	Strong ecosystem protection; rights/activities highly restricted	Valmiki is Bihar's sole national park
Wildlife sanctuary	Protection with some regulated rights/uses possible	Numerous state sanctuaries
Biosphere reserve	Core, buffer and transition zones integrating conservation and sustainable use	Nilgiri, Nanda Devi etc.

- Red Data Book assesses conservation status; **IUCN** categories include EX, EW, CR, EN, VU, NT, LC, DD and NE.
- Ramsar Convention concerns wetlands; CITES regulates international trade in listed wild species; Convention on Biological Diversity links conservation, sustainable use and benefit sharing.
- **Project Tiger** began in 1973; **Project Elephant** in 1992. Ex-situ programmes complement, not replace, habitat conservation.

PYQ LENS · 69th: Red Data Book is maintained/published internationally by
IUCN

PYQ LENS · 71st: Asian elephant IUCN status
Endangered

PYQ LENS · 71st: World Ozone Day
16 September

Bihar ecology and exam-oriented map

- **Valmiki National Park**, within Valmiki Tiger Reserve in West Champaran, is Bihar's only national park. It lies along the Indo-Nepal border in the Himalayan Terai landscape.
- Bihar conservation questions may mix national park, tiger reserve, wildlife sanctuary, Ramsar site, species and state-symbol terminology; identify the legal/site category precisely.
- Kanwar/Kabar Tal is a major wetland and Bihar's first Ramsar site. Other Bihar wetland designations and counts can change; verify current government/Ramsar lists near the exam.
- Vikramshila Gangetic Dolphin Sanctuary on the Ganga in Bhagalpur is associated with conservation of the Gangetic river dolphin.

PYQ LENS · 71st: Bihar's only national park
Valmiki National Park

PYQ LENS · 71st: State with the largest number of wildlife sanctuaries in the listed 2023 comparison

Madhya Pradesh

PYQ LENS · 71st: Bihar state animal, according to exam-oriented option terminology

Bull/Ox; modern secondary lists also identify gaur, so consult the official key for the exact paper

EDITOR ADDITION · Current-fact guardrail

Protected-area counts, Ramsar designations, IUCN assessments and government schemes are dynamic. Static biology concepts in this book are consolidated; facts marked “current” should be checked against official sources immediately before the examination.

Pollution and environmental health

EDITOR ADDITION · Added-scope marker

This pollution table is assistant-added to integrate extended ecology questions from the supplied workbooks.

Issue	Mechanism / indicator	Major response
Air pollution	PM _{2.5} /PM ₁₀ , SO ₂ , NO _x , ozone, CO and toxics harm lungs/ecosystems	Clean fuels, emission control, public transport and monitoring
Water pollution	Sewage, pathogens, nutrients, metals and organics lower quality	Source control and staged treatment; sanitation
Eutrophication	N/P enrichment → algal bloom → decomposition → low dissolved oxygen	Reduce fertiliser/sewage runoff
Biomagnification	Persistent toxin concentration rises at higher trophic levels	Ban/control persistent pollutants; classic DDT/mercury cases
Ozone depletion	Stratospheric O ₃ destroyed by radicals from CFC-related compounds	Montreal Protocol; UV-B protection
Greenhouse warming	CO ₂ , CH ₄ , N ₂ O and others retain outgoing infrared energy	Mitigation plus adaptation
Acid rain	SO ₂ /NO _x form sulfuric/nitric acids	Reduce precursor emissions

- **BOD** is oxygen consumed by microbes while decomposing biodegradable organic matter under defined conditions; high BOD generally indicates high organic pollution. **COD** measures oxidisable load chemically and is typically higher/faster to determine.
- Primary sewage treatment is physical; secondary uses microbes to reduce organic load/BOD; tertiary/advanced steps remove remaining nutrients/pathogens/contaminants according to need.

PYQ LENS · 69th: A major human activity increasing atmospheric CO₂

Deforestation (also fossil-fuel combustion, depending options)

PYQ LENS · 67th: Gas central to anthropogenic greenhouse warming among standard choices

Carbon dioxide

PYQ LENS · 65th: Ozone layer protects from

Ultraviolet radiation

Adaptations and behavioural ecology

EDITOR ADDITION · Added-scope marker

This adaptation summary is assistant-added to connect ecology/evolution PYQs with common organismal examples.

- Adaptation is a heritable feature favoured by selection in an environment; **acclimatisation** is an individual, often reversible physiological adjustment.
- Desert plants: thick cuticle, sunken stomata, reduced leaves/spines, succulent tissues, CAM photosynthesis and deep/spreading roots. Desert animals may be nocturnal and conserve water with concentrated urine.
- Aquatic hydrophytes may have aerenchyma, reduced supporting tissue and stomata only on upper surface in floating leaves. Xerophytes show water-saving traits; halophytes tolerate saline habitats and may use salt glands or succulence.
- Migration is regular directional movement; hibernation is winter dormancy; aestivation is hot/dry-season dormancy. Camouflage, mimicry and warning coloration alter predator-prey interactions.

UNIT 20

PRIORITY A

TOPIC-FIRST · DUPLICATES MERGED

Rapid Revision Atlas & Exam Traps

Consolidated from **All lectures, consolidated** · PYQ theme: Microbes, Disease & Immunity; Plant Form, Function & Classification; Nutrition & Biomolecules; Genetics, DNA & Biotechnology; Ecology, Environment & Conservation

Last-day one-liners

Prompt	Answer
Structural and functional unit of life	Cell
Powerhouse of cell	Mitochondrion
Protein factories	Ribosomes
Packaging/post office	Golgi apparatus
Intracellular digestive vesicles	Lysosomes
Fluid-mosaic model	Singer and Nicolson, 1972
70S ribosome	Prokaryotic cytosol and mitochondria/chloroplasts
80S ribosome	Eukaryotic cytosol
Plant middle lamella	Calcium pectate rich
Photosynthetic plastid	Chloroplast
Starch-storing plastid	Amyloplast
DNA base absent from RNA	Thymine
RNA base replacing thymine	Uracil
Start codon	AUG
Stop codons	UAA, UAG, UGA
DNA replication stage	S phase
Mitosis	Equational division
Meiosis I	Reductional division
Crossing-over stage	Pachytene of prophase I
Chiasmata visible	Diplotene
Mendel's law from monohybrid cross	Segregation
Trisomy 21	Down syndrome
XXY	Klinefelter syndrome
45,X	Turner syndrome
Evolution by natural selection	Darwin and Wallace
Bryophytes	Amphibians of plant kingdom
Pteridophytes	Vascular cryptogams
Gymnosperm seed	Naked; no fruit
Angiosperm speciality	Double fertilisation
Apple	Accessory/false fruit
Clove	Dried flower bud
Ginger/potato	Modified stem
Sweet potato/carrot	Modified root
Water and mineral tissue	Xylem
Food-translocation tissue	Phloem

Prompt	Answer
Photosynthesis C ₄ example	Sugarcane/maize
CAM example	Pineapple/cacti
Ripening hormone	Ethylene
Stress/dormancy hormone	ABA
Growth/phototropism hormone	Auxin
Liver/muscle carbohydrate	Glycogen
Most energy-dense macronutrient	Fat, ≈9 kcal/g
Night blindness	Vitamin A deficiency
Beriberi	Vitamin B ₁ deficiency
Pellagra	Niacin/B ₃ deficiency
Scurvy	Vitamin C deficiency
Rickets	Vitamin D deficiency
Co in a vitamin	Vitamin B ₁₂ /cobalamin
Haemoglobin metal	Iron
Thyroid nutrient	Iodine
Largest gland	Liver
Largest endocrine gland	Thyroid
Small intestine	Complete digestion and most absorption
Bile	Made by liver; stored/concentrated in gall bladder; no enzyme
Functional kidney unit	Nephron
Main human nitrogen waste	Urea
Fertilisation site	Ampullary-isthmic oviduct junction
Emergency contraceptive	Levonorgestrel
Insulin source	Pancreatic islet beta cells
Normal pacemaker	SA node
Universal RBC donor shorthand	O negative
First primary-response antibody	IgM
Most abundant serum antibody	IgG
Placenta-crossing class	IgG
Mucosal/secretory class	IgA
Allergy/parasites class	IgE
Malaria vector	Female Anopheles
Dengue vector	Aedes
Kala-azar vector	Sandfly
Sleeping sickness vector	Tsetse fly
Plague vector	Rat flea

Prompt	Answer
TB bacterium	Mycobacterium tuberculosis
AIDS virus	HIV
Ringworm	Fungal disease, not worm
Antibiotic in disputed item	Chloramphenicol
Cisplatin metal	Platinum
Bee keeping	Apiculture
Silk production	Sericulture
Lac source	Insect
Adult human bones	206 (usual count)
Foot bones	26
Smallest bone	Stapes
Shoulder joint	Ball-and-socket
Myopia correction	Concave lens
Hearing receptor	Organ of Corti
Balance organ	Semicircular canals + vestibule
Speech production	Broca's area
Ecological energy pyramid	Always upright
10% law	Lindeman, approximate
Mutualism	+/+
Commensalism	+/0
Parasitism/predation	+/-
Competition	-/-
GPP – respiration	NPP
Bihar's only national park	Valmiki National Park
World Ozone Day	16 September

Key-sensitive / ambiguous questions — do not memorise blindly

Item	Safe exam response	Why flagged
Who discovered DNA?	Miescher discovered DNA; Watson-Crick modelled double helix	Verb and option set alter key
Insulin classification	Protein/peptide hormone	Workbook answer-field conflict corrected
Antibiotic drug	Chloramphenicol in 60–62 item	Quinine is antimalarial; sulfonamide terminology varies in another item
Blood-cancer isotope	Co-60 for supplied BPSC key; know P-32 nuance	Clinical uses and exam key differ
COVID vaccine mechanism	Immune exposure to antigen/instructions; vector uses harmless carrier	Curated/source explanation conflict
Spore formation	More than one listed group can form spores	Fungi, algae, yeast and fern all can
Groundnut biofertiliser	Rhizobium/Bradyrhizobium	Azospirillum is grass/cereal associated
Virus-host interaction	Endoparasitism, a subtype of parasitism	Overlapping options
Mycorrhiza effect	Both mineral uptake and disease protection	Combined option is strongest
Bihar state animal	Use official key; option may say Bull/Ox while lists may say Gaur	Terminology/source conflict
Current park/Ramsar/IUCN counts	Verify immediately before exam	Dynamic fact

How to attack biology MCQs

- **Define first.** Is the stem asking discovery of substance, structure, function, site, vector, causal agent, deficiency or exception?
- **Mark universals.** Words such as always, only, all and never often signal an overstatement; seek known exceptions.
- **Separate process levels.** Breathing ≠ cellular respiration; excretion ≠ egestion; pollination ≠ fertilisation; immunity ≠ antibiotic action.
- **Trace a pathway.** For heart, nephron, digestive tract, sperm, nerve and xylem, write the ordered route mentally.
- **Match agent to disease.** Vector transmits; pathogen causes; reservoir maintains; vaccine prevents; antibiotic targets bacteria.
- **Use option granularity.** Prefer the more specific answer if it does not contradict a broader one—unless both appear as separate overlapping choices, when the item may be defective.
- **Respect official keys but learn corrections.** The PYQ appendix preserves the supplied answer and a study answer/status so you can distinguish exam history from science.

EDITOR ADDITION · Editorial exam method

The five-step method immediately above is an assistant-added revision aid, not a claim from the lecture PDFs.

Appendix A · Complete reviewed PYQ audit

Every biology-linked question selected from the two supplied workbooks appears below. “Supplied answer” preserves the workbook field; “study answer” is what to learn. Any restoration, correction, ambiguity, key sensitivity or assistant-added answer is visibly badged.

EXAM KEY · Scope rule

CORE = direct biological concept/application. EXTENDED = ecology, wildlife, conservation, agri-biology, nutrition programmes or public-health boundary. This distinction prevents environmental-current-affairs questions from inflating the apparent core-biology trend.

56-59th BPSC (2015)

15 reviewed question(s) · workbook order retained

Q19 · EXTENDED · SOURCE KEY

Sultanpur Bird Sanctuary is located at

Chandigarh · Bharatpur · Gurgaon · Gandhinagar

Study answer: Gurgaon

Q20 · EXTENDED · SOURCE KEY

The forest area in India per person (in average hectare) is

0.23 · 0.34 · 0.2 · 0.29

Study answer: 0.2

Q41 · CORE · SOURCE KEY

Typhoid and Cholera are typical examples of

Infectious diseases · Airborne diseases · Waterborne diseases · None of these

Study answer: Waterborne diseases

Q42 · CORE · RESTORED

Energy is stored in liver and muscles in the form of:

Carbohydrate · Fat · Protein · Glycogen

Study answer: Glycogen

Supplied answer: Glycogen · **Editorial note:** Question text restored from the repository's curated biology bank; the XLSX row has missing text.

Q43 · CORE · SOURCE KEY

Who discovered bacteria?

Fleming · Lamble · Temin · Leeuwenhoek

Study answer: Leeuwenhoek

Q44 · CORE · KEY SENSITIVE

The reason for large-scale diversity among the organisms is

adaptation · Cooperation · Mutation · Polyploidy

Study answer: Adaptation (exam-key answer)

Supplied answer: adaptation · **Editorial note:** Adaptation is the expected BPSC answer; mutation/recombination generate heritable variation, on which selection acts.

Q45 · CORE · SOURCE KEY

What is vermiculture?

The science of raising worms · The science of studying animals · The science of studying fishes · The science of killing worms

Study answer: The science of raising worms

Q47 · CORE · SOURCE KEY

A plant bud is

An embryonic shoot · An embryonic leaf · An endosperm · A seed

Study answer: An embryonic shoot

Q48 · CORE · SOURCE KEY

The average blood flow through kidneys per minute is

1000 cc · 1200 cc · 200 cc · 500 cc

Study answer: 1200 cc

Q49 · CORE · SOURCE KEY**The first human heart transplant was performed in**

America · England · South Africa · France

Study answer: South Africa**Q50 · CORE · SOURCE KEY****The smallest organism, capable of autonomous growth and reproduction, is ...**

Virus · Bacteria · Mycoplasma · [Missing in source]

Study answer: Mycoplasma**Q134 · CORE · RESTORED****Cloves are obtained from which of the following parts of the plant?**

Dried leaves · Dried stems · Dried seeds · [Missing in source]

Study answer: Dried flower buds**Supplied answer:** [Missing in source] **Editorial note:** The workbook option/answer row is truncated; clove is the dried flower bud.**Q135 · CORE · SOURCE KEY****Oncogenes are associated with**

Tuberculosis · Hepatitis · Cancer · Typhoid

Study answer: Cancer**Q137 · CORE · SOURCE KEY****Who synthesized the DNA in vitro?**

Arthur Kornberg · Robert Hooke · Edward Jenner · Joseph Lister

Study answer: Arthur Kornberg**Q142 · CORE · SOURCE KEY****Which of the following can cause cancer as well as cure it depending upon its intensity and use?**

Tobacco · Alcohol · Ionised radiation · Ultraviolet rays

Study answer: Ionised radiation

60-62nd BPSC (Combined)

11 reviewed question(s) · workbook order retained

Q31 · CORE · SOURCE KEY

Which of the following pairs is not correctly matched?

Vitamin B Oranges · Vitamin D - Cod-liver oil · Vitamin E - Wheat germ oil · Vitamin K - Alfalfa · None of the above/More than one of the above

Study answer: None of the above/More than one of the above

Q33 · CORE · SOURCE KEY

Which of the following is responsible for controlling hereditary characters of the living cells?

Enzyme · Hormone · RNA · DNA · None of the above/More than one of the above

Study answer: DNA

Q34 · CORE · SOURCE KEY

Which of the following vitamins is known as ergocalciferol ?

Vitamin D · Vitamin D · Vitamin B · Vitamin B · None of the above/More than one of the above

Study answer: Vitamin D

Q35 · CORE · CORRECTED

Which of the following pairs is not correctly matched?

Retinol - Xerophthalmia · Tocopherol - Beriberi · Cyanocobalamin - Anaemia · Ergocalciferol- Rickets · None of the above/More than one of the above

Study answer: Tocopherol-Beriberi is the mismatched pair

Supplied answer: None of the above/More than one of the above · **Editorial note:** Tocopherol is vitamin E; beriberi is due to thiamine (vitamin B1) deficiency.

Q37 · CORE · CORRECTED

Which one of the following is an antibiotic drug?

Quinine · Sulphaguanidine · Chloramphenicol · Aspirin · None of the above/More than one of the above

Study answer: Chloramphenicol

Supplied answer: Quinine · **Editorial note:** The workbook key points to Quinine; chloramphenicol is the antibiotic among the listed choices.

Q39 · CORE · CORRECTED

Insulin is

Fat · Vitamin · Carbohydrate · Protein · None of the above/More than one of the above

Study answer: Protein (peptide hormone)

Supplied answer: None of the above/More than one of the above · **Editorial note:** The workbook letter maps to 'None'; insulin is a peptide/protein hormone.

Q43 · CORE · SOURCE KEY

Which of the following hormones play a role in release of milk from mammary glands?

Adrenaline · Thyroxine · Progesterone · Oxytocin · None of the above/More than one of the above

Study answer: Oxytocin

Q45 · CORE · SOURCE KEY

In human body, Vitamin A is stored in

Liver · Skin · Lung · Kidney · None of the above/More than one of the above

Study answer: Liver

Q46 · CORE · SOURCE KEY**Which of the following vitamins is used as an antidote to anticoagulant poisons?**

Vitamin A · Vitamin D · Vitamin E · Vitamin K · None of the above/More than one of the above

Study answer: Vitamin K**Q48 · EXTENDED · AMBIGUOUS****Which of the following is present in maximum in acid rain?**HCl · HNO₃ · H₂SO₄ · H₂CO₃ · None of the above/More than one of the above**Study answer:** Sulfuric acid is generally the dominant acid; nitric acid also contributes**Supplied answer:** None of the above/More than one of the above · **Editorial note:** The workbook chooses the combined/none option. The singular word 'maximum' makes H₂SO₄ the standard response, although both H₂SO₄ and HNO₃ occur.**Q50 · CORE · KEY SENSITIVE****Which of the following radio isotopes is used in the treatment of blood cancer [leukemia]?**

Iodine -131 · Sodium-24 · Phosphorus-32 · Cobalt-60 · None of the above/More than one of the above

Study answer: Cobalt-60 (exam key)**Supplied answer:** Cobalt-60 · **Editorial note:** The supplied key selects Co-60; P-32 has also had haematological uses, so follow the official/key context.

63rd BPSC

4 reviewed question(s) · workbook order retained

Q87 · CORE · SOURCE KEY

The element excreted through human sweat is

sulfur · iron · magnesium · zinc · None of the above/More than one of the above

Study answer: None of the above/More than one of the above

Q91 · CORE · SOURCE KEY

In human system, insulin controls the metabolism of

fats · carbohydrates · proteins · nucleic acids · None of the above/More than one of the above

Study answer: carbohydrates

Q92 · CORE · AMBIGUOUS

The antibiotic among the following is

penicillin · aspirin · paracetamol · sulfadiazine · None of the above/More than one of the above

Study answer: More than one: penicillin and sulfadiazine are antibacterial drugs

Supplied answer: None of the above/More than one of the above · **Editorial note:** The broad term 'antibiotic' makes more than one option defensible; the supplied key selects option E.

Q93 · CORE · SOURCE KEY

Which of the following is mostly carbohydrate?

Wheat · Barley · Rice · Maize · None of the above/More than one of the above

Study answer: Rice

64th BPSC

8 reviewed question(s) · workbook order retained

Q5 · CORE · SOURCE KEY

Who discovered DNA?

James Watson and Francis Crick · Gregor Mendel · Johannsen · Hargovind Khorana · None of the above/More than one of the above

Study answer: None of the above/More than one of the above

Q6 · CORE · SOURCE KEY

What is triglyceride?

Protein · Carbohydrate · Fat · Mineral · None of the above/More than one of the above

Study answer: Fat

Q7 · CORE · SOURCE KEY

Electroencephalogram (EEG) is used in monitoring

heart · liver · pancreas · brain · None of the above/More than one of the above

Study answer: brain

Q8 · CORE · SOURCE KEY

Body temperature is regulated by

thalamus · hypothalamus · cerebellum · medulla · None of the above/More than one of the above

Study answer: hypothalamus

Q13 · CORE · SOURCE KEY

Human body contains

about 70% water · 20%-30% water · 10%-20% water · 30%-40% water · None of the above/More than one of the above

Study answer: about 70% water

Q15 · EXTENDED · SOURCE KEY

The main component of greenhouse gases is

carbon dioxide · methane · nitrous oxide · ozone · None of the above/More than one of the above

Study answer: None of the above/More than one of the above

Q17 · CORE · SOURCE KEY

The powerhouse of cell is

cell wall · mitochondrion · ribosome · nucleus · None of the above/More than one of the above

Study answer: mitochondrion

Q18 · CORE · SOURCE KEY

AIDS is caused by

water · bacteria · virus · fungus · None of the above/More than one of the above

Study answer: virus

65th BPSC

13 reviewed question(s) · workbook order retained

Q96 · EXTENDED · SOURCE KEY

The main component of greenhouse gases is

carbon dioxide · methane · nitrous oxide · ozone · None of the above/More than one of the above

Study answer: None of the above/More than one of the above

Q99 · CORE · SOURCE KEY

What is the formula of glucose?

C₆H₂₂O₁₁ · C₆H₁₀O₅ · C₆H₁₂O₆ · C₆H₆O₆ · None of the above/More than one of the above

Study answer: C₆H₁₂O₆

Q101 · CORE · SOURCE KEY

The book, The Origin of Species was written by

Linnaeus · Lamarck · Mendel · Darwin · None of the above/More than one of the above

Study answer: Darwin

Q102 · CORE · SOURCE KEY

The interaction between algae and fungi to form lichen is called

parasitism · mutualism · commensalism · conversion · None of the above/More than one of the above

Study answer: mutualism

Q103 · CORE · SOURCE KEY

The actual location or place where an organism lives is called

habitat · ecosystem · niche · biome · None of the above/More than one of the above

Study answer: habitat

Q104 · CORE · SOURCE KEY

Solar energy is converted into ATP in

mitochondria · chloroplast · ribosome · peroxisome · None of the above/More than one of the above

Study answer: chloroplast

Q105 · CORE · SOURCE KEY

The by-product of photosynthesis is

CO₂ · O₂ · energy · sugar · None of the above/More than one of the above

Study answer: O₂

Q106 · CORE · SOURCE KEY

The study of bones is done under the branch of science called

geology · serology · orology · osteology · None of the above/More than one of the above

Study answer: osteology

Q107 · CORE · SOURCE KEY

Which of the following does not have any enzyme in its cells?

Lichen · Virus · Bacteria · Algae · None of the above/More than one of the above

Study answer: Virus

Q108 · CORE · SOURCE KEY

In which organ of the human body are the lymphocytes formed?

Liver · Bone marrow · Spleen · Pancreas · None of the above/More than one of the above

Study answer: Bone marrow

Q109 · CORE · SOURCE KEY**Which of the following hormones contains iodine?**

Testosterone · Adrenaline · Thyroxine · Insulin · None of the above/More than one of the above

Study answer: Thyroxine**Q110 · CORE · SOURCE KEY****AIDS is caused by**

helminth · bacteria · fungus · virus · None of the above/More than one of the above

Study answer: virus**Q118 · CORE · SOURCE KEY****Body temperature is regulated by**

medulla · thalamus · hypothalamus · cerebellum · None of the above/More than one of the above

Study answer: hypothalamus

66th BPSC

13 reviewed question(s) · workbook order retained

Q1 · CORE · SOURCE KEY

The radiations used in the treatment of muscle ache are

infrared · microwave · UV · X-ray · None of the above/More than one of the above

Study answer: infrared

Q11 · CORE · SOURCE KEY

Milk is a poor source of

calcium · protein · vitamin C · carbohydrate · None of the above/More than one of the above

Study answer: vitamin C

Q13 · CORE · SOURCE KEY

Glucose is converted to ethyl alcohol by the enzyme

maltase · invertase · zymase · diastase · None of the above/More than one of the above

Study answer: zymase

Q21 · CORE · SOURCE KEY

The vitamin which is effective in blood clotting is

vitamin A · vitamin B · vitamin D · vitamin K · None of the above/More than one of the above

Study answer: vitamin K

Q22 · CORE · SOURCE KEY

The female sex hormone is

estrogen · androgen · insulin · oxytocin · None of the above/More than one of the above

Study answer: estrogen

Q23 · CORE · SOURCE KEY

The Nobel Prize for developing treatment of Parkinson's disease was given to

Arvid Carlsson · John F. Enders · Robert B. Laughlin · Walter Kohn · None of the above/More than one of the above

Study answer: Arvid Carlsson

Q24 · CORE · SOURCE KEY

The malfunctioning of thyroid gland is due to the deficiency of

vitamin A · calcium · iodine · iron · None of the above/More than one of the above

Study answer: iodine

Q25 · CORE · KEY SENSITIVE

The sensitive area of the human tongue to bitterness is

tip · middle part · posterior part · edge · None of the above/More than one of the above

Study answer: Posterior tongue in the old sensitivity map; all basic tastes are broadly distributed

Supplied answer: posterior part · **Editorial note:** The classic tongue map is an oversimplification. Posterior sensitivity to bitterness does not mean bitterness is sensed only there.

Q26 · CORE · SOURCE KEY

Ginger is a stem and not root because

it stores food material · it has nodes and internodes · it grows horizontally in the soil · it lacks chlorophyll · None of the above/More than one of the above

Study answer: it has nodes and internodes

Q27 · CORE · SOURCE KEY**The sweetest sugar among the following is**

fructose · glucose · maltose · sucrose · None of the above/More than one of the above

Study answer: fructose**Q28 · CORE · SOURCE KEY****Among the following, which is not a true fruit?**

Apple · Grape · Date · Plum · None of the above/More than one of the above

Study answer: Apple**Q29 · CORE · SOURCE KEY****Legumes are highly nutritious because they are rich in**

fat · protein · oil · starch · None of the above/More than one of the above

Study answer: protein**Q30 · CORE · SOURCE KEY****Clove, a spice, is obtained from which part of the plant?**

Fruit · Stem · Root · Flower bud · None of the above/More than one of the above

Study answer: Flower bud

67th BPSC

15 reviewed question(s) · workbook order retained

Q55 · CORE · SOURCE KEY

What are salt-loving plants called?

Glycophytes · Halophytes · Xerophytes · Mesophytes · None of the above/More than one of the above

Study answer: Halophytes

Q56 · CORE · KEY SENSITIVE

Fungi are plants that lack

chlorophyll · sunlight · oxygen · carbon dioxide · None of the above/More than one of the above

Study answer: Chlorophyll (historical option)

Supplied answer: chlorophyll · **Editorial note:** Fungi lack chlorophyll, but modern classification does not regard fungi as plants; they form a separate kingdom.

Q57 · CORE · SOURCE KEY

Pollination is best defined as

growth of pollen tube in ovule · visiting flowers by insects · transfer of pollen grain from anther to stigma · germination of pollen grains · None of the above/More than one of the above

Study answer: transfer of pollen grain from anther to stigma

Q58 · CORE · SOURCE KEY

Plants receive their nutrients mainly from

light · soil · chlorophyll · atmosphere · None of the above/More than one of the above

Study answer: soil

Q59 · CORE · SOURCE KEY

Pine, Fir, Spruce, Cedar, Larch and Cypress are the famous timber-yielding plants of which several also occur widely in the hilly regions of India. All these belong to

monocotyledons · dicotyledons · angiosperm · gymnosperm · None of the above/More than one of the above

Study answer: gymnosperm

Q60 · CORE · SOURCE KEY

From the following pairs, find the one which is correctly matched.

Tetanus—BCG · Malaria—Chloroquine · Scurvy—Thiamine · Tuberculosis—ATS · None of the above/More than one of the above

Study answer: Malaria—Chloroquine

Q61 · CORE · SOURCE KEY

The animal without red blood cells is

snake · peacock · frog · earthworm · None of the above/More than one of the above

Study answer: earthworm

Q62 · CORE · SOURCE KEY

The ultimate substance to which the carbohydrates are degraded, is

glucose · maltose · amino acid · glycerol · None of the above/More than one of the above

Study answer: glucose

Q63 · CORE · SOURCE KEY

Most of the enzymes are

acids · alkalis · proteins · lipids · None of the above/More than one of the above

Study answer: proteins

Q64 · CORE · SOURCE KEY**Who was the first scientist/doctor, who attempted vaccination?**

Elie Metchnikoff · Edward Jenner · Joseph Lister · Robert Koch · None of the above/More than one of the above

Study answer: Edward Jenner**Q93 · EXTENDED · SOURCE KEY****At which place in Bihar, North India's first bird ringing station has been recently opened?**

East Champaran · West Champaran · Bhagalpur · Saran · None of the above/More than one of the above

Study answer: Bhagalpur**Q94 · EXTENDED · SOURCE KEY****Which of the following in Bihar started the Vulture's Conservation Plan?**

Valmiki Tiger Reserve · Kanwar Lake Bird Sanctuary · Kaimur Tiger Reserve · Rajgir Wildlife Sanctuary · None of the above/More than one of the above

Study answer: Valmiki Tiger Reserve**Q100 · EXTENDED · SOURCE KEY****Recently China has claimed rights over territories around the 'Sakteng Wildlife Sanctuary' claiming it as a 'disputed region'. This Wildlife Sanctuary is located in**

Nepal · Bangladesh · Arunachal Pradesh · Bhutan · None of the above/More than one of the above

Study answer: Bhutan**Q106 · EXTENDED · SOURCE KEY****What is the name of the campaign launched to ensure complete COVID-19 vaccination (in June 2022)?**

Pradhan Mantri Vaccine Campaign · Garib Kalyan Vaccine Campaign 2.0 · Har Ghar Dastak Campaign 2.0 · Atmanirbhar Vaccine Campaign 2.0 · None of the above/More than one of the above

Study answer: Har Ghar Dastak Campaign 2.0**Q148 · EXTENDED · SOURCE KEY****Consider the following statements : 1. In India, the Project Tiger was launched in 1973. 2. The theme for National Biodiversity in 2022 is 'Building a shared future for all life'. 3. The Project Tiger is a tiger conservation programme. Which of the above statements is/are correct?**

2 and 3 only · 1, 2 and 3 · 1 only · 1 and 3 only · None of the above/More than one of the above

Study answer: 1, 2 and 3

68th BPSC

14 reviewed question(s) · workbook order retained

Q66 · EXTENDED · SOURCE KEY

Which animal has been approved as a food animal by the Food Safety and Standards Authority of India (FSSAI) in November 2022?

Himalayan yak · Mountain goat · Himalayan tahr · More than one of the above · None of the above

Study answer: Himalayan yak

Q70 · EXTENDED · SOURCE KEY

Which month of the year is celebrated as Rashtriya Poshan Maah or National Nutrition Month under the Poshan Abhiyaan of the Government of India?

May · September · July · More than one of the above · None of the above

Study answer: September

Q97 · EXTENDED · SOURCE KEY

Which one of the following is the world's richest region from a marine biodiversity perspective, comprising 21 islands with estuaries, beaches, forests of the nearshore environment, seagrasses, coral reefs, salt marshes and mangroves?

Gulf of Mannar Biosphere Reserve · Nanda Devi Biosphere Reserve · Sunderbans Biosphere Reserve · More than one of the above · None of the above

Study answer: Gulf of Mannar Biosphere Reserve

Q123 · EXTENDED · SOURCE KEY

Which gas contributes to global warming through its greenhouse effect?

Nitrous oxide · Methane · Carbon dioxide · More than one of the above · None of the above

Study answer: More than one of the above

Q131 · CORE · SOURCE KEY

The stomata open or close due to change in the

protein composition of cells · amount of water in cells · position of nucleus in cells · More than one of the above · None of the above

Study answer: amount of water in cells

Q132 · CORE · SOURCE KEY

Chlorophyll absorbs _____ wavelengths of the sunlight.

green and blue · green and red · red and blue · More than one of the above · None of the above

Study answer: red and blue

Q133 · CORE · SOURCE KEY

Which of the following is not a correct match?

Thick, hard stem with branching near base : Tree · Weak stem which cannot stand upright : Creeper · Petiole : Attaches leaf to stem · More than one of the above · None of the above

Study answer: Thick, hard stem with branching near base : Tree (This describes a shrub)

Q134 · CORE · SOURCE KEY

Water reaches great heights in trees because of suction pull caused by

absorption · transpiration · evaporation · More than one of the above · None of the above

Study answer: transpiration

Q135 · CORE · SOURCE KEY

Which among the following is a character of chloroplast which makes them qualified to self-replication?

Presence of DNA only · Absence of RNA · Presence of both DNA and RNA · More than one of the above · None of the above

Study answer: Presence of both DNA and RNA

Q136 · CORE · SOURCE KEY

The parts of the skeleton that are not as hard as bones and which can be bent are called as

carpals · cartilage · vertebrae · More than one of the above · None of the above

Study answer: cartilage

Q137 · CORE · SOURCE KEY

In lichen, a chlorophyll-containing partner, ____ live together.

fungus and a bacterium · alga and a virus · alga and a fungus · More than one of the above · None of the above

Study answer: alga and a fungus

Q138 · CORE · SOURCE KEY

Female Anopheles can be distinguished from female Culex because it sits

at an angle with substratum · parallel to the surface of substratum · at right angle to the surface of substratum · More than one of the above · None of the above

Study answer: at an angle with substratum

Q139 · CORE · SOURCE KEY

The most important cell type associated with the immunity of the body is

platelets · lymphocytes · RBCs · More than one of the above · None of the above

Study answer: lymphocytes

Q140 · CORE · SOURCE KEY

Which one of the following statements is incorrect about muscles in human body?

Muscles work in pairs. · Muscles become shorter, stiffer and thicker when contracted. · Muscles can only push the bone. · More than one of the above · None of the above

Study answer: Muscles can only push the bone.

69th BPSC

16 reviewed question(s) · workbook order retained

Q3 · CORE · SOURCE KEY

Match List-I with List-II : (Cloth) a. Linen b. Coir c. Mohair d. Down (Origin) 1. Coconut plant 2. Flax plant 3. Feathers of duck and geese 4. Angora goat

a-1, b-3, c-2, d-4 · a-4, b-3, c-1, d-2 · a-2, b-1, c-4, d-3 · a-4, b-1, c-3, d-2

Study answer: a-2, b-1, c-4, d-3

Q4 · CORE · SOURCE KEY

Consider the following statements : 1. Taeniasis is an intestinal infection caused by three species of tapeworm. 2. When cysts develop in the brain, the condition is referred to as neurocysticercosis (NCC). Which of the above statements is/are correct?

Only 1 · Only 2 · Both 1 and 2 · None of the above

Study answer: Both 1 and 2

Q7 · CORE · SOURCE KEY

Which of the following are not the tastes of the tongue? 1. Sweet 2. Bitter 3. Salty 4. Spicy 5. Umami 6. Sour 7. Pungent

2, 5 and 7 · 1, 3 and 4 · 4 and 7 · 3 and 6

Study answer: 4 and 7

Q8 · CORE · SOURCE KEY

Which of the following pairs are correctly matched? 1. Anorexia : Sleep disorder 2. Insomnia : Eating disorder 3. Dyspnoea : Shortness of breath 4. Anosmia : Partial or full loss of smell

Only 1 and 2 · Only 2 and 3 · Only 3 and 4 · All of the above

Study answer: Only 3 and 4

Q18 · CORE · VERIFIED

How do COVID vaccines stimulate an immune response?

By introducing live attenuated SARS-CoV-2 virus · By introducing a real SARS-CoV-2 virus · By introducing a harmless piece of SARS-CoV-2 virus · By introducing antibodies against SARS-CoV-2 virus

Study answer: A harmless antigenic piece/instruction from SARS-CoV-2

Supplied answer: By introducing a harmless piece of SARS-CoV-2 virus · **Editorial note:** Vaccines train immunity without introducing a disease-causing dose of the real virus.

Q20 · CORE · SOURCE KEY

The DNA double helix structure was discovered by

James Watson and Francis Crick · Rosalind Franklin and Maurice Wilkins · Linus Pauling · Gregor Mendel

Study answer: James Watson and Francis Crick

Q22 · CORE · VERIFIED

How do vector vaccines work to provide immunity?

By introducing a weakened or inactivated virus into the body · By directly attacking and destroying pathogens in the body · By placing the virus in a modified version of a different virus · By entering directly into the cells and enabling them to create spike proteins

Study answer: A modified harmless carrier virus delivers genetic instructions

Supplied answer: By placing the virus in a modified version of a different virus · **Editorial note:** This is the defining idea of a viral-vector vaccine.

Q25 · CORE · SOURCE KEY

What is the expanded form of the term 'mRNA' that has been widely discussed since the beginning of the pandemic?

Messenger Ribonucleic Acid · Mutant Ribonucleic Acid · Modified Ribonucleic Acid · Mnemonic Ribonucleic Acid

Study answer: Messenger Ribonucleic Acid

Q30 • CORE • SOURCE KEY**Input and output nerves meet at**

liver • central nervous system • heart • None of the above

Study answer: central nervous system**Q46 • EXTENDED • SOURCE KEY****Which State Government launched the State-Level Committee for Vulture Conservation (SLCVC), proposed by the national action for the protection of vultures in India?**

Tamil Nadu • Sikkim • Assam • Meghalaya

Study answer: Tamil Nadu**Q47 • EXTENDED • SOURCE KEY****Ten captive-bred Asian Giant Tortoises (*Manouria emys*), one of the largest tortoise species in mainland Asia, were recently released into which National Park of India?**

Ntangki National Park • Periyar National Park • Pench National Park • Khangchendzonga National Park

Study answer: Ntangki National Park**Q53 • EXTENDED • SOURCE KEY****Tillyardembi Fossils of the world's first plant pollinators, called Tillyardembiids, were discovered recently in which country?**

Greece • India • Russia • China

Study answer: Russia**Q54 • EXTENDED • SOURCE KEY****The 'Lisbon Declaration', which made headlines, is associated with the conservation of which entity?**

Air • Mountains • Oceans • Freshwater glaciers

Study answer: Oceans**Q86 • EXTENDED • SOURCE KEY****Which of the following National Parks/Wildlife Sanctuaries are in Bihar? 1. Valmiki National Park 2. Dudhwa National Park 3. Gajner Wildlife Sanctuary 4. Bhimbandh Wildlife Sanctuary**

1 and 4 • 2 and 3 • 1 and 3 • 2 and 4

Study answer: 1 and 4**Q87 • EXTENDED • SOURCE KEY****Consider the following statements regarding the 'Gangetic Dolphin' : 1. Categorized as 'endangered' under IUCN's Red List. 2. Does not have a crystalline eye lens. 3. Navigation and hunting by echolocation. 4. Recognized as India's National Aquatic Animal. Which of the above statements are correct?**

Only 1, 2 and 4 • Only 2, 3 and 4 • Only 1, 3 and 4 • All of the above

Study answer: All of the above**Q93 • EXTENDED • SOURCE KEY****Which one of the following is the world's richest region from a marine biodiversity perspective, comprising twenty-one islands with estuaries, beaches, forests... coral reefs, salt marshes and mangroves?**

Gulf of Mannar Biosphere Reserve • Nanda Devi Biosphere Reserve • Sunderbans Biosphere Reserve • Nilgiri Biosphere Reserve

Study answer: Gulf of Mannar Biosphere Reserve

70th BPSC

12 reviewed question(s) · workbook order retained

Q3 · CORE · SOURCE KEY

Which of the following acid is secreted in the stomach?

Sulphuric acid · Nitric acid · Phosphoric acid · Hydrochloric acid

Study answer: Hydrochloric acid

Q31 · CORE · KEY SENSITIVE

Which component of fertilizer is used for stimulating early growth purpose?

Potassium · Nitrogen · Oxygen · Phosphorus

Study answer: Nitrogen (wording-dependent)

Supplied answer: Nitrogen (Commonly associated with early vegetative growth) · **Editorial note:** The workbook selects nitrogen for early vegetative growth; phosphorus is especially associated with early root development.

Q39 · EXTENDED · SOURCE KEY

In which city of Bihar the recently inaugurated National Dolphin Research Centre is located?

Darbhangha · Patna · Munger · Gaya

Study answer: Patna

Q41 · CORE · SOURCE KEY

Which of the following bacterial disease that is called as splenic fever?

Typhoid · Anthrax · Cholera · None of the above

Study answer: Anthrax

Q45 · CORE · SOURCE KEY

Which of the following radiation is used to get relief from body aches?

Infra-red radiation · UV radiation · Visible radiation · None of these

Study answer: Infra-red radiation

Q56 · CORE · AMBIGUOUS

In which organisms does spore formation occur as a mode of reproduction?

Fungi · Algae · Yeast · Fern

Study answer: More than one listed organism can form spores

Supplied answer: Fungi (Fern is also correct, but Fungi is universally associated with this fundamental mode) · **Editorial note:** Fungi, many algae, yeast and ferns can all form spores in relevant life cycles; the workbook's single choice is defective.

Q66 · CORE · SOURCE KEY

Which of the following is the fluid part of the blood?

WBC · Plasma · Blood platelets · RBC

Study answer: Plasma

Q79 · CORE · SOURCE KEY

What is the structural and functional unit of life?

Cell · Organ · Tissue · Organism

Study answer: Cell

Q120 · CORE · SOURCE KEY

Which of the following the chlorophyll in photosynthesis is used for?

Reduction of CO₂ · Breaking down water molecules · No function · Absorbing light

Study answer: Absorbing light

Q134 · CORE · SOURCE KEY**Which part of alimentary canal receives bile from the liver?**

Stomach · Oesophagus · Small intestine · Large intestine

Study answer: Small intestine**Q143 · CORE · SOURCE KEY****The process of breaking down organic matter by micro organisms is called**

Respiration · Photosynthesis · Decomposition · Fermentation

Study answer: Decomposition**Q148 · EXTENDED · SOURCE KEY****Which of the following is a major cause of air pollution?**

Disposal of plastics · Burning of fossil fuels · Deforestation · Excessive use of pesticides

Study answer: Burning of fossil fuels

71st BPSC (2025)

20 reviewed question(s) · workbook order retained

Q14 · EXTENDED · ADDED ANSWER

Which organisation recognized the Nagi bird sanctuary Site as an Important Bird and Biodiversity Area (IBA)?

UNESCO · World Wildlife Fund · Bird Life International · More than one of the above

Study answer: BirdLife International

Supplied answer: not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q40 · EXTENDED · ADDED ANSWER

Which of the following statement/s is/are NOT CORRECT about the U-WIN portal of Ministry of Health and Family welfare?

It provides QR-based e-vaccination certificates · It facilitates the creation of Ayushman Bharat Health Accounts (ABHA)... · The portal is accessible in 11 regional languages · None of the above

Study answer: None of the statements is incorrect

Supplied answer: not supplied · **Editorial note:** All three listed U-WIN features are supported; answer absent in supplied workbook.

Q112 · EXTENDED · ADDED ANSWER

Kanha National Park is established in -

Uttarakhand · Assam · Madhya Pradesh · Maharashtra

Study answer: Madhya Pradesh

Supplied answer: not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q114 · EXTENDED · ADDED ANSWER

Which of the following parks/sanctuaries come under the project tiger in Bihar?

Valmiki National Park · Udaypur Wildlife Sanctuary · Bhimbandh Wildlife Sanctuary · None of these

Study answer: Valmiki National Park / Valmiki Tiger Reserve

Supplied answer: not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q115 · EXTENDED · KEY SENSITIVE

Which of the following is the state animal of Bihar?

Cow · Bull · Buffalo · Horse

Study answer: Bull/Ox (exam-oriented answer)

Supplied answer: not supplied · **Editorial note:** Secondary lists conflict between Bull/Ox and Gaur; retain the exam-oriented option and revise from the official key used for your paper set.

Q134 · CORE · ADDED ANSWER

Which metal compound is used for treatment of carcinoma?

Platinum · Chromium · Iron · Chlorine

Study answer: Platinum compound (e.g., cisplatin)

Supplied answer: not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q135 · CORE · ADDED ANSWER

Which compound is used in contraceptive pills?

Cholecalciferol · Levonorgestrel · Venlafaxine · Cetirizine

Study answer: Levonorgestrel

Supplied answer: not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q138 · CORE · ADDED ANSWER**Full form of PHBV biodegradable polymer:**

Polyhydroxy butyl vaniline · Poly hydroxy butane veratric acid · Poly (3 - Hydroxy butyrate-co-3 hydroxy valerate) · Poly hydroxy butyric acid veratric acid

Study answer: Poly(3-hydroxybutyrate-co-3-hydroxyvalerate)**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q139 · CORE · ADDED ANSWER****Hormones are secreted by:**

Sebaceous glands · Reproductive glands · Exocrine glands · Endocrine glands

Study answer: Endocrine glands**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q140 · CORE · AMBIGUOUS****Which one of the following term characterizes the interaction between herpes simplex virus and a human?**

Parasitism · Symbiosis · Endosymbiosis · Endoparasitism

Study answer: Parasitism; more specifically intracellular endoparasitism**Supplied answer:** not supplied · **Editorial note:** Options A and D overlap; answer absent in supplied workbook.**Q141 · CORE · ADDED ANSWER****Which class of immunoglobulin is the first to be produced in response to an infection?**

IgG · IgA · IgM · IgE

Study answer: IgM**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q142 · CORE · ADDED ANSWER****The biological membranes are composed of:**

Lipids and proteins in varying combinations which are specific to each species, cell type and organelle · Lipids only · Proteins only · None of the above

Study answer: Lipids and proteins in species/cell-specific proportions**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q143 · CORE · ADDED ANSWER****Which of the following is a non-sense codon?**

UAA · AUG · UUU · UGC

Study answer: UAA**Supplied answer:** not supplied · **Editorial note:** UAA is one of the three stop/nonsense codons; answer absent in workbook.**Q144 · CORE · ADDED ANSWER****The species which are in danger of extinction are known as?**

Vulnerable · Rare · Endangered · Extinct

Study answer: Endangered**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q145 · EXTENDED · ADDED ANSWER****Which of the following is a Kharif crop?**

Sugarcane · Barley · Gram · Wheat

Study answer: Sugarcane**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Q146 · CORE · ADDED ANSWER**Xylem in plants is responsible for following function:**

Transport of water and dissolved minerals · Transport of foods · Transport of oxygen and other gases · Transport of essential amino acids

Study answer: Transport of water and dissolved minerals**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q147 · CORE · ADDED ANSWER****Who was first to use the characteristics of vascular tissues in the classification of plants?**

Engler and Prantl · Bentham and Hooker · AP de Candolle · More than one of the above

Study answer: A. P. de Candolle**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q148 · CORE · ADDED ANSWER****The part which we eat in Mango is:**

Epicarp · Mesocarp · Endocarp · None of the above

Study answer: Mesocarp**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q149 · EXTENDED · ADDED ANSWER****Ozone day is celebrated on:**

16th September · 5th June · 21st April · 5th December

Study answer: 16 September**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.**Q150 · CORE · ADDED ANSWER****In mycorrhizal association, the advantage of plant is:**

Protection · Food · (A) and (B) both · Increased mineral absorption and diseases protection

Study answer: Increased mineral absorption and protection from disease**Supplied answer:** not supplied · **Editorial note:** Answer absent in supplied workbook; content answer added by editor.

Appendix B • Original visual provenance

The notes use only reviewed figures extracted from the supplied lecture PDFs. No web image was inserted. When one original figure was reused for a closely related concept, its original source remains stated in the caption.

Asset	Source lecture	Page
Eukaryotic Cell	L1. SYLLABUS ANALYSIS & CELL.pdf	4
Prokaryotic Cell	L1. SYLLABUS ANALYSIS & CELL.pdf	4
Heterotrophic Nutrition	L10. Nutrition, Part - 01.pdf	2
Carbohydrate Structures	L10. Nutrition, Part - 01.pdf	3
Protein Formation	L10. Nutrition, Part - 01.pdf	4
Digestive System	L13. Digestive System, Part - 01.pdf	3
Tooth	L13. Digestive System, Part - 01.pdf	4
Absorption Summary	L13. Digestive System, Part - 01.pdf	5
Digestive Glands	L14. Digestive System, Part - 02.pdf	3
Human Heart	L15. Circulatory System, Part - 01.pdf	3
Vertebrate Hearts	L15. Circulatory System, Part - 01.pdf	3
Blood Components	L16. Circulatory System, Part - 02.pdf	3
Cardiac Conduction	L17. Circulatory System, Part - 03.pdf	2
Blood Vessels	L17. Circulatory System, Part - 03.pdf	4
Capillary Bed	L17. Circulatory System, Part - 03.pdf	4
Alveolus	L18. Respiratory System- 01.pdf	2
Inhalation Process	L18. Respiratory System- 01.pdf	2
Respiratory System	L18. Respiratory System- 01.pdf	2
Breathing Mechanics	L18. Respiratory System- 01.pdf	3
Lung Lobes	L19. Respiratory System- 02.pdf	2
Cell Membrane	L2. Cell Organelles.pdf	2
Endoplasmic Reticulum	L2. Cell Organelles.pdf	3
Mitochondrion	L2. Cell Organelles.pdf	3
Nephron	L20. Excretory System & Reproductive System.pdf	4
Reproductive System	L20. Excretory System & Reproductive System.pdf	6
Endocrine Glands	L21. Human Endocrine System.pdf	2
Brain	L22. Human Endocrine System.pdf	4
Neuron	L23. Nervous System - 02.pdf	2
Nerve Impulse	L23. Nervous System - 02.pdf	3
Reflex Arc	L23. Nervous System - 02.pdf	3
Plant Kingdom Classification	L24. Plant Kingdom.pdf	2
Bryophyte Classes	L24. Plant Kingdom.pdf	3
Pteridophytes	L24. Plant Kingdom.pdf	3
Gymnosperms	L24. Plant Kingdom.pdf	5
Root Systems	L25. Plant Morphology.pdf	2
Underground Stems	L25. Plant Morphology.pdf	3
Leaf Parts	L25. Plant Morphology.pdf	4
Plant Modifications	L25. Plant Morphology.pdf	4

Asset	Source lecture	Page
Dengue Vector	L27. Infectious and Non-Infectious Diseases, Part - 02.pdf	4
Hiv Cycle	L27. Infectious and Non-Infectious Diseases, Part - 02.pdf	5
Hiv Structure	L27. Infectious and Non-Infectious Diseases, Part - 02.pdf	5
Sars Cov 2	L27. Infectious and Non-Infectious Diseases, Part - 02.pdf	5
Asexual Plant Reproduction	L28. Plant Physiology.pdf	2
Plant Physiology	L28. Plant Physiology.pdf	2
Flower Structure	L28. Plant Physiology.pdf	3
Photoperiodism	L28. Plant Physiology.pdf	5
Chromosome Structure	L29. Genetics and Evolution.pdf	3
Colour Blindness Test	L29. Genetics and Evolution.pdf	4
Mendel Traits	L29. Genetics and Evolution.pdf	4
Sex Determination	L29. Genetics and Evolution.pdf	4
Lamarck Darwin	L29. Genetics and Evolution.pdf	5
Homologous Analogous	L29. Genetics and Evolution.pdf	6
Golgi Apparatus	L3. Cell Organelles II.pdf	2
Chloroplast	L3. Cell Organelles II.pdf	3
Lysosome	L3. Cell Organelles II.pdf	3
Coelom Types	L30. Animal Diversity.pdf	2
Annelida	L30. Animal Diversity.pdf	3
Platyhelminthes	L30. Animal Diversity.pdf	3
Porifera	L30. Animal Diversity.pdf	3
Aschelminthes	L30. Animal Diversity.pdf	4
Echinodermata	L30. Animal Diversity.pdf	4
Mollusca	L30. Animal Diversity.pdf	4
Ecosystem Components	L31. Ecology Microbes(Bacteria & Virus).pdf	2
Food Chain Web	L31. Ecology Microbes(Bacteria & Virus).pdf	2
Ecological Pyramids	L31. Ecology Microbes(Bacteria & Virus).pdf	3
Aquatic Biomass Pyramid	L31. Ecology Microbes(Bacteria & Virus).pdf	4
Energy Flow	L31. Ecology Microbes(Bacteria & Virus).pdf	4
Centrosome	L4. Nucleus Centrosome, Pollination.pdf	2
Nucleus	L4. Nucleus Centrosome, Pollination.pdf	3
Dna Rna	L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf	2
Dna Structure	L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf	2
Rna Structure	L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf	3
Rna Types	L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf	3
Transcription Translation	L6. Nucleic Acid-DNA RNA and Protein Synthesis.pdf	4
Mitosis Meiosis	L7. Cell Division (Mitosis & meiosis).pdf	2
Mitosis Stages	L7. Cell Division (Mitosis & meiosis).pdf	2
Meiosis Stages	L7. Cell Division (Mitosis & meiosis).pdf	4
Collenchyma	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	2
Leaf Anatomy	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	2
Parenchyma	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	2

Asset	Source lecture	Page
Phloem	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	3
Sclerenchyma	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	3
Xylem Transport	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	3
Annual Rings	L8. Plant tissue - Meristem, Permanent tissue, vascular tissue , annual rings.pdf	4
Adipose Tissue	L9. Animal tissues and Skeleton system.pdf	2
Epithelium Types	L9. Animal tissues and Skeleton system.pdf	2
Connective Tissues	L9. Animal tissues and Skeleton system.pdf	3
Muscle Types	L9. Animal tissues and Skeleton system.pdf	3
Long Bone	L9. Animal tissues and Skeleton system.pdf	4
Rib Cage	L9. Animal tissues and Skeleton system.pdf	5
Skull	L9. Animal tissues and Skeleton system.pdf	5
Vertebral Column	L9. Animal tissues and Skeleton system.pdf	5

Appendix C • Sources, method & limitations

Primary source corpus. All 33 PDFs in EDUTERIA BIO NOTES, from L1 through L32, plus the supplied 56th–59th and 60th–71st BPSC question workbooks. Two L22 PDFs are byte-identical; both were inventoried but their content was not duplicated.

Consolidation method. Page-delimited text and embedded figures were extracted, every lecture was read, recurring concepts were grouped, duplicate statements were merged, unique details were retained, and selected visuals were linked to a page-level manifest. Questions were reviewed individually rather than selected solely by keyword.

Editorial additions/corrections. Blue and coral boxes visibly identify externally supplied context and scientific corrections. For standard verification the editor used mainstream textbook/agency-level biology, including NCERT Class XI–XII Biology, WHO public-health terminology, IUCN conservation categories, and official programme descriptions. The 71st U-WIN item was checked against the official BPSC final-key context and Government of India programme features.

Limitations. A supplied answer key can contain errors; several are explicitly badged. Protected-area counts, IUCN status, government schemes and health guidance can change after this September 2026 edition. Verify dynamic facts and the exact official key for the examination you are preparing for.

KEY-SENSITIVE • Health disclaimer

This is an examination study aid, not medical advice. Disease descriptions and treatments are deliberately concise; diagnosis, vaccination and medicines require current professional guidance.

END • Revise Unit 20, then revisit every coral key-sensitive card.